From Posts to Prediction: Using Influence-weighted Investor Sentiment for Next-day Return Forecasting

Authors: Bubu Zhang (202008949) and Sebastian Eriksen Grunnet-Jensen (202303137)

Supervisor: Martin Rud Langerhuus

June 2025

Department of Economics

Characters incl. blanks and figures: 192,132

Abstract

This study examines whether investor sentiment extracted from social media becomes more informative for return forecasting when adjusted for user influence. The underlying assumption is that incorporating engagement-based influence metrics increases the predictive value of sentiment variables. If this proves to be true, weighting sentiment by follower count and engagement should improve short-term return forecasts compared to using raw sentiment alone. To evaluate this, a novel influence-weighted sentiment variable is constructed from X posts, using raw sentiment as the baseline for comparison. Sentiment is classified using the large language models RoBERTa and FinBERT, while user influence is quantified based on follower count, likes, and reposts. The variable is tested against unweighted sentiment in forecasting next-day S&P 500 index returns across several machine learning models and feature sets. Results show that weighting sentiment by influence enhances predictive and directional accuracy on a statistically significant level, indicating that market reactions are more strongly driven by sentiment from influential users than by lower-engagement content. These findings contribute to behavioral finance by highlighting the importance of sentiment diffusion dynamics. The results also offer practical implications for financial forecasting, demonstrating that incorporating user influence into sentiment analysis can make alternative data more actionable for predicting asset returns.

Figures and Tables

Figures:

Tables:

Table of Contents

1. Introduction

In today's highly competitive environment, forecasting stock returns remains a central challenge in financial economics, with direct implications for asset pricing, portfolio construction, and trading strategy (Cochrane, 2005). Traditional frameworks such as the Capital Asset Pricing model (CAPM) and the Fama-French framework, have long been central for academic and applied financial analysis. Typically, these models are estimated using Ordinary Least Squares (OLS) regression, under strict assumptions of rational investors, efficient markets, and stable risk–return relationships (Brealey et al., 2022). However, persistent market anomalies undermine these assumptions (Banz, 1981; Bondt & Thaler, 1985, and the inability of CAPM to capture unsystematic risk and investor-specific preferences (Sharma, n.d.) have cast doubt on their sufficiency. Behavioral finance offers another view on the role of irrational decision-making driven by investor sentiment, by acknowledging that investor emotions, ranging from panic to enthusiasm can lead to mispricing in the market (Shiller, 2003). Thus, when conventional frameworks can no longer account for the observed anomalies, a paradigm shift toward more flexible, data-driven methods becomes essential (Kuhn, 1970; Gleißner & Ernst, 2019).

Simultaneously, finance is experiencing a data revolution, with over 90% of global data being generated in the last two years alone (G2, 2025), much of it being unstructured and drawn from novel sources such as earnings calls, news, satellite imagery, and social media (Wigglesworth, 2019). Traditional models struggle to process this volume and variety, prompting widespread adoption of machine learning (ML) and natural language processing (NLP) (Najafabadi et al., 2015). ML uncovers nonlinear patterns in high-dimensional data (Gu et al., 2020; Ndikum, 2020), and by integrating unstructured signals, such as sentiment, ML has already demonstrated significant improvements in forecast accuracy (Chen et al., 2023). This evolution aligns with the Adaptive Market Hypothesis, which emphasizes bounded rationality, learning, and dynamic environments (Lo, 2004).

Sentiment analysis, entailing quantifying investor mood from text has emerged as a promising NLP application. Platforms like X (formerly Twitter) offer real-time insights in investor moods, and studies show that these signals have predictive power for stock returns (Tetlock, 2007; Bollen et al., 2011a). In response, researchers have begun incorporating sentiment factors into asset-pricing models, seeking to capture the behavioral drivers behind market moves. However, most sentiment models in literature treat all posts equally, ignoring differences in reach and engagement. In reality, a post's influence can depend on follower count and user interactions (likes, reposts, replies). Engaging content from a high-profile account, or a repost is more likely to have an impact on investor sentiment than a low-visibility post. Yet existing models rarely weight sentiment signals by diffusion potential, leaving a critical gap in how investor sentiment is quantified and integrated into return forecasts (Lachana & Schroeder, 2024). This thesis addresses that gap by developing an influence-weighted sentiment variable, to more accurately capture how sentiment propagates through social media and impacts market outcomes. This is then tested to see if the performance on return forecasting is better than with raw sentiment. If this improves forecasting accuracy even by a small margin, it can have significant implications for asset managers and traders. The purpose of this study is motivated by potential practical implications, to see whether the weighted-sentiment variable could be a better tool to capture and predict market movements, contributing to more informed and data-driven decision-making.

1.1 Problem statement and Research questions

The central problem currently concerns that the developing literature, using sentiment from social media as signals for return forecasting, does not take into consideration any measures of how sentiment can diffuse. This may misrepresent the true economic impact of social media content on stock returns. In particular, aggregating sentiment without weighting can be misleading, because while counting the number of X posts expressing a given sentiment provides useful information, there are also quiet indicators, such as likes, replies and reposts which also reflect sentiment. A post with more likes, shows agreement with a sentiment and posts that are reposted can have an effect on how many people can be influenced. Without considering this aspect of social media, extracting sentiment from posts risks being statistically significant but economically spurious. To address these issues, the overarching research question is posed:

How does the predictive performance of influence-weighted sentiment variables for forecasting S&P 500 index returns compare to that of raw sentiment alone?

The following sub-questions are designed to guide the analysis:

> **Sub-RQ1**: How can LLMs be used to extract investor sentiment from X posts, and how can engagement and influence indicators be incorporated into a novel sentiment variable?
>
> **Sub-RQ2**: To what extent do influence-weighted sentiment variables improve next-day return forecasting accuracy over unweighted (raw) sentiment?

Sub-RQ1 is essential because it defines and constructs the influence-weighted sentiment score (IWSS) metric itself. Without a procedure for labeling posts (via fine-tuned LLMs) and then weighing those labels by follower counts, likes, reposts, and replies, there is no IWSS to test. Once IWSS is properly extracted and validated, Sub-RQ2 directly evaluates whether IWSS adds value beyond raw sentiment. Together, they answer the overarching question by first ensuring that IWSS is correctly defined and then comparing its predictive performance to that of raw sentiment.

By answering these research questions, the study will contribute to expanding literature on sentiment-based asset pricing. Further, by integrating insights from behavioral finance and computational methods it aims to demonstrate how modern modelling techniques can enable a more adaptive and realistic framework to test if sentiment is useful for forecasting stock returns in the modern data-driven financial market.

Note, that although the influence-weighted sentiment variable (IWSS) introduced in this thesis is grounded in established theories, and some financial motivations are explained, the primary focus is empirical rather than theoretical. In particular, the construction of IWSS draws on concepts from behavioral finance and common factors and models from finance are used, but the focus of the study is on evaluating its predictive accuracy for next-day S&P 500 returns. Thus, readers should view this work as a data-driven, machine-learning based investigation anchored in finance, rather than as a pure finance-theory paper.

1.2 Thesis Outline

The subsequent chapters of this thesis are structured to answer the research questions. Chapter 2 reviews key literature on asset pricing, behavioral finance critiques, and the emergence of ML/NLP and social media sentiment in financial forecasting. Chapter 3 describes the four-phase methodology encompassing data collection/cleaning, sentiment extraction via fine-tuned LLMs, IWSS feature engineering, and return forecasting with OLS, Random Forest, XGBoost, and LSTM models. Chapter 4 presents the empirical results and discusses them including model performance metrics and significance tests comparing raw sentiment and influence-weighted sentiment. Chapter 5 summarizes contributions, outlines practical implications, while Chapter 6 suggests avenues for extending IWSS for future research. Lastly, Chapter 7 concludes the thesis by re-addressing the research questions and the problem statement.

2. Literature Review

Understanding how and to what extent investor sentiment can improve stock return forecasting, especially when drawn from social media platforms like X and adjusted for user influence, directly frames the core of this study. This focus aligns with the central research question, acknowledging the increasing integration of sentiment into financial modeling. Addressing the systematic comparison of unweighted forecasting performance versus influence-adjusted sentiment signals requires a foundation in behavioral finance, which emphasizes the role of sentiment-driven inefficiencies in financial markets. This review also builds on the application of recent advances in forecasting methodology, including the application of more complex machine learning and deep learning techniques. Furthermore, literature on sentiment extraction is reviewed, from early rule-based approaches to more advanced transformer-based language models, evaluating the effectiveness of quantifying sentiment in financially meaningful ways.

2.1 Theories and Concepts

Investor sentiment is defined as the collective belief of investors about future returns and investment risks that are not substantiated by fundamental data (Baker & Wurgler, 2007). According to behavioral finance, sentiment is a key driver of market inefficiencies that can instigate market bubbles and crashes as investors can behave irrationally (Shiller, 2003). Under the Efficient Market Hypothesis (EMH), all available information is instantly reflected in prices so any informational edge would be immediately arbitraged away, leaving no predictable excess return (Fama, 2004). However, documented anomalies such as momentum effects (Jegadeesh & Titman, 1993) and market overreactions (Brown & Cliff, 2001) challenge the strong-form EMH, paving the way for Behavioral Finance theories.

Behavioral finance frameworks like Prospect Theory (Kahneman & Tversky, 1979) provide psychological explanations for these deviations, suggesting that people make decisions based on perceived gains and losses relative to a reference point. Investors thus tend to dislike losses more than they like equivalent gains (loss aversion) and tend to outweigh small probabilities. This results in biased and inconsistent behaviors. For instance, in financial markets, sentiment can explain anomalies in stock returns. In practice, overly optimistic sentiment can inflate valuations of lottery-type stocks (low probability of high payoffs), leading to the assets underperforming relative to the expectations (Barberis & Huang, 2008). Conversely, negative sentiment amplifies loss aversion, contributing to higher demanded risk premiums for volatile assets, which helps explain the equity premium puzzle, where equities historically deliver higher average returns compared to safer assets like Treasury bills (Benartzi & Thaler, 1995). However, the prospect theory has been criticized as Barberis (2013) notes, investors can anchor on different reference points (purchase price, peer performance, past returns), so there is no single baseline to judge outcomes, making it unclear whether a given price change constitutes a gain or a loss. This ambiguity underscores the need to specify, in any empirical model, how reference points are chosen or inferred when extracting sentiment signals.

Construal Level Theory (CLT) provides a complementary perspective, by explaining how temporal framing affects decision-making. CLT suggests that distant or long-term events are processed abstractly, emphasizing long-term benefits, while immediate events are processed concretely, highlighting specific risks (Trope & Liberman, 2010). Eyal et al. (2004) demonstrate that individuals focus more on pros when considering long-term decisions but shift to cons when evaluating short-term choices. This is because people think broader and more positively about the future, meaning they are more likely to commit to long-term actions. Applied to investors, a bullish (positive) social media post framed around “five-year growth” encourages risk-tolerant, long-horizon positions, whereas a warning about “tomorrow’s earnings” can trigger short-term caution. However, the study’s use of diverse undergraduate samples raises questions about whether expert investors follow the same cognitive framing, suggesting a gap in translating CLT findings directly into market behavior.

Empirical research supports the behavioral assertion that sentiment influences market behavior. Tetlock (2007) finds that negative tone in financial news forecasts modest price declines, while Bollen et al. (2011) show that aggregate Twitter mood can statistically anticipate movements in the Dow Jones Industrial Average (DJIA). Thus, studies collecting sentiment from social media or news sources consistently indicate that positive or negative public mood correlates with subsequent stock trends, confirming that collective buying and selling, driven by sentiment, can move prices away from fundamental values. Without these empirical validations, there would be little justification for incorporating sentiment into predictive models.

These theories provide the foundations for how sentiment can explain market inefficiencies. The remaining sections of the literature review will provide a detailed examination of existing research. Existing literature on stock return forecasting will be examined, followed by a comprehensive review of sentiment analysis techniques for extracting sentiment from text. Lastly, the review will conclude with research that focuses on how sentiment is incorporated into predictive models for stock return, and how this study can build on the gaps and limitations of the existing literature.

2.2 Forecasting methods for stock returns

Stock return forecasting traditionally builds on top of models like ARIMA and GARCH, which analyze time series movements and volatility. These models enable the identification of trends, autocorrelation, and fluctuations in the market over time. Additionally, regression-based models like CAPM and Fama-French are utilized to explain returns through factors like market risk, company size, and value. Along with the increasing amounts of data and processing power, more advanced methods from machine learning and deep learning have been adopted to model complex and non-linear relationships in financial data.

2.2.1 Traditional methods

Models like ARIMA and GARCH, as well as regression-based models like CAPM and Fama-French, have long been prominent methods within the landscape of traditional forecasting. These models provide valuable insight into tendencies, volatility, and factor-driven returns. Despite machine learning showing promising results, traditional forecasting methods are still important due to their interpretability, stability, and documented effectiveness in financial analysis.

2.2.1.1 Time-series models

Traditional stock return forecasting methods have traditionally employed time-series models like ARIMA and GARCH, to capture autocorrelations and volatility clustering. ARIMA (Autoregressive Integrated Moving Average) models capture linear autocorrelation in time series data. The autoregressive (AR) component reflects how past returns influence current returns, the moving average (MA) component accounts for sudden shocks, and the integration (I) term enforces stationarity (Wang, 2024). This makes the model suitable for modeling the impact previous behavior has on current behavior, which is observable in stock returns.

On the other hand, GARCH (Generalized Autoregressive Conditional Heteroskedasticity) models explicitly models volatility clustering. A general overview of GARCH describes that volatility in time-series depends on past errors and past volatility. When used in conjunction with financial time series, returns tend to show volatility clustering, meaning that large fluctuations in price are likely to be followed by large fluctuations, and small fluctuations by small fluctuations (Wang, 2024). These models can be stacked into a single ensemble, improving short-term return predictions compared to using each model in isolation. To evaluate a stacked ensemble model like this, Wang (2024) showcases how the prediction errors are reduced, when applying the model in contexts like equity forecasting for firms like Amazon and Caterpillar, underscoring the potential of hybrid approaches that integrate trend and volatility dynamics. However, because both ARIMA and GARCH assume linear relationships, their performance often deteriorate when applied to more complex or high-frequency datasets, where the relationships may be too complex for basic linear models to capture.

2.2.1.2 Regression-based models

Another common approach to forecasting stock returns involves regression-based models, focusing on how observable factors drive returns, either across time (time-series regressions) or across assets at a given point (cross-sectional regressions). The Ordinary Least Squares (OLS) method creates the foundation for many return-predictive frameworks, enabling researchers to estimate linear relationships between returns and a set of explanatory variables. Within time-series regression, OLS has been utilized for return-predictability testing, using variables such as yields, interest rates, and lagged returns. However, due to financial data exhibiting a high noise-to-signal ratio, the out-of-sample effectiveness of OLS has been questioned (Shih et al., 2024).

In cross sectional settings, regression models are central to understanding the drivers of return differences across assets. The Capital Asset Pricing Model (CAPM) was one of the first models linking expected returns to market risk through a single beta coefficient. However, the model has multiple empirical deficiencies, such as its inability to explain the effects of size and value. This turned into the development of the Fama-French three-factor model, introducing firm size (Small Minus Big) and book-to-market value (High Minus Low) as additional explanatory variables (Fama, 2004). Later, the model has been extended to a five-factor model, where profitability and investment factors contribute to the explanatory power. Although these factor models are grounded in economic rationale, they are still estimated using OLS or common econometric techniques, assuming stable, linear relationships between factors and returns (Perold, 2004).

2.2.1.2 Extensions to Machine Learning Models

Traditional regression-based factor models, such as Fama–French, have been re-estimated using ML techniques to boost predictive performance. Diallo et al. (2019) demonstrate this by applying Bayesian-optimized support-vector regression (BSVR) to both three- and five-factor Fama French Factors. These include size, value, profitability, investment, and market risk. Their results are mainly positive, with the three-factor model achieving out-of-sample correlations above 90 percent on portfolio returns, even outperforming the five-factor version in some cases. Although Diallo et al.'s study focuses on a limited set of industry returns, it emphasizes how ML can extract stronger signals from a modest factor set and avoid overfitting that might accompany additional factors.

Gu et al. (2020) extend ML applications much more broadly by employing Random Forests (RF), Neural Networks (NN), and elastic-net regularization on a universe of over 30,000 stocks and 900 predictors. Their findings show that non-linear models consistently yield positive out-of-sample R^2 and therefore attain better predictive performance than the OLS-regression models.

Backtesting showed the highest cumulative returns for Neural Networks, using the same factors. This study in particular highlights the strength of using ML methods, both in managing more variables with growing literature contributing to what can help predict returns and in capturing nonlinear dependencies that traditional OLS models fail to detect.

2.2.2 Machine Learning and Deep Learning Methods

As seen in the previous section, modeling often equals utilizing powerful tools to uncover patterns in complex data (e.g. financial) and generating predictive insights. Machine learning and deep learning techniques are increasingly used within stock return forecasting. The following section highlights some foundational models and modern deep learning techniques, outlining how past literature has utilized them for financial prediction tasks.

2.2.2.1 Machine Learning Methods

Several key supervised learning algorithms have been applied to predict stock prices or stock returns with help from historical market data and various features. A select set of machine learning methods are outlined below, chosen for their relevance to the predictive task:

1. **Random forest (RF)** is an ensemble supervised learning method widely used in the field of machine learning. Instead of relying on a single decision tree, RF builds multiple trees using two key techniques, one is bagging, where each tree is trained on a random sample (with replacement) of the original training data. Then, at each split, a random subset of features is considered, which reduces correlation among trees and lower overall variance. During prediction, each decision tree "votes," and the forest's final output is typically the majority class (for classification) or the average prediction (for regression). By aggregating many weak learners, RF often achieves higher predictive accuracy and greater robustness to overfitting than a single tree. In financial applications, RF can capture complex, nonlinear interactions among inputs such as technical indicators, fundamental ratios, and macroeconomic variables without specifying an explicit functional form. For example, if one tree asks a sequence of binary questions (e.g., "Is volatility above threshold X?" or

"Does the price-to-earnings ratio exceed Y?") to split the data into subsets, another tree might find a different combination of splits altogether. The ensemble of all these diverse trees thus learns a flexible decision boundary in high-dimensional space (IBM, 2024a).

2. **XGBoost**, also known as eXtreme Gradient Boosting, is a distributed, open-source machine learning library, and an advanced implementation of gradient boosted decision trees with the same general framework. Like Random Forest, XGBoost is an ensemble learning method, but instead of bagging multiple trees in parallel, it builds trees sequentially, where each new tree corrects errors made by the previous ensemble. This design prioritizes speed and performance, making it a good model to apply for larger-scale datasets. It supports a multitude of programming languages, but specifically for some of XGBoost's features in python, that make it stand out from the normal gradient boosting, involve a range of key features. These features include parallel and distributed computing, a cache-aware prefetching algorithm which helps reduce the runtime for large datasets, handling of missing data, and built in regularization to prevent overfitting (Kavlakoglu & Russi, 2024).

3. **Neural networks (NN)** are programs within machine learning inspired by the human brain, designed to capture complex patterns and nonlinear relationships within data. They consist of layers of interconnected nodes, starting with an input layer, one or more hidden layers, and an output layer. Every node handles input by taking weights and bias, summing them up, and passing that result through an activation function (e.g., ReLU or sigmoid), which then determines the output. If a certain threshold is exceeded in this process, it activates the node and sends the data to the next layer in the network. During training of the NN, the weights and biases go through backpropagation, where they adjust based on prediction errors, minimizing loss. Over successive training, this process helps improve the accuracy of prediction. A key advantage of NNs is that their hidden layers automatically learn and combine relevant features, reducing the need for extensive manual feature engineering. This capacity to learn nonlinear mappings and extract hierarchical representations makes NNs particularly well-suited for modeling complex financial data (IBM, 2025).

Many existing studies show increased performance by using these ML methods. Wang (2022) conducted research on Chinese A stocks with 640 days of data, generating indicators like RSI, MACD, CCI, and KDJ as functions for modeling. The results emphasize that ML models like RF, XGBoost, and Neural Networks outperform simply using OLS, with XGBoost showing the best ability to predict whether returns will be positive or negative. Further, Omar et al. (2022) forecast the Karachi-100 index using daily closing prices and autoregressive lags (one-day, one-week, one-month). They show RF and XGBoost outperform ARIMA and GARCH in terms or predictive accuracy based on prediction error (MSE) and ability of the input features to explain the target variable (R^2). The Neural Network also performs well but requires extensive hyperparameter tuning to achieve good results. Similarly, Owen & Oktariani (2020) enhance U.S. equity price history with StockTwits sentiment and news-headline indices. ML models consistently reduce out-of-sample MSE by 10–15%, additionally ML models such as XGBoost showed the highest amount of excess returns (Sharpe ratio of ~1.2).

In summary, the existing literature emphasizes that supervised ML models can help improve predictive accuracy in return forecasting compared to traditional and simple regression models. Methodologically, studies use a mix of historical pricing, technical indicators, and new data sources to train such models and evaluate them on their prediction accuracy and financial parameters. Performance wise all three models usually surpass traditional benchmarks like linear or ARIMA models, however their relative performance rank varies depending on context. RF and XGBoost offer speed and stability and often yield strong results with less data or simpler functions. Deep neural networks, especially advanced architecture, have delivered higher accuracy, but with the disadvantage of requiring more data and greater complexity. Comparative analyses highlight that no model alone dominates universally. Each has its own strengths, making it more suitable for specific market conditions or datasets. The latest innovations move towards combining models and data sources to utilize their complementary strengths and make complex models more interpretable and in compliance with investment targets. The growing consensus is that a thoughtful selection (or fusion) of models, controlled by the details of the problem and informed by these empirical findings, is key to improving stock return forecasting in practice.

2.2.2.2 Deep learning methods

Stock return prediction with deep learning has increasingly drawn intense research attention in recent years. Among different architectures LSTM-networks (Long Short-Term Memory) and transformer models have proved to be two leading approaches in return forecasting.

LSTM networks (Long Short-Term Memory) are a class of recurrent neural networks designed to mitigate the vanishing-gradient problem and capture long-term dependencies in sequential data. By using gated memory cells (input, output and forget gates), the LSTM can store relevant information over extended horizons, which can be helpful for addressing the vanishing gradient problem and capture long-term dependencies. They can help improve forecasting accuracy by preserving and propagating information over longer time horizons (IBM, 2024b). This property can be useful for financial forecasting, holding long-term information like trends, volatility regimes, or momentum effects that may be missed by standard NNs. In stock return forecasting, LSTMs have shown strong performance when datasets are modest in size, since their recurrent structure limits parameter growth and focuses on recent history.

Transformer models are a type of deep learning model introduced in 2017, which has quickly become a popular model within NLP and other machine learning tasks. The advantage of transformer models is their ability to process sequential inputs parallel, making them more efficient than earlier models such as RNN's. Two of the central components in Transformer models are positional encoding, and self-attention. Positional encoding allocates a unique value for each word to preserve the sequential order. Self-attention evaluates each word in relation to other words in the sequence, which helps the model to focus on the most relevant parts of the input (IBM, 2024b). However, they tend to be more computationally expensive, and risk overfitting to smaller datasets.

Empirical comparisons in literature have shown mixed results. Lahboub & Benali (2024) evaluate both architectures on a small dataset of daily stock prices for multiple companies and report that the LSTM achieves an R^2 above 0.95, whereas the transformer sometimes produces negative R^2 values which could be attributed to the transformer's high data demand in a limited-sample context. In contrast, Wang & Yuan (2023) compare a pure transformer, an LSTM, and a hidden

Markov model on Chinese new-energy stock index data, finding the transformer achieves the lowest forecasting errors and highest directional accuracy. These contrasting findings in literature imply that transformers excel when more informative history is available, whereas LSTMs can still perform well on smaller or more stable environments (Lahboub & Benali, 2024).

However, both architectures are prone to overfitting, although it is more pronounced in transformer models due to their complexity. Transformers by nature have more regularization parameters such as dropout, early stopping, model checkpoint callbacks, and data augmentation which are essential to prevent overfitting. Lahboub & Benali (2024) say that LSTMs with fewer parameters can also sometimes attain good results, even with shorter training histories. However, they also benefit from techniques like dropout to improve robustness. An interesting observation from Kabir et al. (2025) is that Transformers might require more feature engineering or hybrid designs in the financial world, e.g. by combining them with CNN's or using them to complement instead of replacing LSTM's, to reach their full potential. This is partly due to financial time series being noisy and non-stationary, and that a purely attention-based model might lock onto spurious patterns unless it's run by inductive biases or additional data.

Despite considerable progress, the current deep learning methods for stock prediction face several gaps and challenges before they can be fully applied for real-world asset management. Overfitting remains a central concern, especially with limited financial data. While researchers are aware of this and use techniques like cross-validation or early stopping, there is still a tendency in literature to report in-sample or same-period test results that are likely to overestimate the true performance. More work with robust model validation is needed, for example by using longer out-of-sample periods, more market contexts, and even stress-testing models on extreme events (Suárez-Cetrulo et al., 2023).

2.3 Sentiment Classification Extraction

Investor sentiment is gaining traction in research on being a key factor in influencing financial behavior. To capture and quantify sentiment in textual data, researchers apply NLP techniques aiming to identify and measure emotions triggered by specific stimuli such as news headlines, earnings reports, or social media content (Frijda, 1994). Sentiment is typically classified in two ways, either in a multi-dimensional manner with multiple emotional states (i.e., calm, alert, happy, etc.) (Bollen et al., 2011) or using a two-dimensional model based on valence (positive, negative or neutral) and arousal, indicating high or low (Smailović et al., 2013). Although both models are valid, financial studies tend to exclusively focus on the valence categorization of the sentiment due to its simplicity. This study will follow this approach and use the valence categorization of sentiment.

2.3.1 Lexicon/Dictionary-based approaches

Traditional sentiment analysis approaches are lexicon-based, relying on predefined sets of words, phrases and rules. This involves matching terms to lists or dictionaries that indicate positive or negative sentiment. Various tools such as OpinionFinder, TextBlob, and Vader are widely used to automate this process (Wilson et al., 2005). This is a popular method in finance-related studies, using static dictionaries like Harvard-IV or Loughran-Mcdonald to label texts from social media and headlines of news articles (Tetlock, 2007; Sul et al., 2016).

A major benefit of this method contributing to its popularity is that it does not require any manual labelling and categories based on word matching. However, despite the wide usage, the lexical method struggles to capture context and cannot detect nuances like sarcasm or negations, which can lead to misclassification. For instance, the sentence: “Oh great, another policy” could be misclassified as positive due to the word “great” (Sul et al., 2016). Additionally, the static nature of the dictionaries would be unable to keep up with evolving social media slang or domain-specific jargon in financial discourse (Barbaglia et al., 2023).

2.3.2 Machine Learning Approaches

To address the limitations of lexical approaches, researchers increasingly use supervised machine learning techniques that can learn sentiment patterns from labelled data. Existing literature shows that models such as Support Vector Machines (SVMs) and random forest are commonly used and consistently outperform lexicon-based tools, as they are better at handling financial language and informal expressions (Kamil & Shah, 2024). Most sentiment analysis for financial texts use supervised machine learning, where datasets are pre-labelled with valence sentiment categories. ML models typically achieve higher classification accuracy, and lower false positives as they are trained on pre-labelled text (Machová et al., 2022). However, ML methods are resource intensive and require domain-expertise and substantial manual labelling of texts, potentially leading to inconsistent labels (Dhaoui et al., 2017). Additionally, ML models suffer from poor transferability, often performing poorly on data outside of their training domain (Nandwani & Verma, 2021).

2.3.3 Deep Learning Approaches

Deep learning models emerge as a more robust alternative to ML models by reducing the reliance on manual labelling and enhancing the ability to capture complex patterns and sequential dependencies in text. Long Short-Term Memory (LSTMs) in particular have shown higher classification accuracy in sentiment analysis by capturing long-term dependencies in text. For instance, Yin (2024) developed an LSTM model trained on Chinese social media data that achieved 98.21% classification accuracy, significantly outperforming ML models. While its generalizability to finance may be limited due to the wide range of topics that could be covered, other studies have produced evidence to support the superior performance of LSTM over traditional ML in financial contexts. Wang et al. (2020) also find that LSTMs outperform ML methods like XGBoost with an accuracy score of around 95%. Further it performs better across other evaluation metrics like the F1 score and AUC which account for data imbalance, which is important for unstructured text data from financial articles. Nevertheless, deep learning models still face challenges in handling sarcasm and irony and require substantial computational power.

The most recent advancement in sentiment analysis uses large language models (LLMs) such as transformer-based models. Transformer models are a type of neural network architecture that is designed to capture textual relationships and dependencies in the data. BERT (Bidirectional Encoder Representations from Transformers) builds on this architecture, leveraging its attention mechanisms to understand the bidirectional context of words (considering both past and future context) within a sentence, enabling a better contextual understanding. BERT models are typically pre-trained on a large corpus and can be finetuned for specific tasks and are better at capturing long-range dependencies compared to LSTMs (Su, 2024). Further, BERT-based models handle complex sentiment expressions including sarcasm more effectively (Elmitwalli & Mehegan, 2024).

There are enhanced variations of BERT. FinBERT for example is specifically trained on financial texts and has shown more accurate classification of investor sentiment compared to other models, including ML and deep learning models (Araci, 2019). RoBERTa is an optimized version of BERT which is trained on even more data to improve performance and generalizability. However, the findings in literature are divided when it comes to which BERT model is superior in terms of performance. Karanikola et al. (2023) found that RoBERTa outperforms both traditional ML models 7% in accuracy and surpassed the performance of FinBERT in accuracy and F1 scores. Conversely, Kumar and Chaturvedi (2024) found FinBERT to outperform RoBERTa in similar financial contexts, exhibiting superior knowledge on financial jargon. Hence, while literature shows that BERT-based models do outperform other models for sentiment extraction, there are mixed findings that highlight the data-specific nature of deep learning models. Therefore, this study will adopt a BERT-based approach. However, as literature suggests model performance is highly data-dependent, the study emphasizes empirical validation and testing to determine the most suitable model to analyze sentiment in X posts.

2.4 Sentiment as a Predictor of Stock Return

The previous sections establish forecasting methodologies and sentiment extraction techniques from text. The following section will review the empirical studies on how to combine the methods and address the research questions on how sentiment can impact stock returns. There is a strong theoretical and empirical foundation suggesting that investor sentiment affects decision-making and subsequently stock returns. Extensive research explores how sentiment signals, derived from traditional media, online forums, and social media, can be used to predict asset price movements.

Early research uses traditional media, such as news articles and message boards as a proxy for investor sentiment. Studies found correlations between negative sentiment of wall street journal articles and declines in stock returns (Tetlock, 2007). Similarly, Antweiler and Frank (2004) showed that message board sentiment is significantly related to stock returns. However, the magnitude of the effect is small, limiting its impact for traders. The study controlled for firm-specific factors, time, and market index movements, however, other omitted variables like macroeconomic factors were not accounted for which could affect the sentiment-return relationship.

With the emergence of real-time platforms, studies have begun incorporating social media sentiment. Social media platforms contain a significant amount of data which can be utilized by the average person. Platforms like X are now widely used among investors, where their posts are often tagged with ticker symbols. This offers a proxy for investor sentiment under the assumption that authors of such tweets are active market participants. Bollen et al. (2011) linked Twitter mood to movements in the Dow Jones Industrial Average using Granger causality and neural networks, resulting in over 87% predictive accuracy. However, these findings were later questioned due to methodological weaknesses and lack of out-of-sample validation (Lachanski & Pav, 2017). Subsequent studies have increasingly incorporated lexical approaches to sentiment analysis of X posts alongside traditional asset pricing frameworks, for instance although sentiment index alone is correlated with stock returns, the explanatory power using regression was limited. Moreover, when using traditional asset pricing models like the Fama-French three-factor model, sentiment significantly improved the model's ability to explain returns, also when controlling for other financial indicators like volatility (Azar & Lo, 2016). Spikes in tweet activity and sentiment have been linked to abnormal stock returns, especially around earnings or unexpected news (Ranco et al., 2015), demonstrating the growing importance of social media in sentiment-based forecasting.

With the exponential increase of large-scale alternative data, there has been a growing emphasis in recent studies on using advanced machine learning and deep learning techniques to more effectively model the relationship between sentiment and stock returns and extract sentiment. Supervised sentiment models trained on financial news have demonstrated improved out-of-sample return predictions. The improved sentiment variables were then integrated into Fama-French style regressions which yielded a high accuracy. To take it one step further, backtesting the study achieved high Sharpe ratios (>4) when converting the output of their model into a trading signal (Ke et al., 2019). The adoption of LLMs such as BERT, OPT, and ChatGPT has further enhanced the accuracy of sentiment extraction. When combined with a simple regression model based on previous day return, sentiment modeled by LLMs achieved the highest explanatory power and Sharpe ratio for backtested long/short strategies compared to lexical approaches (Kirtac & Germano, 2024).

Deep learning forecasting methodologies have also been explored in recent studies, combined with the more sophisticated sentiment extraction techniques to enhance return predictions. Numerous studies use sentiment variables extracted from FinBERT, as it has consistently demonstrated enhanced performance in literature. By incorporating FinBERT-based sentiment with technical indicators like moving averages, researchers have improved return forecasts over longer horizons. For instance, LSTM models have shown to be effective in capturing temporal patterns in online media data like X and news articles, demonstrating strong performance when comparing predicted and actual out-of-sample returns (Das et al., 2018). LSTMs also generally outperform traditional models like ARIMA, and further including a sentiment variable derived from FinBERT improves predictive accuracy more. The combination of FinBERT and LSTM manages to outperform ARIMA and LSTM without the sentiment variable, as well as a base BERT combined with LSTM on shorter time frames. This suggests that the forecasting model itself significantly impacts performance, as deep learning approaches can capture nuanced details beyond sentiment alone. Although, the predictive strength prove to be stronger on short term rather than long term horizons (Zeng & Jiang, 2023).

A transformer-based model that combines FinBERT-derived sentiment from daily news headlines with technical indicators has also shown promise in predictive accuracy for long-term horizons. The model shows using transformers outperforms LSTM over longer timeframes by leveraging self-attention mechanisms to capture long-term dependencies (Kaeley et al., 2023). However, its advantage diminishes over shorter horizons. Furthermore, it is important to consider that both Zeng and Jiang (2023) and Kaeley et al. (2023) extract sentiment based on a single daily headline, which may not fully capture the sentiment conveyed in a full article and reduce the reliability of a sentiment forecast. While X posts that will be used in this current study differ significantly from news headlines in language and length, the findings of these studies still are relevant. This limitation is particularly relevant in financial markets, where information evolves rapidly and investor sentiment can shift significantly within short timeframes. In such dynamic environments, relying on brief or isolated textual inputs may lead to sentiment signals that are incomplete or lagging. Although X posts are quite different linguistically from news headlines, these results are still important. They emphasize the importance of sentiment as a predictive variable and highlight how the method of extraction, as well as the sophistication of the modeling framework employed, influences forecasting accuracy.

While early literature found only modest correlations between sentiment and returns, recent advances leveraging domain-specific language models, event-based structuring, and financial regressions have produced more robust results, both statistically and economically. This sets the stage for further innovation in asset pricing models through the inclusion of sentiment derived from dynamic, high-frequency sources such as X, which may offer more accurate and real-time insights into market expectations and investor behavior.

2.5 Research Gaps

While the literature increasingly supports the use of sentiment in forecasting stock returns, several key research gaps remain. One key gap is methodological inconsistency, which makes it difficult to compare sentiment-based forecasting results across studies. Sentiment-driven return models draw on widely varying data sources, from news articles and earnings reports to discussion forums and social media (Azar & Lo, 2016; Ke et al., 2019). Additionally, they also employ different extraction techniques, for instance, using lexicon-based dictionaries or transformer architecture like RoBERTa and FinBERT to extract sentiment from text. Evaluation approaches also diverge, with some researchers reporting regression metrics (e.g., RMSE) and others focusing on classification accuracy. Kearney and Liu (2014) specifically note that this heterogeneity in information sources, sentiment-analysis methods, and modeling frameworks leads to reproducibility challenges and prevents unified inference. When one study uses news headlines to form a dataset with a custom dictionary and another uses X data with a transformer like FinBERT, differences in forecasting performance may arise from the data or extraction method, rather than from a genuine improvement in modeling sentiment. Similarly, conflicting evaluation criteria mean that a model which achieves the best performance in one metric may not hold up under another. As a result, readers cannot reliably determine which approaches truly outperform others. To address this, the study will standardize both data and evaluation, using the publicly available S&P 500 and X (formerly Twitter) datasets and compare raw-sentiment and influence-weighted sentiment models using consistent metrics (next-day directional accuracy, RMSE). This allows for a robust comparison and clarifies which approach delivers genuine predictive gains.

Another key limitation revolves around what is generally considered a sentiment signal. Most models treat sentiment as an indicator, assuming that each post, tweet, or article has equal impact. However, this overlooks the fact that some platforms and users are more well-known which can skew sentiment diffusion. For example, larger news platforms like the New York Times and Bloomberg have a bigger audience and farther reach. Similarly, social media platforms like X can have a small number of highly followed users that can disproportionately affect sentiment (Barabási, 2018). Influence can thus play a large role in sentiment diffusion which could have an impact on the sentiment variable itself. Understanding the impact of sentiment thus requires more than just looking at the text itself. It should also entail who writes the text and how the information spreads, looking at the credibility and reach of users, especially on social media platforms. A comparative analysis by Cha et al. (2010) reveals that a large follower base does not mean there is high engagement or message propagation, instead they found that users who consistently engage with their audience and concentrate on specific topics tend to exert greater influence through retweets and mentions. This supports the “million follower fallacy,” which argues that follower count alone is an inadequate proxy for influence. This can be inflated by social norms or bots on X. Influence is better reflected through engagement metrics, such as how often a user’s content is shared or discussed.

The way information spreads will also have clear implications for return predictability. Users and media with high influence have a large reach, thus, one would expect they would have a larger impact in the diffusion of sentiment. However, their messages would diffuse more rapidly, which would lead to a quicker assimilation in the asset prices as demonstrated in studies (Peress, 2012). While the referenced study focuses on individual stocks, the current study examines a market index. However, similar effects may persist, given that the index is commonly accessed through instruments like SPY and may still have implications for investors monitoring the stock market. The gradual information diffusion theory (Hong & Stein, 1999) suggests that mispricing occurs because information is incorporated into prices gradually, as some investors respond slowly while others amplify trends by reacting to past price movements. Influence on social media can accelerate this process by amplifying the trends contributing to price movements at the market index level. Therefore, incorporating user influence and metrics into sentiment modeling is crucial, as treating all sentiment equally can overlook key differences in how information originates and spreads. Accounting for influence could substantially enhance the reliability and effectiveness of sentiment-based financial forecasting. Although influence and engagement metrics could be incorporated directly as predictor variables in the model for stock returns, this study will instead construct a weighted sentiment variable. This approach will more effectively capture the relationship between sentiment and influence, which aligns with the objective of examining their role in sentiment diffusion rather than modelling their effects on stock returns.

2.6 Synthesis of Findings in Literature

Figure 1 synthesizes key findings into two parallel pipelines, forecasting models on the left and sentiment-extraction methods on the right, each advancing through progressively more sophisticated stages before converging in combined approaches for return prediction.

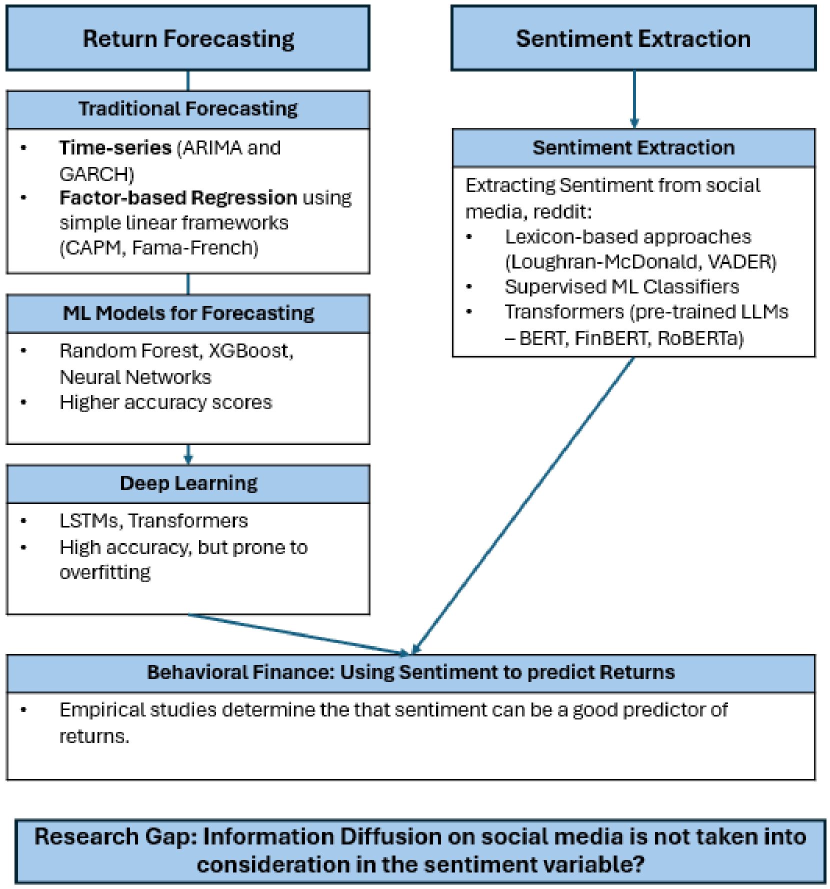


Figure 1: Synthesis of Existing Literature

In the forecasting pipeline, the literature begins with traditional time-series techniques (ARIMA and GARCH) that capture linear autocorrelation and volatility clustering. It then incorporates factor-based regressions (CAPM, Fama–French) to explain returns using variables such as market beta, size, book-to-market ratios and more alternative data. Next, studies introduce "shallow" machine learning models (Random Forest, XGBoost, Neural Networks), which capture nonlinearities and interaction effects and consistently outperform the simpler linear models on out-of-sample accuracy. Finally, recent research evaluates deep-learning architectures (LSTMs and Transformers) which can encode longer dependencies and broader feature sets. Empirical evidence suggests LSTMs perform very well out-of-sample, however, as a newer technique needs to be validated more thoroughly as both have the tendency to overfit without careful regularization.

On the sentiment extraction side, methods start with lexicon or dictionary-based approaches (e.g., Loughran-McDonald, VADER) that match words to static sentiment lists. While these require no labelled data, they often misclassify domain-specific jargon and fail to detect sarcasm. To address this, supervised machine learning classifiers (SVMs, Random Forests) are trained on labelled financial texts or social media posts, improving accuracy but demanding extensive annotation and risking poor out-of-domain transfer. More recent work uses transformer-based models (BERT, FinBERT, RoBERTa) which are fine-tuned on financial corpora or social-media data, yielding the best performance by understanding context, slang, and nuance, although with high computational costs and data requirements.

Both pipelines (lexicon-based methods and machine learning-based models) converge as studies combine sentiment variables with return-forecasting models. Based on behavioral finance concepts, investor sentiment could have an impact on improving return forecasts. Research has shown that many sentiment extraction techniques were combined with a variety of different forecasting methods. Across these studies it is determined that sentiment can be a good predictor with positive correlations as well as high accuracy in predicting returns, although with varying levels. When back tested some studies even show a potential for profitability with Sharpe Ratios above 1. However, nearly all existing research treats each text unit as equally informative, overlooking how social-media posts diffuse through networks of varying influence. This gap motivates the thesis: first, to extract investor sentiment from X posts using LLMs and weight those signals by user influence metrics (Sub-RQ1), second, to test whether the resulting influence-weighted sentiment improves return forecasts relative to unweighted sentiment (Sub-RQ2).

3. Methodology

This chapter will look at the method that will be used for this study. This research adopts a deductive, multi-phase research quantitative design to evaluate whether an influence-weighted sentiment variable can improve next-day return prediction for the S&P 500 index. The methodology is structured around 3 main objectives as aligned with the sub-research questions. There are three primary objectives, aligned with the study's sub-research questions:

1. To extract investor sentiment from social media using LLMs (Sub-RQ1a)

2. To construct a novel sentiment variable that incorporates influence to capture diffusion in social media (Sub-RQ1b)

3. To evaluate the predictive power of the new variable in forecasting next-day stock returns (Sub-RQ2).

This study combines social media and financial data, applying natural language processing, feature engineering and supervised learning techniques. A time-based data split (train, validation, test sets) ensures out-of-sample integrity and avoids information leakage to maintain validity of the results.

3.1 Research Process

The four-phase approach is described as follows:

1. Data collection and cleaning of X Posts in preparation for sentiment analysis,
2. Sentiment analysis to extract sentiment from the cleaned posts using LLMs (objective 1),
3. Variable construction informed by preliminary data exploration (objective 2), and
4. Return forecasting and evaluation, where the calculated returns are mapped to the sentiment variable (objective 3).

Figure 2 summarizes this four-phase approach, illustrating how the research methodology proceeds from the collection of raw data (section 3.2) to sentiment extraction (section 3.3) then to construction the influence-weighted variables (section 3.4) and finally to return forecasting and evaluation (section 3.5).

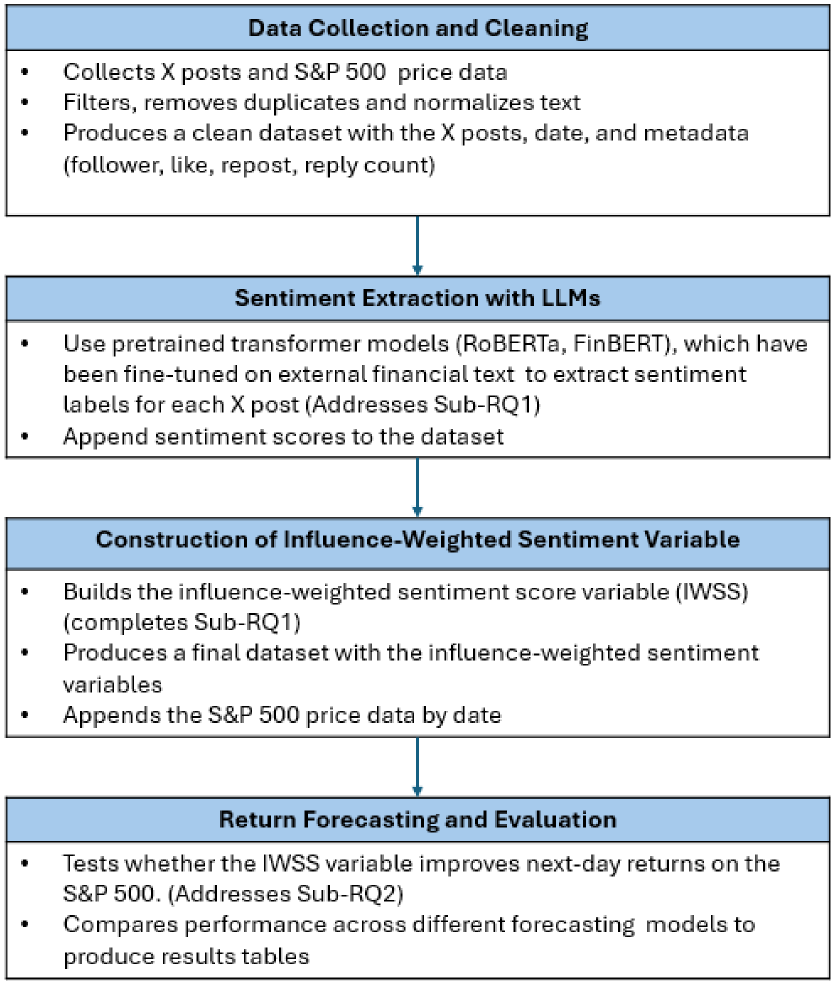


Figure 2: Outline of the four-phase approach and intended outcomes.

Guided by Bryman's (1984) framework for quantitative research (Bryman, 1984), this study follows a hypothetico-deductive epistemology. This approach proposes that knowledge progresses through generating theories and hypotheses, then using empirical testing to reject or accept the hypothesis. The hypotheses that shape this study follows the research question, that influence-weighted investor sentiment can enhance the prediction of next-day returns on the S&P 500 index. This hypothesis can be either accepted or rejected following the methodology described. Specifically, the hypothesis is accepted if the influence-weighted sentiment model constructed shows an improvement in out-of-sample R^2 and directional accuracy compared to the model with only raw sentiment score. If this condition is not met, the hypothesis is rejected. This testable and replicable framework aligns with the principles of empirical testing, ensuring clear falsifiability and scientific transparency.

3.2 Data Collection and Cleaning

Following the first step of the research process, this sub-section describes the data collection and cleaning process, which is essential to the reliability of the study. The study builds on data collected on X posts and historical price data, and this step ensures data quality and alignment, crucial for extracting accurate sentiment signals and mapping them to help predict next-day returns.

Two primary sources of data are used to build the dataset for this study, social media posts and historical price information gathered from January 1, 2022, to December 30, 2024. Social media data was obtained by scraping posts related to the S&P 500 from X (formerly Twitter) along with their metadata (e.g., author, follower, like count). This allowed for the extraction of investor sentiment and the construction of an influence-weighted sentiment variable to address Sub-RQ1. Further, historical S&P 500 index prices (tracking instruments) were retrieved for the same period via the Yahoo Finance API (yFinance) to extract price information like closing prices to calculate returns, as well as additional financial information like trading volume and volatility to address Sub-RQ2. The years 2020 and 2021 were deliberately excluded due to the atypical and extreme effects of the COVID-19 pandemic on financial markets. Zhu et al. (2023) found that investor sentiment during that period was already at a seemingly constant low due to heightened fear and recession-related concerns and did not meaningfully predict market-returns hence why it is excluded from the time frame. Moreover, as the scope of this study is quite narrowly focused on next-day returns, consistent negative sentiment like this potentially limits the variation in how the market responds, reducing the negative or positive influence on the model's predictions. By concentrating on data from 2022–2024, more typical investor behavior and market dynamics are captured. After collection, both datasets were thoroughly checked to ensure data quality, such as duplicate removal, timestamp alignment, and handling of missing values before being merged into a single dataset that pairs influence-weighted sentiment measures with corresponding next-day S&P 500 returns. This alignment ensures accurate mapping of sentiment signals to subsequent market performance.

3.2.1 Collecting X Posts and Metadata

To collect data from X posts, a no-code web scraping tool called Octoparse was used. The scraper used X's search function to extract public posts that contain the hashtag #SPX500, which refers to the S&P500 index. While using the conventional $SPX ticker was considered, technical and capacity limitations with Octoparse made this approach unfeasible.

The study focuses on the S&P 500 index instead of individual stocks due to constraints in data collection, financial resources and computational capacity. Scraping data for all 500 constituents would have required significantly more processing power and storage, making such an approach too large for the scale of the study. Additionally, individual stocks may not be mentioned with equal frequency on X, potentially leading to biases in the dataset as there may be an overrepresentation of specific stocks. Due to Octoparse's limited customizability, a scraper built using the Selenium WebDriver library was developed to extract follower counts of each user profile in the dataset. However, due to automated scraping restrictions introduced by X, it was difficult to maintain a stable application. To overcome this, 48 consecutive scraping runs were conducted, resulting in 48 separate datasets that were merged into one. Follower counts were integrated into the main dataset using a VLOOKUP formula to match them with corresponding user handles. However, there are some ethical considerations regarding the responsible collection of social media data. The collection concerned scraping X posts without using the official API, which technically violates the platform's Terms of Service (TOS), as it prohibits crawling without prior written consent. However, no authentication measures were bypassed, and only publicly available data was collected. The scraping tool also never logs in, meaning it does not consent to be bound by X's TOS, making it perfectly acceptable from a legal standpoint (Fatenaite, 2025). The collection of social media data is strictly for academic, non-commercial research purposes, without redistributing content or storing any user-identifiable information. The anonymity of users has been thoroughly preserved by removing any identifying information, ensuring the legality of handling the data (Fatenaite, 2025). Efforts have also been made to minimize server load and without overwhelming the website by limiting the scraping rate to avoid interfering with the platform's regular operations.

The original scraped dataset consisted of over 60,000 posts. However, rows with missing values in the X-post content resulted in a dataset of 48,363 entries. After scraping for follower counts, a further 147 rows were removed as some accounts had been deleted so follower count could not be obtained. An additional 76 rows were removed due to duplicate posts. During the scraping, the link to the post was also included in the text content. This was removed to reduce noise and enhance textual integrity. Further only relevant fields for sentiment analysis and constructing influence metrics were retained as the Octoparse scraper automatically scraped for many other fields. The retained columns from X posts are listed and briefly described in Table 1.

Column	**Description**
Author_Handle	The username for the author of the post
Date	The date including timestamp the X post was posted
X_Post	The content of the X post
Reply_Count	The number of replies the X post got
Repost_Count	The number of times the post was reposted
Like_Count	The number of likes the post got
Follower_Count	The number of followers the author of the post has

Table 1: Variables retained from X data collection.

3.2.2 S&P 500 Historical Price Data

As aforementioned, the historical price data for the S&P 500 tracking instruments is gathered through the Yahoo Finance API on python (yFinance). To scrape the data the period was defined earlier, retrieving daily data for the ticker (^GSPC) for the S&P 500 index. The data collected were historical OHLC (Open, High, Low, Close) prices, as well as volatility via the VIX index and trading volume. The OHLC prices will help calculate the daily returns, which will be used both as the target variable (next-day returns) as well as an explanatory variable in the final models (current day returns). Returns were calculated specifically, using the adjusted as they adjust for corporate actions so that the returns reflect the true economic gain or loss (Ganti, n.d.).

Having obtained clean, time-stamped X posts with follower-count and S&P 500 OHLC data, sentiment labels can now be extracted and thereafter combined with engagement indicators, constituting the first step toward constructing the influence-weighted sentiment score (IWSS).

3.3 Sentiment Analysis

To address the first objective (Sub-RQ1a), addressing the first part of Sub-RQ1, investor sentiment is extracted from X posts using Hugging Face's pre-trained transformer models, FinBERT and RoBERTa, which process each post and assign it a sentiment label. These models are a type of large language model (LLM) capable of understanding, learning, and generating text. They were selected for their ability to handle the complex, informal, and context-dependent nature of social media text, which traditional sentiment analysis techniques often fail to capture as determined in the literature review.

3.3.1 Pretrained Models

Manually labeling sentiment in informal, noisy financial social-media posts requires substantial time and resources. To address this, large language models (LLMs) are leveraged to automatically classify each X post as positive, negative, or neutral. Specifically, two pre-trained transformers are used:

1. FinBERT, which is trained on financial texts and excels at domain-specific terminology, and,

2. RoBERTa, which is trained on a large corpora of X posts and handles informal language well.

Each model has complementary strengths, FinBERT having financial expertise, while RoBERTa is better suited to the linguistic style of X posts. Thus, both models are evaluated to develop a task-specific sentiment classifier for financial X posts. The resulting labels will then be encoded and used later in constructing the influence-weighted sentiment score (IWSS) variable.

3.3.1.1 FinBERT Model

The FinBERT model is pre-trained on a financial corpus of over 4.9 billion tokens, which include earnings calls, SEC filings and financial reports is very good at classifying text as positive, negative, or neutral, making it a powerful tool for financial NLP research and practice (Huang et al., 2022). The version of FinBERT used from hugging face is the Yi Yang version (yiyanghkust/finbert-tone) as it has strong empirical performance in classifying financial tone and is widely adopted in academia.

3.3.1.2 RoBERTa Model

RoBERTa is a more general language model. The cardiffnlp/twitter-roberta-base-sentiment version of the model is used as it is pre-trained specifically on X data. This specialization makes it well-suited to process noisy posts and includes hashtags, emojis and abbreviations which FinBERT cannot catch. Although RoBERTa aligns more closely with the style of social-media text, it may not identify financial jargon as effectively as FinBERT.

3.3.2 Zero-Shot Inference

As both FinBERT and RoBERTa were originally trained on datasets that differ slightly from the X posts extracted in this study, the models were evaluated and fine-tuned based on the pre-labeled TimKoornstra/financial-tweets-sentiment dataset from Hugging Face. This dataset consists of over 38,000 preprocessed financial tweets labeled as positive (17,368), neutral (12,181), and negative (8,542), and covers a wide range of content related to financial markets and individual stocks.

Manually labelling this study's 48,000 X posts collected is impractical and prone to human errors. Instead, the TimKoornstra dataset offered a preprocessed, domain-relevant alternative for fine-tuning as the aligns very closely to the financial X posts collected. Despite some class imbalance, studies indicate that rebalancing the datasets results in very small gains and is not well researched for deep learning (Henning et al., 2023). Thus, the dataset is used in its original form. This dataset closely resembles the language and content of the X Posts used in the study, making it a suitable benchmark for both evaluation and training.

Before any fine-tuning, FinBERT and RoBERTa is applied in zero-shot inference model directly to the TimKoornstra dataset. This involves feeding the tweets through each model without further training and comparing the predicted labels to the ground truth. This provides an assessment of how well each model can generalize to financial tweet sentiment without task-specific fine tuning. The zero-shot evaluation gives an initial sense of how pre-trained models generalize to S&P 500-related X posts. Fine-tuning is then applied to align model performance with the linguistic characteristics of our data, directly supporting the creation of a reliable sentiment variable for Sub-RQ1.

3.3.3 Fine-tuning Transformer Models

To improve the performance of the two transformer-based models FinBERT and RoBERTa, both models are fine-tuned on the pre-labeled TimKoornstra dataset to get a better model that aligns more closely with the linguistic and contextual patterns found in the X posts collected in this study. The FinBERT model needs to adapt to the informal and dynamic language found in social media platforms like X (formerly Twitter), while the RoBERTa model should pick up on more financial based sentiment. Both models were fine-tuned using the Hugging Face Transformers library.

The TimKoornstra dataset contains categories of positive, neutral, and negative X post sentiment. To prepare for training and compatibility with Hugging Face, the categories must be converted to numerical labels for the model to learn. For the FinBERT model, neutral = 0, positive = 1, negative = 2, whilst for the RoBERTa model, negative = 0, neutral = 1, positive = 2 so each matches the configuration of their respective model.

Then, the data is split into three parts, 60% for training the model, 20% for validation of its performance during training and then lastly 20% for testing on unseen data. This is crucial for an unbiased model to avoid overfitting. Thereafter, each X post is transformed into a format that the model can understand. This involves converting the words in each post into a series of numbers (tokens), with all tweets made the same length for consistency. Longer tweets are shortened, and shorter ones will be padded to match a fixed length (512 characters). This ensures uniformity during training.

The models are then trained using the Hugging Face Trainer Framework. Training is conducted over three epochs, which has shown to be effective for fine-tuning BERT-based transformer models (Devlin et al., 2019). The learning rate was set to 0.00002, which is set low enough to maintain stability, but high enough to learn effectively. Batch size was set to 32 for both training and evaluation to ensure efficient utilization of hardware resources. However, experiments were also conducted using a smaller batch size of 16. While smaller batches increase training time due to more frequent parameter updates, they have been shown to improve generalization by avoiding sharp local minima in the loss landscape, a phenomenon described by Keskar et al. (2017). In practice, using a batch size of 16 led to slightly more stable performance across validation checkpoints, though results remained broadly consistent.

Model performance is evaluated at the end of each training epoch to monitor learning progress and avoid overfitting and model checkpoints are saved after each epoch to retain the best epoch. At the conclusion of training, the best-performing model is automatically restored based on balanced accuracy, a metric that compensates for class imbalance by averaging recall across all classes (Buda et al., 2018). This ensures that model selection prioritizes consistent performance across all categories rather than favoring majority-class accuracy.

After the models are fine-tuned, the model that performed best on the validation set is retained to use for final testing. The performance metrics assessed are accuracy, to show how often the model is correct, balanced accuracy which adjusts for difference in number of examples in each sentiment category, and Precision, recall, and F1-score, which provide a fuller picture of how well the model handles each sentiment type. This process is summarized below in Figure 3.

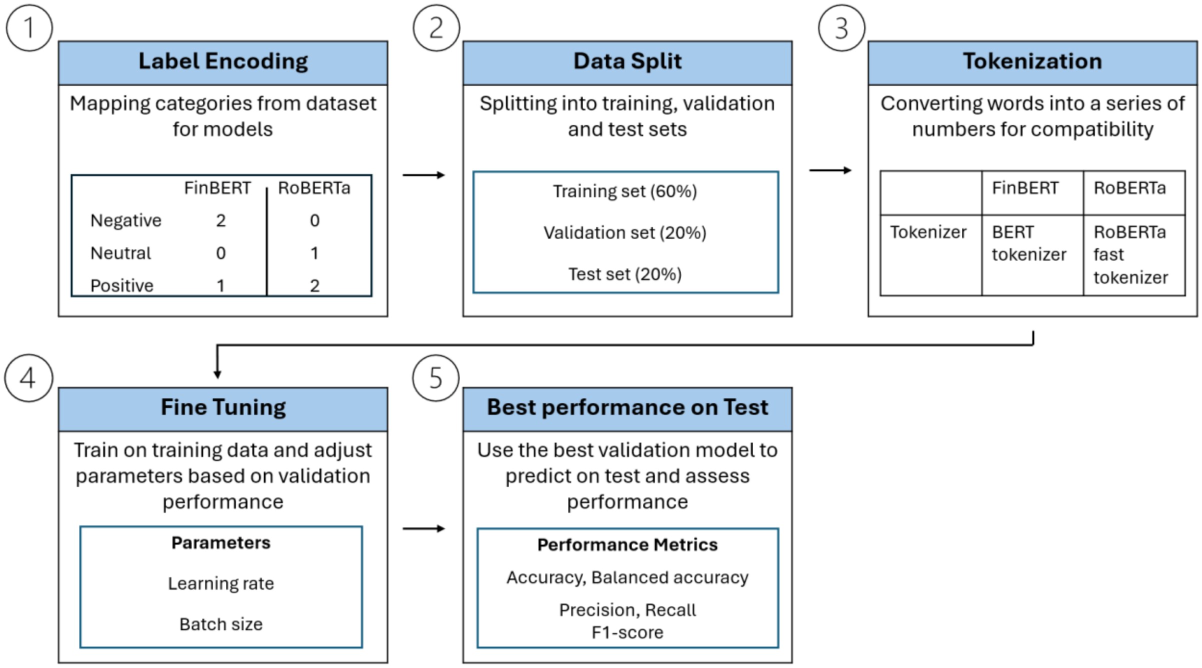


Figure 3: Process of Sentiment Classification

The evaluation of the performance metrics allows for a direct comparison between the models. To select the best model, the metrics accuracy, and f1-score are emphasized. This is because accuracy reflects overall correctness, while F1-score balances precision and recall, which is particularly important in the presence of class imbalance as observed in the TimKoornstra dataset. The model that demonstrated the strongest out-of-sample performance based on these metrics is then selected for sentiment prediction on the X posts collected of this study.

3.3.3 Sentiment Prediction on X Posts

After identifying the best-performing model (using accuracy and F1 as criteria), that model is applied to all 48,019 cleaned and preprocessed X posts to generate sentiment labels. These labels will be used to construct the influence-weighted sentiment score. This will later be combined with financial data with stock returns, and other financial metrics. Each post in this dataset is tokenized by the best models tokenizer and then applied to each tokenized X post. The model outputs a sentiment label (positive, negative, or neutral), which is stored in a new column alongside the corresponding X Post. This procedure ensures that sentiment extraction is performed consistently and efficiently across all X Posts, using a model that has been empirically validated and fine-tuned on domain-relevant data.

Thereafter, the new sentiment-labeled dataset is split chronologically into training (~60%), validation (~20%) and test (~20%) subsets to prevent data leakage and preserve the time-dependent nature of asset returns, given that the two datasets will be combined at a later stage. Calculating returns inherently reflects the percentage change in price from one period to the next, and therefore requires chronological ordering (Fan & Yao, 2017). This structure allows the model to learn from historical patterns, validate performance on recent but unseen data, and test predictive accuracy in a forward-looking, real-world context.

3.4 Feature Engineering: Construction the Influence-weighted Sentiment Score (IWSS)

With the dataset divided into training, validation and test sets, the focus shifts to Objective 2 (Sub-RQ1b): constructing an influence-weighted sentiment score (IWSS). Raw sentiment classification captures a post's emotional tone but says nothing about its reach or impact. To address this, engagement metadata collected such as likes, replies and reposts are incorporated directly into the sentiment score. The IWSS combines raw sentiment with indicators of social influence, reflecting both the emotional tone of a post and its potential reach (audience size and amplification). IWSS is designed to improve S&P 500 return prediction by weighing each post's sentiment according to how widely and rapidly it spreads.

The construction of IWSS involves preprocessing, transformations, modeling steps, and weight estimations, all of which are components of the feature engineering process. The construction of IWSS follows three key stages:

1. An exploratory analysis to inform preprocessing of the engagement score,
2. Design of the variable structure combining sentiment score and engagement measures,

3. Estimation of engagement weights

All feature construction and weight estimation procedures are conducted using the training dataset only, with the same transformations applied to the validation and test sets to avoid data leakage and preserve model generalizability.

3.4.1 Exploratory Analysis for Feature Preprocessing

As a preparatory step for constructing the IWSS variable, an exploratory analysis was conducted on the training set to understand the scale, variance, and distribution of each engagement and reach metric. Relationships among likes, replies, reposts, and follower counts, as well as their association with sentiment labels were examined. These insights guided any necessary data transformations like log-scaling, and preprocessing steps such as outlier removal, to reduce noise and establish meaningful weighting strategies.

As highlighted in the literature review, there is limited research on constructing a sentiment variable that incorporates both sentiment and its diffusion through social media. However, there are existing studies that suggest that engagement indicators such as reposts, likes, replies and follower count can help determine online visibility and spread of posts on platforms like X which is why they were selected (Anger & Kittl, 2011). To address any skewness and scale differences among the engagement metrics, various data transformation techniques are evaluated. This is expected in the data as deep metrics like replies and reposts are likely to have a much narrower range compared to metrics like follower and like count. Box-Cox transformation and logarithmic scaling are applied based on their ability to stabilize variance and reduce the impact of extreme values. Additionally, correlation analysis and variance inflation factors (VIF) were used to assess multicollinearity among the engagement indicators.

One issue that arises is the presence of bots. To address this, in the preprocessing steps, posts that are potential bot-automated were removed through a time-sensitive duplicate detection method, focusing on users that frequently post highly similar content within a short time interval as they can artificially inflate sentiment (Ferrara et al., 2016; Varol et al., 2017) reducing the validity of the model. Posts from the same user were flagged as duplicates if they occurred within five minutes and exhibited at least 90% similarity, as measured using fuzzy matching based on Levenshtein distance. This calculates the minimum number of single-character edits required to change one word into another using the python library rapidfuzz (Pykes, 2025). This method is applied consistently across training, validation, and test sets.

3.4.2 Variable Design: The IWSS Structure

The structure of IWSS is modelled after concepts from the Bass Diffusion Model (Bass, 1969), which distinguishes between external and internal channels of influence based on how new products or ideas are adopted. In adapting the framework, external influences are described as mass exposure, which is proxied by the author's follower count (F), representing the potential audience size for each post. While view-count was initially collected in the metadata, they are not used as they there were many inconsistencies in the scraped data that made it unreliable. For instance, posts registering likes and reposts could also have zero views, thus, follower count is selected as a more stable indicator of exposure. To prevent extremely large or potentially inflated followings (such as bots or inactive accounts) from dominating the weights, a logarithmic transformation, log(F + 1) is applied.

Internal influence reflects peer-to-peer diffusion which under the bass diffusion model is like word-of-mouth. In this study, it is measured through engagement metrics which mirror the word-of-mouth, measuring interaction between users such as likes, reposts and replies. Existing literature (Anger & Kittl, 2011) emphasizes that follower reach alone does not fully capture a post's social influence; user interactions play a critical role in how sentiment diffuses through networks. By assuming that higher engagement correlates with greater likelihood of sentiment propagation among peers, potentially including market participants, these engagement metrics serve as valid proxies for a post's diffusion potential.

On top of this, it may be important to account for differences in the depth of engagement, therefore and engagement score are weighed as follows:

$$E = \omega_1 \times Likes + \omega_2 \times Reposts + \omega_3 \times Replies$$

The weights $(\omega_1, \omega_2, \omega_3)$, can be calculated to represent the relative importance of likes, replies, and reposts, respectively. Liking a post is a relatively passive form of engagement, replies indicate deeper interaction as it shows a response to a post, and reposts expand reach by exposing the content to reposters followers (Nurollahian et al., 2023).

Sentiment (S) itself is interpreted as the "message" that propagates through these social channels. Sentiment scores are mapped from classifications of the best LLM as previously mentioned into the set $S \in \{-1, 0.1, 1\}$, corresponding to negative, neutral, and positive sentiment labels, following existing methodologies (Azar & Lo, 2016; Giachanou & Crestani, 2016). By combining these factors of exposure and interaction, the IWSS formulation is as follows:

$$IWSS = S \times E \times log(F + 1)$$

Note, that the neutral sentiment is set to 0.1 instead of 0 as this avoid completely nullifying the signals from the influence metrics. This gives neutral posts a small non-zero baseline to ensure that even neutral content can still contribute to the overall daily sentiment score without overstating its impact.

This structure and formulation directly address Sub-RQ1 by demonstrating how influence indicators (follower count and engagement) can be integrated with sentiment to produce a single, continuous variable that approximates the diffusion-weighted impact of social media content on investor sentiment. However, it is important to note that the structure assumes that likes, reposts and replies are valid proxies for a post's diffusion in a way that affects investor behavior following the deductive approach. Although this is a reasonable assumption, it is beyond the scope of this thesis to test. Overall, by constructing IWSS in this manner, the analysis captures not only the investor sentiment but also estimates of how widely and rapidly that sentiment may propagate, thereby creating a variable well suited for forecasting next-day market returns.

3.4.3 Estimating Engagement Weights

To derive the weights used in the construction of the engagement score E, two approaches are estimated, a global weights ratio approach and a data-driven method using a Random Forest classifier. These methods combine likes, reposts, and replies into a single weighted score, capturing each metric's relative importance in predicting sentiment on X. Weights are not assumed equal because likes, replies, and reposts may carry different informational value and diffusion impact. Once extracted, the weights are fixed and applied to the validation and test datasets to ensure consistency and avoid data leakage. Equal weights will be used as a benchmark. Method 1 (Global Ratio) leverages the proportion of each engagement indicator for interpretable weights. Method 2 (Random Forest-based) attempts to take a more data-driven approach, by deriving weights based on their relation to sentiment to produce varying weights on the engagement metrics.

3.4.3.1 Global Weights Ratio Approach

The global weights approach applies Box-Cox transformation to the engagement indicators, following recommendations from the exploratory analysis. To ensure consistency and avoid data leakage, the Box-Cox transformation parameters (λ) are estimated using only the training dataset and then applied uniformly to the validation and test sets. This transformation is intended to reduce the effect of distributional skew and mitigate the potential influence of outliers. Raw like, repost, and reply counts may otherwise be disproportionately affected by extreme values, potentially distorting their contribution to the final engagement score.

Once transformed, weights are calculated as each metric's share of total engagement, for example:

$$w_{Likes} = Likes/(Likes + Reposts + Replies)$$

These weights are fixed after estimation and consistently applied across all data partitions. This method is a simple way to combine the engagement indicators into a single score.

3.4.3.2 Random Forest Classifier Approach

The second weighting strategy for estimating engagement weights involves training a Random Forest Classifier to assess the relative importance of each engagement indicator in classifying sentiment. It is particularly suitable here due to its ability to handle non-linear relationships and its tolerance for skewed data distributions, as identified during the exploratory data analysis Further, RF is a good investigative tool for feature importance interpretation (Probst et al., 2018). The model is trained exclusively on the training set, using the engagement metrics as input features and the sentiment categories (positive, neutral, or negative) as the target variable. This is a more data-driven approach compared to the ratio method, as it leverages supervised learning to weight the engagement ratios and sentiment score.

To ensure stability and reproducibility, the Random Forest is configured with carefully chosen hyperparameters, including 200 estimators (trees), a minimum of 5 samples required to split an internal node, at least 2 samples per leaf, and a fixed random seed (42). These settings balance model complexity and generalization. Training on the full dataset maximizes learning to ensure the model captures the relationships necessary for reliable importance estimation.

Three different techniques were implemented to extract feature importance.

1. **Impurity-based feature importance**

   The impurity-based feature importance method calculates the importance of features based on the total decrease in node impurity (measured by Gini impurity) across all trees in the ensemble. Features that consistently lead to more homogeneous splits are assigned higher importance scores. However, this method is known to be biased toward variables with higher variance as there are more possible splits, which can overstate their true contribution (Hastie et al., 2009; Molnar, 2022). Given that engagement metrics differ significantly in scale and variance, relying solely on impurity importance could misrepresent true feature contributions.

2. **Permutation-based feature importance**

   To address the potential bias, permutation importance is also calculated on the training data. Permutation importance is a model-agnostic method that randomly shuffles the values of each feature and measures the resulting decrease in model performance, typically in terms of accuracy (Molnar, 2022). The idea is that if permuting a feature significantly worsens prediction performance, it must be an important predictor. Permutation importance is less affected by scale and less reliant on frequency of splits, offering a more unbiased assessment of actual predictive contribution (Scikit-learn, 2023). The permutation importance is calculated using the permutation_importance() function from the sklearn.inspection module in scikit-learn with the 10 repetitions per feature to ensure the stability of the importance estimates, and fixed random state for reproducibility. A larger drop in accuracy indicates higher feature importance, as the method disrupts meaningful patterns in the data, directly impacting model performance. However, a limitation of permutation feature importance is that the importance is linked to the error of the model, however, it does not explain how features influence the prediction, only how much it affects the loss. To complement this, SHAP analysis is conducted for a deeper interpretation of feature effects.

3. **SHAP Analysis**

   SHAP (SHapley, Additive exPlanations) analysis is conducted to derive weights that capture the marginal contribution of each engagement feature to sentiment. SHAP values provide a theoretically grounded way to attribute a model's prediction to its input features (Lundberg & Lee, 2017). Using the TreeExplainer algorithm from the python library 'SHAP', SHAP values are calculated for each engagement metric across all samples in the training set. The mean absolute SHAP value per feature is then extracted to derive a normalized set of feature importance weights. Unlike other methods, SHAP captures both the direction and magnitude of a feature's contribution.

The weights obtained from these techniques are fixed after the results of the training set and applied without modification on validation and test sets. This ensures that the weighting strategy does not contain any forward-looking bias by being trained on the hold-out validation and test sets when being used for forecasting. Thereafter, the weights are applied to the IWSS function as described in section 3.4.2 combining the engagement score calculated from the weights with the follower metric and sentiment score.

The final output of the feature engineering process contained five IWSS features, each defined by a different weighting strategy. *IWSS_Equal* is the baseline calculated with the equal weighting (0.33) for the engagement metrics. *IWSS_Ratio* is calculated using the ratio-based approach, *IWSS_Impurity*, *IWSS_Permutation*, and *IWSS_SHAP* are calculated using the random forest weights using the impurity, permutation and SHAP approaches respectively.

3.4.4 Aggregating IWSS to Daily Variables

Once IWSS features for each X post are constructed, they are aggregated for each trading day. The scores are aggregated to form daily sentiment measures using the sum of IWSS and raw sentiment values per trading day. This approach assumes that a greater number of influential posts increases the overall market impact of investor sentiment.

For each trading day, T, the aggregation window included all posts published between 16:00:01 ET on the previous trading day to 16:00:00 ET on day T. Posts made after the market close are assigned to the next trading day, to ensure that sentiment data used in modelling is available prior to the market opens. To account for weekend sentiment, all posts from Friday 16:00:01 to Monday 16:00:00 ET are aggregated to Monday's sentiment scores to account for potential sentiment buildup during non-trading hours. These daily IWSS and sentiment scores are aligned with the same-day market variables, including trading volume, volatility (VIX) and daily returns obtained from the finance API. Daily returns ($return_t$) were calculated as the percentage change in closing prices from the previous trading day, and the next-day return ($return_{t+1}$) used as the model's prediction target was computed as the percentage change from day T's close to day T+1's close. It is assumed that sentiment extracted from X posts on day t can only influence day t + 1 returns, not vice versa, due to timestamp ordering and market close times. This structure ensures that all features including IWSS, are strictly aligned prior to the target variable to avoid forward-looking bias when modelling.

3.5 Return Forecasting and Evaluation

This section addresses Objective 3 of the methodology, thereby supporting sub-RQ2. The aim is to evaluate whether incorporating influence and engagement indicators in the IWSS variable can enhance the forecasting performance of the S&P 500 compared to a raw sentiment score. To answer this, four machine learning algorithms are trained, validated then tested across a set of seven feature sets, each reflecting the prediction of next-day returns.

Prior to model training, exploratory data analysis (EDA) was conducted to assess the integrity and distributional characteristics of the dataset. This included checks for missing values, variable distributions, outliers, and scale differences across features. Although the EDA did not directly inform model specification, it guided preprocessing steps such as standardization and log transformation for skewed features. A full summary of descriptive statistics and diagnostic plots is included in Appendix C.1.

3.5.1 Data split and Hyperparameter tuning strategy

The study adopts a time-based data splitting strategy. The dataset is chronologically split into three disjoint subsets to preserve the temporal order of the financial and sentiment data. The training set (January 2022 - October 2023) is used to fit the model parameters. The validation set (November 2023 - May 2025) is used exclusively for hyperparameter tuning and the test set (June 2024 - December 2025) is used for final performance evaluation. The same time-based split is used when constructing the influence-weighted sentiment variables´. This structure ensures that each model only learns from past data when making predictions on future periods, avoiding information leakage and reflects a forecasting environment close to reality.

The three-way split allows the models to test various combinations of hyperparameters, which are settings that control the model's complexity. For example, the number of trees in a random forest or the learning rate in XGBoost. For each set of parameter values, the model generates forecasts on the validations based only on data from the training set. These predictions are then compared to the actual values using Root Mean squared Error (RMSE), measuring how far off predictions are on average, supporting generalizability and minimizing overfitting. This process is repeated across all hyperparameter combinations and the set that achieves the lowest RMSE on the validation set is selected as the most optimal (see hyperparameter ranges in Appendix A). However, there is some risk of data leakage, as although it shows how well the model can generalize, it informs the tuning process. Thus, the final evaluation is performed on the test set which has been kept separate and not used in training or tuning of the model. This ensures a fair and unbiased assessment of how the models perform on unseen data which is very important in assessing whether influence-weighted sentiment features offer predictive value.

3.5.2 Forecasting Setup and Feature Configurations

To ensure that the conclusions are robust and not model-specific, four different learning algorithms are applied. These are Ordinary Least Squares Regression (OLS), Random Forest (RF), Gradient Boosted Trees (XGBoost), and Long Short-Term Memory networks (LSTM). These models represent varying levels of complexity, from linear models (OLS) to non-linear ensembles (RF, XGBoost), to deep learning architectures capable of capturing temporal dependencies (LSTM). The selection was guided by Gu et al. (2020), who emphasize the importance of comparing models with varying capacities for non-linearity and feature interaction.

This allows a more comprehensive evaluation of how well the sentiment and IWSS features generalize across different learning frameworks. This diversity helps address the research question by demonstrating whether improvements in prediction are consistent regardless of modeling technique, supporting the broader claim of predictive utility for influence-weighted sentiment.

Each feature set is built on a common structure also inspired by Gu et al, (2020) which are standard predictors of short-term returns:

$$r_{t+1} = f(r_t, Vol_t, VIX_t, s_t)$$

Where:

- r_{t+1} is the next-day return (t + 1)
- r_t is the return on the current day (t)
- Vol_t is the trading volume on the current day (t)
- VIX_t is the volatility index on the current day (t)
- $s_t \in \{Sentiment_t, IWSS_{Ratio,t}, IWSS_{Equal,t}, IWSS_{Impurity,t}, IWSS_{Perm,t}, IWSS_{SHAP,t}\}$

Each algorithm is applied to the following seven feature configurations in Table 2 below:

Feature Set	Description
Base + Raw Sentiment	Adds average daily sentiment score
Base + IWSS_Equal	Adds influence-weighted sentiment with equal weights
Base + IWSS_Ratio	IWSS using ratio-based engagement weights
Base + IWSS_Impurity	IWSS using Random Forest impurity-based feature importance
Base + IWSS_Permutation	IWSS using permutation-based feature importance
Base + IWSS_SHAP	IWSS using SHAP-derived engagement feature weights

Table 2: Feature configurations used across the OLS, Random Forest, XGBoost and LSTM models.

In other words, the feature sets that are being evaluated are the baseline (return, volume, volatility (VIX)), and additionally six feature sets will be built on the baseline and extended with one additional sentiment variable (Raw Sentiment, IWSS_Equal, IWSS_Ratio, IWSS_Impurity, IWSS_Permutation, or IWSS_SHAP). This results in a total of 24 model variants (4 algorithms × 6 feature sets), each trained and validated under the same conditions. This structure provides a controlled comparison of the predicted value of investor sentiment against when it is adjusted for influence across various modelling approaches. Note that the primary comparison centers on whether influence-weighted sentiment variants outperform the raw sentiment model. To identify the most effective combination of algorithm and feature set, each model was trained on the training data and tuned using a validation set.

After fine-tuning on the validation set, the best performing model configuration based on the performance metrics are evaluated on the test set. The Diebold-Mariano tests are then conducted to statistically compare predictive performance between Raw Sentiment vs. IWSS features across different models, as well as IWSS features against each other. It is used to evaluate whether forecast errors from two competing models differ significantly, forming a statistical basis to compare predictive accuracy. Figure 4 shows an overview of the forecasting setup.

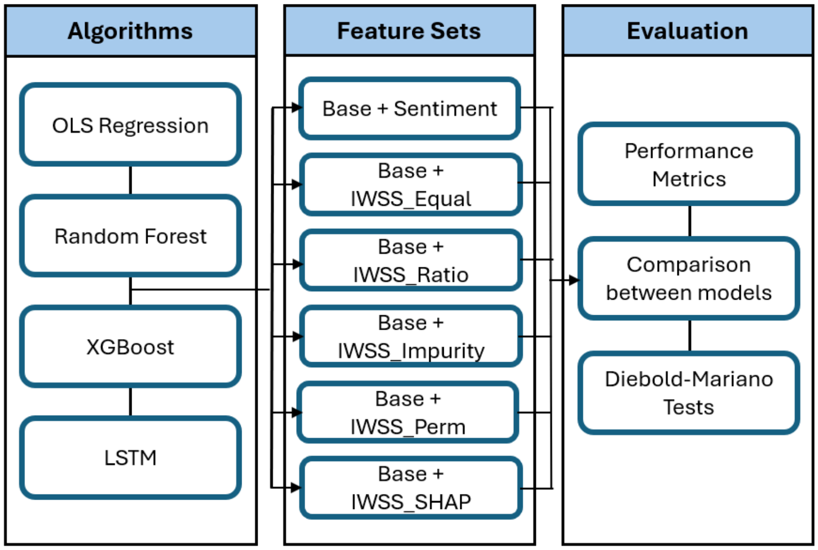


Figure 4: Forecasting set-up showing how each forecasting algorithm/model is applied on each feature set.

A brief exploration of data showed that the variables are on quite different scales (see Appendix B.1). Thus, StandardScaler is used to standardize all features prior to model training. Although some models like Random Forest and XGBoost are scale-invariant, consistent scaling is applied across all models to enable fair comparisons and ensure compatibility with scale-sensitive algorithms such as OLS and LSTM.

3.5.3 Forecasting Models

Different modelling techniques offer different strengths for predicting next-day returns. Linear models like OLS offer a straightforward starting point for evaluating whether more complex approaches offer any valuable improvements. Unlike linear models, Random Forest and XGBoost are capable of capturing non-linear relationships and interactions between variables, while LSTM-models are specifically designed to handle temporal dependencies in sequential data. By applying each of these models with the same features configurations, including both market and sentiment variables, it is possible to evaluate how model complexity and varying structural assumptions impact predictive performance. This systematic comparison directly evaluates whether advanced algorithms outperform simpler linear models in forecasting next-day returns, which helps address

Sub-RQ2 about the extent influence-weighted sentiment variables improve next-day return forecasting accuracy over raw sentiment.

3.5.3.1 OLS Regression

OLS is the simple linear model implemented as a benchmark model to evaluate the added predictive value of more complex machine learning models. The model estimates the relationship between the input features (market and sentiment variables) and next-day returns by minimizing the squared residuals between the predicted and observed values (Wooldridge, 2016). In this study, OLS is implemented using the LinearRegression() class from the sklearn.linear_model module on python which provides tools for different machine learning algorithms. This is applied to all of the model variations.

Although OLS is not designed to handle the non-linearity and noise found in the financial dataset, it is a good baseline comparison for other models to beat. To check if the assumptions of OLS are satisfied, a series of diagnostic tests are conducted. These include checks for linearity, homoscedasticity, normality of residuals, multicollinearity and autocorrelation. It is important to note that violations of these assumptions do not directly impact predictive accuracy (Kuhn & Johnson, 2013; James et al., 2021), however, they are checked to better understand limitations of the model, which can contextualize performance in comparison to non-linear alternatives. The details of each assumption and the corresponding diagnostic method used are found in Appendix E.1.

3.5.3.2 Random Forest

Random Forest (RF) is a non-parametric ensemble learning method based on decision trees. It predicts by constructing multiple trees during training and outputs the mean prediction of the individual trees to reduce overfitting. RF is good at handling multicollinearity as well as non-linear interactions, which is important given that there does not seem to be linear relationships between next-day returns and any of the sentiment variables (see Appendix C.2).

In this study, RF will be implemented using the RandomForestRegressor() class from the sklearn-ensemble module in Python. The algorithm is applied to each of the six feature configurations described in Section 3.5.2. Five hyperparameters are tuned to optimize model performance. The number of trees (n_estimators) controls the ensemble and more trees can help improve performance. Tree depth (max_depth) limits how deep each tree is to prevent overfitting on noise. Further, to regulate how finely a tree can split and manage complexity of overfitting of the model, min_samples_split and min_samples_leaf are used. The last parameter is the maximum features (max_features) that determine how many features are considered at each split, which introduces some randomness and decorrelate trees in the ensemble. These are tuned using a grid search on the validation data to find the set that minimizes RMSE.

3.5.3.3 XGBoost

The XGBoost algorithm was selected as one of the utilized modeling frameworks due to its proven capability and robustness in handling regression problems, particularly when analyzing structured datasets such as financial market data. XGBoost is a tree-based ensemble method, known for its high predictive accuracy, efficiency, and ability to effectively model complex, nonlinear relationships between variables, which are commonly observed in financial forecasting tasks.

Having created an XGBoost model utilizing gradient boosted decision trees, this section aims to present the evaluation metric results for the validation and test set. As done consistently across all models, a range of models were developed, this time using the XGBoost regression framework. Each of the trained models was made using the core financial indicators market volatility (VIX), return, and trading volume. Each indicator has been extended individually by using a variety of sentiment-based features, based on the social media data. The sentiment variables consist of both raw scores and metrics transformed using techniques like SHAP and Equal, trying to capture market sentiment in a structured way.

To avoid overfitting and ensure model robustness, a hyperparameter grid search is conducted to find the best parameters, using time-series cross validation within the training data. This automated approach helps search through different combinations of parameters, aiming to find the most optimal composition of trees, learning rate, tree depth, and feature sampling, by repeatedly training the model with different configurations. After having found the best parameters for each feature set, the models were first evaluated on the validation set, and afterwards on a completely unseen test set. The goal of the models was to determine how well they fit historical patterns (in-sample performance) as well as how reliably they predict future returns (out-of-sample performance).

3.5.3.4 Long-Short Term Memory (LSTM) Neural Network

As previously touched upon, the LSTM model is a specialized form of recurrent neural network capable of learning and capturing temporal dependencies and sequential patterns in data. Unlike the traditional feedforward-neural networks, LSTMs consist of input, forget, and output gates, controlling the flow of information, meaning they effectively model both short-term and long-term dependencies in time series data. This makes them well-suited for financial time series forecasting, where patterns can be highly nonlinear and temporally dependent.

The hyperparameters in the LSTM model, more specifically the dropout rates and L2 regularization values, were specified through validation-based tuning. Firstly, a comprehensive grid search was conducted across all the different combinations of dropout and regularization parameters. To find the most optimal combination of dropout and regularization parameters, observing the combination yielding the lowest loss score helps select the best performing hyperparameters. The selected hyperparameters are subsequently employed for the final evaluation on the test dataset.

The LSTM was specifically chosen due to its documented ability to model complex temporal structures inherent in financial data, such as volatility clustering, autocorrelation, and non-stationary trends. Financial time series often exhibit temporal dependencies, which more simple models may capture insufficiently. LSTM architecture solves these challenges by saving relevant historical information, while irrelevant data is dynamically discarded.

3.5.4 Performance Evaluation

To evaluate the performance of the return forecasting models, six regression-based metrics were evaluated. These were selected to capture different aspects of model performance, including the error magnitude, explanatory power, predictive direction, and generalization ability. The primary focus of the evaluation is on comparing the predictive performance of the models with IWSS features against the models with raw sentiment scores. The key metrics evaluated are RMSE, ROOS, and directional accuracy. To assess whether observed differences in forecasting performance are statistically significant, Diebold-Mariano tests are conducted between each IWSS variant and the raw sentiment benchmark.

3.5.4.1 Performance metrics

Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) are used in this study. MSE measures the average squared difference between predicted and actual returns (GeeksforGeeks, 2025). It is sensitive to large errors, which makes it valuable for financial forecasting as even small deviations can have significant impacts in the MSE. RMSE is the square root of MSE. It retains the same sensitivity to large errors but also expresses the result in the same unit as the predicted variable, in this study it is the daily returns. This makes the RMSE more interpretable in practical terms (Milvus, n.d.). Additionally, RMSE is also very suitable for model comparison, as it allows a direct, unit-consistent comparison across models on the same dataset.

A more practical measure of accuracy is directional accuracy. Directional accuracy measures the proportion of instances in which the model correctly predicts the direction (up or down) of the next-day return. In this context, an "up" prediction corresponds to a positive return, while a "down" prediction reflects a negative return. In financial forecasting, correctly anticipating the direction of market movements is often more valuable than predicting the exact magnitude of those movements, particularly for traders, portfolio managers, and algorithmic strategies that rely on directional signals for buy or sell decisions. As emphasized by Chen et al. (2021), direction-based evaluation metrics are essential for assessing the actionable value of predictive models in real-world market applications, where timing and directional cues can significantly affect profitability and risk exposure.

R^2 is the coefficient of determination, which measures the proportion of variance in the dependent variable (next-day return) explained by the model. It provides insights into the model's explanatory power in regression-based models. However, it is important not to rely too much on this indicator as it is not always reliable, as most of the time, adding additional variables can increase the R^2 (Glantz et al., 2016).

ROOS is the out of sample R squared, adapted from Gu et al. (2020).

$$ROOS = \frac{\sum_{(i,t)\epsilon T_{test}} \left(r_{i,t+1} - \hat{r}_{i,t+1} \right)^2}{\sum_{(i,t)\epsilon T_{test}} \left(r_{i,t+1} \right)^2}$$

The ROOS measures how much a model can reduce forecast errors relative to a naive benchmark that always predicts zero returns. A ROOS > 0 means the model's mean squared error is smaller than always forecasting zero, so it explains some return variation out-of-sample. A ROOS < 0 means that the model does worse than the zero-return forecast. This is a good metric for financial forecasting because it benchmarks against a zero-return instead of historical means which often underperforms a zero-return forecast. This ensures that any positive improvement in ROOS comes from the added influence of sentiment.

This avoids forward-looking bias and provides a more realistic estimate of a model's generalization ability. This is particularly important to address sub-RQ2, regarding if the influence-weighted sentiment variable can improve the predictive accuracy of return forecasts for S&P 500. A model that overfits past data would not be good at forecasting unseen data, which reduces predictive accuracy of the model. While all metrics are reported across training, validation, and test sets to assess learning and overfitting, ROOS and directional accuracy are emphasized in the test set. These two metrics most directly reflect the objectives of Sub-RQ2.

3.5.4.2 Diebold Mariano Tests

To compare the predictive accuracy between the raw sentiment model and each weighted sentiment model, the Diebold-Mariano (DM) test was used. The DM test is designed to assess whether the difference in forecast errors between two competing forecast models is statistically significant based on the loss differential between their prediction errors (Diebold & Mariano, 1995). In this study, the MSE loss function is used to construct the error differential series. Each weighted-sentiment model is compared against the raw sentiment model and to each other, where the null hypothesis assumes equal predictive accuracy, and the alternative suggests that one model outperforms the other using MSE loss functions. To interpret the Diebold-Mariano tests, if a p-value of less than 0.05, it is evidence of a statistically significant difference in predictive performance rejecting the null hypothesis in favor of the alternative one. This statistical test supports Sub-RQ2, by testing whether the factoring in influence in the sentiment variable does improve predictive accuracy of next-day returns compared to what is captured by raw sentiment alone.

3.6 Summary of Methodology

Based on the above, this study follows a robust, four-phase methodology to address the overarching research question, by testing whether an IWSS can help improve next-day S&P 500 return forecasts. First, the data was collected by scraping X posts and using the Yahoo Finance API to extract historical price data. The data are then cleaned to be used for sentiment classification and feature engineering to address Sub-RQ1. To do this, two LLMs (FinBERT and RoBERTa) were fine-tuned on a pre-labeled financial-tweet dataset to extract positive, neutral, and negative sentiment and then used to label the dataset created in this study. The dataset is then chronologically split (60% train, 20% validation, 20% test) to prevent information leakage. Then, four IWSS features were constructed by combining each post's sentiment score with a weighted engagement metric (likes, reposts, replies) and log(followers + 1), employing equal-weight, global-ratio, and Random Forest-derived (impurity, permutation, SHAP) weighting strategies, completing the answer to Sub-RQ1. To prepare data for forecasting, the daily IWSS and raw sentiment features were aggregated per post to a daily sentiment signal and merged with same-day financial variables extracted from Yahoo Finance. Finally, OLS, Random Forest, XGBoost, and LSTM models were trained on six feature sets (raw sentiment and IWSS variants) with time-based splits and hyperparameter tuning to minimize RMSE. The test performance measured by RMSE, ROOS, R^2, directional accuracy, and Diebold-Mariano tests demonstrates the method's comprehensiveness and assesses whether IWSS consistently outperforms raw sentiment in forecasting next-day returns, addressing Sub-RQ2.

4. Data Analysis and Results

The findings from the predictive models in the next day return forecasting revealed many observations regarding the impact of weighted sentiment compared to raw sentiment on forecasting returns. This chapter will address the overall research question and each of the sub-research questions by drawing conclusions from the results of the sentiment analysis on the extracted X posts, as well as from the return predictions based on a combination of the sentiment analysis data and financial data obtained from Yahoo Finance. The first section will present the findings related to the LLMs used in sentiment forecast, which explores sub-RQ1:*How can LLMs be used to extract investor sentiment from X posts, and how can engagement and influence indicators be incorporated into a novel sentiment variable?* This identifies which language model is most suitable for extracting investor sentiment from the collected data. The second section will address sub-RQ1, but focus on the second part about creating a weighted sentiment variable. This section will discuss the results of the different weighting strategies and what each represents. The last section will address sub-RQ2: *To what extent do influence-weighted sentiment variables improve next-day return forecasting accuracy over raw sentiment alone?* This section will present the model performances of the base model, the model with raw sentiment and the models with the IWSS variables across four learning algorithms. A comparative analysis between the IWSS model and sentiment score model will be conducted to determine if there is an improved performance.

4.1 FinBERT and RoBERTa's Comparative Results

Addressing the first part of sub-RQ1, the effectiveness of how LLMs can be used to extract investor sentiment primarily depends on the choice of model and extraction method. To do this, the performance of the two pre-trained LLMs FinBERT yiyanghkust/finbert-tone (Yang, 2020) and RoBERTa cardiffnlp/twitter-roberta-base-sentiment (CardiffNLP, 2020) was evaluated. Both were initially tested under zero-shot conditions (directly on the test set), then fine-tuned using the labeled TimKoornstra dataset. Additionally, in the dataset to be used for constructing influence-weighted sentiment 106 randomly selected posts were manually labelled to validate the best performing model. Performance was evaluated using precision, recall, F1-score, and loss-based learning trends.

4.1.1 Zero-shot Results

Initially, FinBERT and RoBERTa were directly applied on the test set without any fine-tuning to see how well they performed as a baseline. FinBERT achieved a rather poor accuracy of 26.5%, which was worse than random guessing (33.3% for a 3-class problem). In contrast, RoBERTa performed significantly better, having achieved an accuracy of 61.4%. This indicates RoBERTa was much more reliable for classification, and highlights its superior capability in handling social media texts like on X.

	FinBERT			**RoBERTa**		
	Precision	Recall	F1-score	Precision	Recall	F1-score
Negative	0.7532	0.1702	0.2777	0.7815	0.3991	0.5284
Neutral	0.1524	0.9455	0.2625	0.2542	0.8033	0.3861
Positive	0.8702	0.1561	0.2648	0.9052	0.6067	0.7265

Table 3: Comparison table for zero-shot FinBERT and RoBERTa

The results in Table 3 indicate that FinBERT's imbalance between precision and recall severely limited its reliability in classifying sentiment. For both negative and positive sentiment, FinBERT demonstrated high precision (0.75 and 0.87 respectively) but very low recall (0.17 and 0.16), leading to weak F1-scores of 0.27 and 0.26. This suggests that while FinBERT was often correct when assigning these labels, it missed the majority of actual negative and positive posts. For the neutral class, the opposite pattern was observed. FinBERT achieved an extremely high recall(0.95), but low precision (0.15), resulting in a low F1-score of 0.26. This indicates that it correctly identified most neutral posts but misclassified many non-neutral posts as neutral, reflecting a strong overprediction bias toward the neutral class.

In contrast, RoBERTa achieved a much better balance across all classes. For negative sentiment, it maintained a similar precision (0.78) but more than doubled the recall (0.39), yielding a higher F1-score of 0.53. For positive sentiment, it improved on both precision (0.91) and recall (0.61), resulting in a strong F1-score of 0.78. Even for neutral sentiment, RoBERTa demonstrated better precision (0.25) while maintaining high recall (0.80), improving the F1-score to 0.31. However, RoBERTa still exhibits similar struggles regarding neutral and negative classes, where precision and recall remain relatively lower compared to the positive class. This suggests that both models find it more difficult to consistently distinguish between neutral and negative sentiment in the informal language of X posts.

Despite this, the results highlight that RoBERTa is better at capturing the three sentiment classes at a zero-shot setting. This advantage is likely due to its pretraining on social media text, which may be more similar to the X posts about financial data as it is adjusted to understand more informal and context-dependent language, compared to FinBERT's domain-specific training on formal financial documents. This is consistent with literature, as Gururangan et al. (2020), highlights the importance of pre-training alignment with the target domain. Thus, further fine-tuning was conducted to improve the results so both models align better with the informal tones of X posts, while also training on perhaps more finance-specific language as the X posts are selected based on if they mention the S&P 500 index.

4.1.2 Fine-tuned Model Results

Following established best practices, both models were fine-tuned on the TimKoornstra dataset for three epochs. This training duration aligns with best practices for large language models, which are prone to overfitting on smaller datasets. The training and validation loss were monitored to evaluate generalization.

Model	Accuracy	Class	Precision	Recall	F1-score
FinBERT	26.51%	Negative	0.7532	0.1702	0.2777
		Neutral	0.1524	0.9455	0.2625
		Positive	0.8702	0.1561	0.2648
Fine-tuned FinBERT	39.52%	Negative	0.3078	0.6266	0.4128
		Neutral	0.2225	0.6504	0.3316
		Positive	0.8283	0.2824	0.4212
RoBERTa	61.41%	Negative	0.7815	0.3991	0.5284
		Neutral	0.2542	0.8033	0.3861
		Positive	0.9052	0.6067	0.7265
Fine-tuned RoBERTa	64.56%	Negative	0.7881	0.5029	0.614
		Neutral	0.2755	0.8432	0.4153
		Positive	0.929	0.6729	0.7805

Table 4: Comparison of evaluation metrics across zero-shot and fine-tuned RoBERTa and FinBERT

Based on Table 4, RoBERTa's training loss decreased from 0.53 to 0.21 indicating that the model is learning. However, validation loss increased from 0.70 to 0.79 across the epochs. This could suggest that there may be overfitting early on. However, validation accuracy and balanced accuracy all improved during training, suggesting that the model was learning to generalize better, despite validation loss increasing (see Appendix D.1). FinBERT also showed a decrease in training loss (0.68 to 0.21) indicating strong fitting to the training data. In contrast to RoBERTa, FinBERT's validation loss decreased until epoch 2 and then increased to the final epoch (see Appendix D.2). This suggests FinBERT peaked in generalization based on loss and began overfitting thereafter.

4.1.3 Model summary and sample evaluation

In summary, both RoBERTa and FinBERT models seemed to improve with fine-tuning, which improved classification across all sentiment classes (see Figure 5). This demonstrates that LLMs can effectively extract investor sentiment from X posts, particularly when they are not only pretrained on relevant data but also fine-tuned on domain-specific tasks aligning with the literature

(Gururangan et al., 2020) and answering sub-RQ1. Despite these improvements, both models seem to struggle the most with negative and neutral posts. This is likely due to the imbalance in class distribution in the training dataset, as more positive than negative and neutral classes were present in the data.

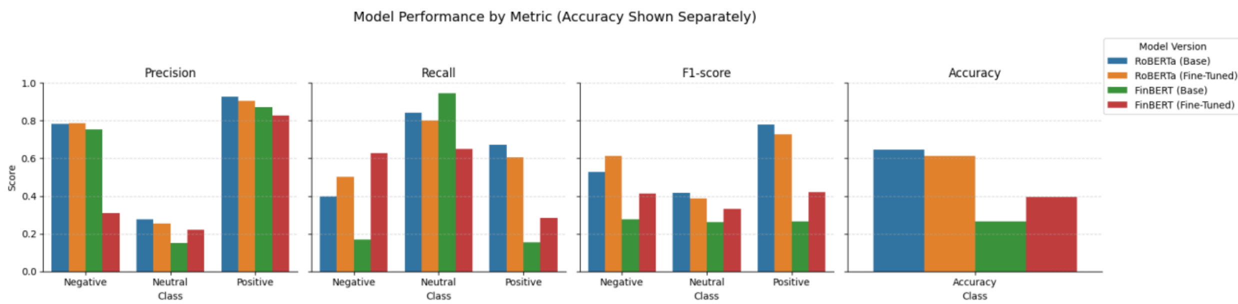


Figure 5: Comparison of Model Performance of RoBERTa and FinBERT LLMs

As visually presented in Figure 5 the fine-tuned RoBERTa model consistently outperformed FinBERT across all metrics, indicating that even though the X posts were based on financial subjects, the linguistic style may be more important. RoBERTa's pretraining on social media data gives it a linguistic advantage, while FinBERT, although trained on financial language, does not capture the same patterns in informal social media texts. These findings are consistent with prior work, which also found RoBERTa to be more effective for sentiment classification in financial social media (Karanikola et al., 2023).

To verify the performance of the fine-tuned RoBERTa, a small-scale diagnostic evaluation was conducted using a manually labeled sample of 106 X posts from the collected dataset. Among these, 43 were manually labelled Neutral, 38 positives, and 29 negative reflecting a moderately balanced distribution. Although this sample is not large enough for statistical generalization or further fine-tuning, it serves as a useful indication of the model's applicability to this domain-specific dataset. The fine-tuned RoBERTa achieved an accuracy of 80.19%, which is very high, suggesting a strong initial performance based on this random sample and thereby the collected data to be used for IWSS construction. As shown in table 5, the model exhibited consistently high class-wise performance yielding F1-scores of 0.794 (neutral), 0.769 (negative), and 0.835 (positive).

These results are proof of the model's capability to identify sentiment nuances even in the domain-specific S&P 500-related X posts.

	Fine-tuned RoBERTa		
	Precision	**Recall**	**F1-score**
Negative	0.7941	0.7941	0.7941
Neutral	0.8621	0.6944	0.7692
Positive	0.7674	0.9167	0.8354

Table 5: Evaluation metrics tested on collected dataset

For comparison, the other transformers were also tested on this sample. Base RoBERTa achieved 60.38%, fine-tuned FinBERT reached 66.98%, and base FinBERT was the worst performer with only 16.04% accuracy (Appendix D.2). Based on this comparative evaluation, fine-tuned RoBERTa was selected as the sentiment classifier used to classify the sentiment of X posts collected in this study based on the S&P 500 index. These classifications will be used for constructing the influence-weighted sentiment variable (IWSS). This decision directly supports the first sub-research question by showing how LLMs, when properly adapted, can reliably extract sentiment from investor communications on X enabling the development of a sentiment signal that is both content- and context-aware.

4.2 Constructing the Influence-weighted Sentiment Variable

The effectiveness of sentiment extraction using the fine-tuned RoBERTa LLM has been conducted, thus, the analysis shifts focus to the second part of sub RQ1: *how engagement and influence indicators can be incorporated into a novel sentiment variable*. The focus is on constructing an influence-weighted sentiment score that reflects the diffusion and reach of the sentiment signals created by an X post. As outlined in the methodology, this is achieved by combining the sentiment (encoded as numbers) with engagement score and reach based on the influence indicators of likes, reposts, replies and follower count. The formula for IWSS combines the sentiment label (S) with a composite engagement score (E) and a logarithmic follower term, as previously outlined. Here, we focus on how the engagement weights (E) were derived through data-driven methods by first conducting an exploratory analysis on a training set of the extracted sentiment and engagement indicators of each X post to ensure that the data-driven methods are as reliable as possible.

4.2.1 Preliminary Analysis for the construction of IWSS

To ensure the influence indicators are appropriately prepared for use in the IWSS construction, a preliminary exploratory analysis was conducted on the training set. This step is essential for deriving reliable data-driven engagement weights and ensuring the final sentiment variable reflects both engagement quality and scale.

Descriptive statistics (see Appendix B.1) revealed significant differences in scale and skewness among the influence indicators. Follower count exhibited a much wider range and showed some extreme variance, while reposts had a much narrower range. Further all the variables show rather high skew and kurtosis with skew above 10 and kurtosis over 100. This indicates very heavy-tailed distributions. These discrepancies raise the risk that variables with larger raw magnitudes (e.g., follower count) or highly skewed variables might disproportionately influence any weighting strategies, irrespective of their actual predictive value or behavioral relevance.

To address this, various transformation techniques were evaluated (see Appendix B.1). Among these, the Box-Cox transformation was especially effective in making the data less extreme and more balanced, reducing skewness and kurtosis across engagement metrics, while a log transformation was optimal for follower count. This is important to try and compare the different types of engagement fairly. These transformations produced approximately normalized distributions with skew and kurtosis closure to 0 as would be ideal, ensuring that no single metric dominated due to scale. This step is essential not only for fair comparison between indicators, but also to meet statistical assumptions used in later modeling and to maintain interpretability in the weighting procedure.

Additionally, a large proportion of X posts had zero engagement across all indicators (see Appendix B.2). As these posts do not have measurable influence, they were excluded from the training set and subsequently the validation and test sets to ensure consistency. Posts with zeros in only some indicators were retained. Removing completely inactive posts is particularly important, as it aligns with the definition of the IWSS variable, which is designed to reflect sentiment diffusion. Posts with no engagement cannot propagate sentiment and therefore do not contribute to diffusion dynamics. Including such posts could artificially deflate the daily IWSS values and introduce noise into the weight estimation process, which is critical for constructing an accurate influence-weighted sentiment variable. Keeping these non-engaging posts can also harm the analysis by making the return predictions less accurate. Moreover, since such posts are unlikely to be seen or engaged with, they cannot meaningfully contribute to collective investor sentiment, making their exclusion both methodologically and conceptually justified.

Correlation analysis (see Appendix B.3) moderate relationships between the likes, reposts, replies and follower count, with the highest observed between likes and reposts. However, variance inflation factor (VIF) scores, a metric for testing multicollinearity between variables, show that all scores remain under 5, suggesting that multicollinearity is not a major issue. This is extremely important to the inclusion of these influence indicators in the weighting function described in section 3.4.2, as it confirms each indicator adding distinct and usable information to the final sentiment weight.

The sentiment distribution is relatively balanced, with no dominant class, which helps prevent bias towards one class and supports more reliable weight estimation when modelling (see Appendix B.4). Moreover, although only weak linear associations were observed between sentiment and engagement metrics (after sentiment was label-encoded), this indicates that non-linear models like Random Forests may be better to capture more complex relationships in the final weighted-sentiment variable construction.

After filtering and preprocessing, including the bot-automation detection based on high similarity and timing patterns mentioned in the methodology, the training dataset consists of approximately 32,650 data points and spans the period from January 2022 to October 2023. The validation dataset includes around 7,940 data points, covering the period from November 2023 to May 2024. The testing dataset comprises roughly 7,550 data points, collected between June 2024 and December 2024. These steps lay the foundation for constructing a sentiment weighting function that incorporates engagement metrics on comparable scales, free from redundancy, and reflective of each metric's potential contribution to investor sentiment as observed on X.

4.2.2 Engagement Weight Estimation Results

Following the preliminary analysis and preprocessing, the dataset is prepared for constructing the influence-weighted sentiment variable. The final step is to determine how to assign the weights to the engagement indicators (like, repost and reply count) that form the engagement component of the IWSS variable. Five IWSS variants were engineered, each using a different weighting strategy that reflects different assumptions about how sentiment is quantified and interpreted. Testing these alternative constructions helps to mitigate overreliance on any one conceptualization of influence and allows the final model to be grounded in both behavioral theory and empirical evidence. The assigned weights for each strategy are summarized in Table 6.

IWSS Variant	Like	Repost	Reply
IWSS_Equal	0.333	0.333	0.333
IWSS_Ratio	0.69	0.137	0.173
IWSS_Impurity	0.483	0.284	0.233
IWSS_Perm	0.368	0.368	0.264
IWSS_SHAP	0.324	0.429	0.247

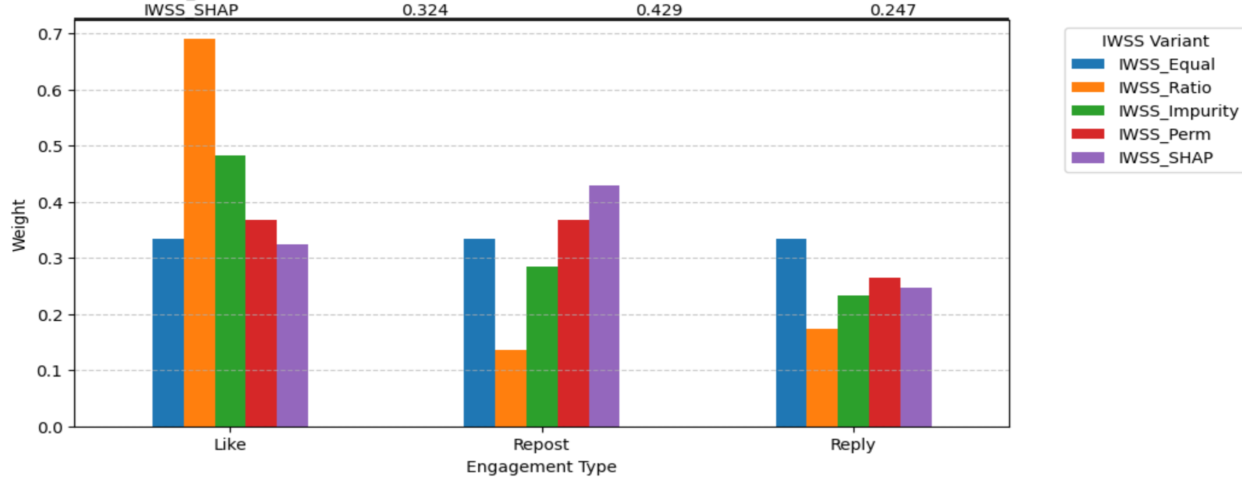


Table 6: Resulting weights of the five IWSS features

Each IWSS feature was derived using the formula outlined in Section 5.4.2, which combines sentiment with engagement level and user reach (follower count). This structure allowed for a clear separation between engagement (how users react to a post) and reach (how many users could potentially see it), enabling the IWSS formulation to capture both the depth and scale of influence in sentiment diffusion. The choice of weighting method noticeably influences the emphasis on engagement indicators. The key variation lies in how engagement indicators are weighted to reflect their relative influence. This construction step is crucial for capturing how different types of user interactions signal attention, approval, or amplification.

The IWSS_Equal variant assigns equal weights (0.33) to all engagement types, assuming that likes, reposts, and replies contribute equally to a post's influence. While this simplifies modeling, it overlooks behavioral differences between the engagement indicators. For instance, it treats likes, which are more passive interactions equally meaningful as a repost, which can amplify content (Cha et al., 2010).

IWSS_Ratio assigns weights based on the relative share of each engagement type in a post's total interactions. Although this reflects within-post dynamics, the approach relies solely on the training data's historical pattern. This makes it potentially unreliable if this were to be used in practice as engagement behaviors can shift during different periods. A model trained during one phase may misrepresent influence during another if engagement patterns evolve. Further, it may be sensitive to skewed behaviors, for instance inflated likes from bots which can potentially distort influence representation.

Data-driven methods derive weights using different feature importance measures. IWSS_Impurity places the greatest weight on likes, likely reflecting impurity-based biases toward high-variance features (Hastie et al., 2009). However, this emphasis may be misleading, especially since likes are often constrained to an author's follower base and can be artificially inflated as they can be bought. Conversely, IWSS_SHAP and IWSS_Permutation more strongly emphasize reposts and replies. SHAP, in particular, highlights reposts as the most influential engagement type (weight = 0.429), suggesting that sentiment influence is best captured not by volume but that deeper interaction and amplifying a sentiment through reposting. This reflects a deeper perspective on sentiment influence, where the spread of sentiment, not just the general investor sentiment, is important.

Overall, the five IWSS variants illustrate how different weightings of engagement shape the construction of sentiment variables. While simpler strategies are easier to implement, they may overlook key behavioral signals. In contrast, data-driven approaches more closely reflect how engagement types such as reposts and replies play distinct roles in amplifying sentiment. These results address the second part of Sub-Research Question 1, demonstrating that influence indicators can be integrated into a sentiment variable that captures not only sentiment, but also its potential reach and impact. This establishes the foundation for comparing the predictive performances of raw sentiment and the newly constructed IWSS variables to see if they can improve prediction forecasts.

4.3 Return Forecasting Model Results

Addressing sub-RQ2 involves evaluating whether the IWSS features improve forecasting performance compared to raw sentiment, and to what extent, by quantifying differences across key performance metrics. The main focus in this section is to compare whether IWSS can perform well compared to raw sentiment across 4 forecasting algorithms (OLS, Random Forests, XGBoost and LSTM). They will be compared and evaluated on the performance metrics R^2, RMSE, MSE, out-of-sample R^2, and directional accuracy to capture explanatory power, error magnitude and ability to predict correctly positive or negative returns. Lastly, the Diebold–Mariano tests are applied to assess which improvements are statistically significant.

4.3.1 Model Specific Analysis and Insights

In this subsection, each forecasting algorithm is first evaluated on individual metrics and Diebold-Mariano test. The results are then summarized in a broader overview

4.3.1.1 OLS Regression

OLS served as a benchmark model across the feature sets, for evaluating the predictive power of each model. As a linear model, no hyperparameters need to be tuned and it offers a useful baseline for seeing if complex machine learning approaches add value. All models were trained on the training set and evaluated on the test set without further fine-tuning. Table 7 summarizes the performance of each metric on the test set as well as the change compared to the baseline feature set of raw sentiment.

Feature Set	R^2	Δ R^2	ROOS	Δ ROOS	RMSE	Δ RMSE	MSE	Δ MSE	Dir. Acc.
Base + Sentiment_score	-0.139929	-	-0.125518	-	0.009082	-	0.000082	-	42.17687
Base + IWSS_Equal	-0.026776	**0.113153**	-0.013796	**0.111722**	0.008619	**-0.000463**	0.000074	**-0.000008**	46.2585
Base + IWSS_Ratio	-0.028458	0.111471	-0.015456	0.110062	0.008626	-0.000456	0.000074	**-0.000008**	46.2585
Base + IWSS_Impurity	-0.027968	0.111961	-0.014972	0.110546	0.008624	-0.000458	0.000074	**-0.000008**	46.2585
Base + IWSS_Permutation	-0.027393	0.112536	-0.014405	0.111113	0.008622	-0.00046	0.000074	**-0.000008**	46.2585
Base + IWSS_SHAP	-0.026939	0.11299	-0.013956	0.111562	0.00862	-0.000462	0.000074	**-0.000008**	46.2585

Table 7: OLS model results with changes across performance metrics

From the results, it is clear that the application of the influence-weighted sentiment scores within the OLS model shows substantial improvements in forecasting accuracy of the raw sentiment score. In particular, the IWSS features consistently showed better R^2 value, ROOS and directional accuracy, as well as lower RMSE and MSE scores demonstrating that with OLS modelling their predictive signal is higher than that of the raw sentiment score. The influence weighting-approach across all managed to boost explanatory power with all models including an IWSS feature improving R^2 by approximately 11.2%. Among these IWSS_Equal exhibited the highest incremental increase of around 0.113, closely followed by IWSS_SHAP (0.1130). Similar gains were observed for ROOS, indicating all models with IWSS features closer to beating the naive benchmark of predicting zero-returns than raw sentiment. The ROOS of -0.1255 for indicates that the raw sentiment score predicted an MSE of 12.55% worse than the predicted mean, whereas for the IWSS features set only misses by 1.5%. Despite these improvements it is important to note that the performance under OLS is overall poor as all ROOS and R^2 are negative indicating that none of the models can beat the naive benchmark of predicting zero-returns or the mean of the training set.

Directional accuracy notably also improved by 4.08% across all models with IWSS, increasing from 42.18% (raw sentiment) to 46.26%. Moreover, IWSS features had lower RMSE (~0.0086) and MSE (0.000074) which is much lower than the raw sentiment score which has an RMSE of 0.0091 and MSE of 0.000082. This suggests that the IWSS features improve predictive accuracy in terms of not only whether it can correctly predict if returns are positive or negative (directional accuracy), the lower magnitude of prediction errors also makes more precise return predictions. Although directional accuracy of the IWSS variables does not beat the 50% random guessing, it is still an improvement on the raw sentiment model.

To statistically validate these differences, Diebold-Mariano (DM) tests were conducted comparing forecast errors between the raw sentiment model and each IWSS model. Further analysis also compares forecast errors between the IWSS models. The DM test evaluates whether the difference in predictive accuracy between two models is statistically significant, based on the MSE loss function. The results (see Appendix E.2) show that all five models with the IWSS feature performed significantly better at a 1% significance level than the raw sentiment model based on the MSE loss.

Overall, although all IWSS features performed well, the simplest method with equal weighting of engagement indicators (IWSS_Equal) performed better than the more complex weighting methods. This indicates under an OLS framework that simpler weighting schemes may be more effective. However, it is important to note that pairwise DM tests between the models do not reveal any significant improvements between the IWSS models (see Appendix E.2), indicating that no one IWSS feature like IWSS_Equal performs significantly better than another.

These results indicate a clear extent of improvement in forecasting when employing influence-weighted sentiment scores over raw sentiment. Importantly, the findings underline the robustness of the IWSS approach itself, rather than the necessity of sophisticated weighting techniques.

Consequently, adopting IWSS, even in its simplest forms, significantly enhances predictive accuracy and provides substantial economic benefits in practical forecasting scenarios.

Assumptions of the OLS models were evaluated (see Appendix E.1). While these violations do not directly impact predictive accuracy, it does provide some insights into the results. The Breusch-Pagan test indicated heteroscedasticity in several models, the Shapiro-Wilk test suggested non-normality of residuals, and the scatter plots showed non-existent linear relationships between the features and next-day returns, violating key OLS assumptions. These violations may help explain the consistently negative R^2 values. It also implies that the linear model structure may not be well-suited to capture the underlying patterns in the data.

4.3.1.2 Random Forest

The Random Forest model was selected as a non-linear alternative to OLS. It is evident based on the results of OLS that a linear model does not fully capture the relationship with time series financial data, which is also expected based on past literature (Gu et al., 2020), suggesting that more complex models are better for forecasting returns. To better model potential interactions involving the influence-weighted sentiment variable, more robust methods such as Random forests are used to assess whether it improves predictive accuracy over raw sentiment.

Hyperparameter tuning was conducted using grid search cross-validation for all Random Forest models. The best parameters were selected based on validation MSE. The best performing IWSS models on the validation set typically favored shallow trees (max_depth of 3), moderate split thresholds (min_samples_split of 5), broader feature consideration at each split (max_features using sqrt) and larger ensemble sizes (n_estimators of 100). This suggests that influence-weighted sentiment introduces weak signals, albeit meaningful, that prefer more conservative and less complex trees to improve generalization and reduce overfitting.

Feature Set	R^2	ΔR^2	ROOS	Δ ROOS	RMSE	Δ RMSE	MSE	Δ MSE	Dir. Acc.	Δ Dir Acc.
Base + Sentiment_score	-0.063052	-	-0.049613	-	0.00877	-	0.000077	-	43.53742	-
Base + IWSS_Equal	0.007223	**0.070275**	0.019774	**0.069387**	0.008475	**-0.000295**	0.000072	**-0.000005**	50.34014	**6.80**
Base + IWSS_Ratio	-0.008149	0.054903	0.004596	0.054209	0.00854	-0.00023	0.000073	-0.000004	46.93878	3.40
Base + IWSS_Impurity	0.000397	0.063449	0.013034	0.062647	0.008504	-0.000266	0.000072	**-0.000005**	46.2585	2.72
Base + IWSS_Permutation	0.005703	0.068755	0.018273	0.067886	0.008482	-0.000288	0.000072	**-0.000005**	46.2585	2.72
Base + IWSS_SHAP	-0.006219	0.056833	0.006502	0.056115	0.008532	-0.000238	0.000073	-0.000004	43.53742	0

Table 8: Random Forest model results with changes across performance metrics

The results in Table 8 show considerably higher performance across all metrics when influence-weighted sentiment is included compared to the raw sentiment feature, specifically within the Random Forest model. A particularly interesting finding is the shift from negative to positive explanatory power observed from using raw sentiment (-0.063) to some of the IWSS features, notably IWSS_Equal with an R^2 of 0.0072, IWSS_Permutation (0.0057), and IWSS_Impurity (0.0004). Although this account is marginal, with only a maximum of a 0.7% increase in explanatory power. However, other metrics also show improvements. Both loss metrics (RMSE, and MSE) show a decrease compared to the model with raw sentiment, with lower RMSE overall. Again IWSS_Equal, IWSS_Permuation and IWSS_Impurity show the lowest values indicating a higher predictive accuracy when forecasting next day returns on S&P 500. Further, all models with IWSS display positive ROOS, indicating that it was able to beat the zero-return forecast benchmark, with the highest being IWSS_Equal which suggests a 2% higher MSE than zero-return. Compared to the raw sentiment model which still underperforms relative to the zero-return forecast (-0.049 ROOS), IWSS is better at capturing predictive signals, whereas the raw sentiment model adds noise.

Directional accuracy in particular reveals strong performances, particularly using the IWSS_Equal feature set. This is because the directional accuracy not only displayed much higher scores than the raw sentiment (43.53%), but it also managed to achieve a 50.34% accuracy. This is particularly impressive as it just slightly beats random guessing (50%) indicating that the random forest model using the IWSS_Equal feature set is better at predicting positive and negative returns than random chance. Although it is a marginal difference, it can have significant impacts on returns in the long run, as they can still better inform buy and sell decisions (Grinold & Kahn, 1999). Conversely, more complex weighting methods like IWSS_SHAP did not show any improvements in directional accuracy, which could suggest that more simple weighting methods better capture any predictive information as IWSS_SHAP may have problems with overfitting to the training set used to calculate its weights as the SHAP scores are more localized.

Based on these findings, it is evident that the IWSS outperforms the raw sentiment feature under Random Forest models. Additionally, the results highlight potential synergy between simpler weighting schemes and the Random Forest model's inherent robustness and flexibility.

The Diebold-Mariano tests (see Appendix E.3) add additional evidence supporting these findings. The test confirms when comparing each IWSS model to the raw sentiment model, that the IWSS models (aside from SHAP) achieved statistically significant improvements in the predictive accuracy over raw sentiment at a 1% significance level. However, the weighting method inherent to the IWSS features did not show many strong effects on predictive accuracy with only two statistically significant differences observed in pairwise comparisons, where IWSS_Permutation outperformed IWSS_Ratio and IWSS_Equal outperforming IWSS_SHAP. This suggests that while weighting can matter in specific cases, its overall influence remains limited.

4.3.1.3 XGBoost

The following results summarize the predictive performance of the feature sets across both the validation and test sets using the XGBoost algorithm. This model was also selected for its ability to handle non-linearity and manage feature interactions efficiently using boosting to sequentially correct errors and minimize bias. Unlike random forest, which builds independent trees using bagging, that relies on bagging, XGBoost builds trees sequentially and optimizes an objective function, making it more effective in handling imbalanced data and preventing overfitting (Nagabhushan, 2021).

Based on the search grid performed on the hyperparameters, the best parameters for each model were selected based on the validation set. The best hyperparameters for all IWSS models contained subsample = 0.6, colsample_bytree = 0.6, max_depth = 5, n_estimators = 100, and learning_rate

= 0.01, while the raw sentiment model and base model favored shallow trees (max_depth of 3) and higher sampling rates (subsample of 0.8 and colsample_bytree = 0.8). This suggests that the IWSS models are deeper and more complex to extract meaningful patterns similar to the deeper trees required for Random Forest (Appendix A.2).

Model	R^2	ΔR^2	ROOS	Δ ROOS	RMSE	Δ RMSE	MSE	Δ MSE	Dir. Acc.	Δ Dir Acc.
Base + Sentiment_score	−0.03960	—	−0.02646	-	0.008673	-	0.000075	-	45.58%	-
Base + IWSS_Equal	−0.01019	**0.02941**	0.00258	**0.02904**	0.008549	**−0.000124**	0.000073	**−0.000002**	48.98%	**+3.40 pp**
Base + IWSS_Ratio	−0.02182	0.01779	−0.00890	0.01756	0.008598	−0.000075	0.000074	−0.000001	51.02%	**+5.44 pp**
Base + IWSS_Impurity	−0.02374	0.01587	−0.01080	0.01566	0.008606	−0.000067	0.000074	−0.000001	44.90%	−0.68 pp
Base + IWSS_Permutation	−0.02974	0.00986	−0.01672	0.00974	0.008631	−0.000042	0.000075	0	48.98%	+3.40 pp
Base + IWSS_SHAP	−0.02270	0.0169	−0.00977	0.01669	0.008602	−0.000071	0.000074	−0.000001	48.98%	+3.40 pp

Table 9: XGBoost model results with changes across performance metrics

The results presented in Table 9 clearly show a trend in improvements in the models containing IWSS features compared to the models with raw sentiment using the XGBoost model. The change in R^2 (ΔR^2) shows that every IWSS model improves upon the raw sentiment baseline, with IWSS_Equal showing the least negative R^2. However, it is important to consider that despite this improvement, the overall explanatory power of these models are very poor, as all of the R^2 values are negative, indicating that these models cannot predict better than the naive benchmark of predicting the mean.

Comparing the ROOS values reveals that all IWSS models have marginal improvement compared to the raw sentiment, producing less negative ROOS values. However, only one model (IWSS_Equal) achieved a positive ROOS, being the only model to reduce error compared to the zero-return forecast, indicating higher predictive value. However, all other models including both IWSS features, and raw sentiment actually increase error relative to the naive benchmark. In other words, IWSS_Equal is the only feature set that adds real predictive power beyond the simplest zero-return forecast.

The error metrics show a decreasing trend when comparing changes in RMSE, and MSE between the IWSS and raw sentiment models. Namely, the declining trend indicates that IWSS models have lower forecasting loss and therefore higher predictive accuracy. This implies that IWSS does achieve higher predictive accuracy and improves out-of-sample robustness by generating cleaner, less noisy sentiment signals that transfer more reliably to unseen data. IWSS_Equal again achieves the best improvement with the lowest RMSE of 0.008549 and MSE of 0.000073.

The directional accuracy shows mostly higher accuracy across all models of the IWSS compared to the raw sentiment model at 45.58%. More notably, IWSS_Ratio managed to achieve a directional accuracy of 51.02% surpassing the 50% random chance threshold, indicating that XGBoost's boosting method of building trees sequentially, may help more accurately predict the direction of a return rather than the exact next-day return. This implies that the engagement-frequency weights may have an impact on the model's ability to capture the market's up and down signal. These weights in particular, emphasized engagement metrics such as likes, which could perhaps be considered important under the XGBoost models.

Overall, these findings provide support for the hypothesis that the influence-weighted sentiment variable yields more accurate next-day return forecasts than using the raw sentiment, particularly when using equal or frequency-based weights. However, based on the Diebold Mariano tests (Appendix E.4), none of the results support the statistical significance for any model comparisons at a 1% or 5% significance level. This means that despite observed improvements in metrics like ΔR^2, ROOS, and directional accuracy, there is no conclusive statistical evidence to confirm the superiority of any IWSS model over raw sentiment or over each other.

Thus, while IWSS_Equal appears to offer the most consistent improvements across metrics, and IWSS_Ratio excels in directional prediction, these findings must be interpreted within the limits of statistical significance. Overall, the results suggest that IWSS models can enhance sentiment signal quality, but no single weighting strategy proves definitively superior using XGBoost models. It is also possible that the complexity of the XGBoost model, with its deep boosting trees, may have been too high relative to the sample size, limiting the model's ability to translate marginal feature improvements into statistically reliable performance gains.

4.3.1.4 LSTM

The LSTM model delivers strong overall performance, particularly in directional accuracy among the four different algorithms tests. As a recurrent neural network, LSTM is designed to capture temporal dependencies, which can be a key advantage when modelling financial data.

A small grid search was conducted on the validation set, in order to optimize LSTM performance. The conducted grid search tested a combination of dropout rates [0.1, 0.2, 0.3] and L2 regularization values [1e-5,1e-4,1e-3]. These values were specifically chosen due to being commonly used ranges that balance regularization strength without overly constraining model capacity. Early stopping with a patience of 10 epochs was applied to prevent overfitting. The best combination of dropout and L2 was determined individually for each of the six feature sets. For example, the optimal configuration for Base+Sentiment_score was dropout = 0.2, L2 = 1e-5. For Base+IWSS_Equal it was dropout = 0.1, L2 = 1e-4, and for Base+IWSS_permutation dropout = 0.1, L2 = 1e-3. Finally, the optimal hyperparameters identified on the validation set were applied to the test set (Appendix A.3).

Feature Set	R^2	ΔR^2	ROOS	Δ ROOS	RMSE	Δ RMSE	MSE	Δ MSE	Dir. Acc.	Δ Dir Acc.
Base + Sentiment_score	-0.178945	-	-0.16404	-	0.009236	-	0.000085	-	48.29932	-
Base + IWSS_Equal	0.004166	**0.183111**	0.016756	**0.180796**	0.008488	**-0.000748**	0.000072	**- 0.0000130**	57.82313	**9.52**
Base + IWSS_Ratio	-0.091983	0.086962	-0.078178	0.085862	0.008888	-0.000348	0.000079	- 0.0000060	41.4966	-6.80
Base + IWSS_Impurity	-0.077594	0.101351	-0.063971	0.100069	0.00883	-0.000406	0.000078	- 0.0000070	48.97959	0.68
Base + IWSS_Permutation	-0.027406	0.151539	-0.014418	0.149622	0.008622	-0.000614	0.000074	- 0.0000110	44.21769	-4.08
Base + IWSS_SHAP	-0.024953	0.153992	-0.011995	0.152045	0.008611	-0.000625	0.000074	- 0.0000110	46.93878	-1.36

Table 10: LSTM model results with changes across performance metrics

The results presented in Table 10 examine the predictive performance of influence-weighted sentiment scores (IWSS) relative to raw sentiment when incorporated into a LSTM model. Looking at R^2, the model with the raw sentiment feature yielded the lowest R^2 (−0.179), indicating that the model cannot beat the naive mean prediction. However, interestingly, the model including IWSS_Equal achieves a positive R^2 of 0.0042. Similarly, this is the same for ROOS, indicating IWSS_Equal is the only model to predict better than just predicting 0 returns. This is the only

IWSS feature to have a positive R^2 and ROOS, which suggests that equal weighting of engagement types effectively enhances the informativeness of sentiment signals when used as input for LSTM architectures. Moreover, although all IWSS variants show a decreasing trend for RMSE and MSE, the lowest values achieved are by IWSS_Equal, indicating higher predictive accuracy in comparison to raw sentiment. The higher performance of the IWSS_Equal suggests that IWSS_Equal enhances the model fit and error metrics. However, it is important to note that the model fit is still quite low, with equally weighted sentiment and financial metrics only explaining 0.4% of the variance in next-day return. This is likely due to omitted variable bias which will be discussed more in Section 6.

Directional accuracy, which measures the model's ability to correctly predict the direction of market movement, is particularly relevant for trading applications. Here the IWSS features produce mixed results. The IWSS_Equal achieved the highest directional accuracy at 57.82% which is 9.52% higher than raw sentiment score. This finding is significant as it beats random guessing by 7% which is difficult to achieve according to literature (Yıldırım, Toroslu, & Fiore, 2021). This finding could be due to the LSTM architecture, indicating that there may be some stable temporal patterns from the equally weighted sentiment, which is not captured by raw sentiment. However, the other IWSS features, including IWSS_SHAP (46.94%), IWSS_Permutation (44.22%), and IWSS_Impurity (48.98%) do not achieve such high accuracy, and also does not outperform the raw sentiment model. This suggests that under LSTM the weighting strategy does have an impact on predictive accuracy on next-day returns.

DM tests comparing each model including IWSS to each other and to the raw sentiment model were conducted to observe if superior model performance was statistically significant (see Appendix E.5). IWSS_Equal, IWSS_SHAP and IWSS_Permutation yields a p-value of less than 0.01, indicating these are significantly better than the raw sentiment forecasts at a 1% significance level. Comparing the IWSS reveals that IWSS Ratio significantly outperforms IWSS_SHAP, and IWSS_PERM at a 1% level, and IWSS Equal is significantly better than IWSS_Ratio. Otherwise, there are no other significant results.

Overall, it seems that under LSTM, IWSS is better across metrics in the next day return prediction as indicated by higher R^2, ROOS, RMSE and MSE, although not all performances are statistically significant. Among the tested models, IWSS_Equal demonstrates the most consistent and statistically supported improvements, both in forecast accuracy and directional prediction, although not at a significant level.

4.4 Summary of Key Findings

Based on the result from the findings from the machine learning and deep learning models, conclusions can be made about the extent of the improvement in forecasting between influence-weighted sentiment scores and raw sentiment.

4.4.1 LLM and Variable Creation Summary

The evaluation comparing the LLMs RoBERTa and FinBERT demonstrated that FinBERT struggled to classify the text both on a zero-shot setting and when fine-tuned to the TimKoornstra dataset. RoBERTa was the model that performed best, 64.6% vs. 39.5% for FinBERT. Although both models exhibit reduced training losses following fine-tuning, only RoBERTa achieves balanced improvements in precision, recall, and F1 across negative, neutral, and positive classes. Moreover, on a small diagnostic assessment on 106 manually labelled X posts, the fine-tuned RoBERTa achieved 80.2% accuracy with F1-scores above 0.79 for all classes. Building on the sentiment predicted with the best LLM (RoBERTa), an influence-weighted sentiment score (IWSS) was constructed by preprocessing engagement indicators (likes, reposts, replies, follower counts) to address extreme skewness by transforming the variables, and then filter to exclude posts with zero total engagement. Correlation and VIF analyses confirmed that multicollinearity was not an issue, and sentiment labels were sufficiently balanced to support reliable weight estimation. Five IWSS features were then engineered based on different weighting strategies: equal weights, ratio-based weights, impurity-based, SHAP, and permutation-based weights. The ratio-based weights put the most emphasis on likes as it is determined by proportion and it is clear that as the feature importance from random forests gets more localized (from impurity-based measures like variance in random forests to SHAP values), the weighting shifts toward deeper engagement metrics, such as reposts, which better reflect sentiment diffusion.These weights were then plugged into the equation:

$$E = \omega_1 \times Likes + \omega_2 \times Reposts + \omega_3 \times Replies$$

To find the engagement score, and then plugged into the main IWSS equation:

$$IWSS = S \times E \times log(F + 1)$$

to create the features to be used in return forecasting.

4.4.2 Return Forecasting Summary

Across all algorithms, incorporating IWSS features delivers both statistically and economically meaningful gains over raw sentiment. Under OLS regressions, adding IWSS lead to improvements in R^2 and ROOS of over 11%, indicating higher explanatory power. Similar improvements are observed across LSTMs, Random Forest and XGBoost, always with at least 1-2%. Moreover, the RMSE and MSE show reductions across algorithms although marginally small. The directional accuracy also consistently shows improvements, although with more mixed results for complex models like LSTM showing an increase of up to 9.5% from the raw sentiment model. Crucially, at least one model with IWSS features (typically IWSS_Equal) always significantly outperforms raw sentiment across all performance metrics according to the Diebold-Mariano tests, confirming that these improvements are statistically meaningful rather than coincidental. Together, these results underscore IWSS as a consistently valuable variable for forecasting next-day returns, delivering improvements across multiple performance metrics and statistical validations.

Additionally, across algorithms, the IWSS feature using equal weighting (IWSS_Equal) emerged as the most powerful predictor of next-day returns. It is the only feature used across models to achieve positive ROOS and R^2 across all algorithms, with the exception of OLS. Further, it consistently produced lower RMSE and MSE than the raw sentiment baseline, indicating improved forecast accuracy. Its directional accuracy is also very strong, exceeding random chance twice under Random Forest, and LSTM algorithms, whereas models with raw sentiment across all algorithms lie under this threshold. This suggests that simpler weighting strategies such as equal weighting on sentiment provide the most robust and reliable next-day forecasts.

One possible explanation is that arbitrary equal weights avoid overfitting, whereas data-driven methods (e.g., IWSS_SHAP, which assigns higher weights to deeper engagement like reposts) may capture noise in the training set that do not generalize. Although IWSS_SHAP and other learned weightings sometimes emphasize high-engagement signals, they never surpass the performance of IWSS_Equal. Importantly, while Diebold–Mariano tests, based on MSE, do not always confirm statistically significant differences between IWSS features in the models, the consistent improvements in directional accuracy, which is key for financial decision-making, under IWSS_Equal and IWSS_Ratio exceed the 50% random-chance threshold under multiple algorithms. This indicates that there is a practical value of weighting sentiment, especially factoring in engagement metrics that indicate its potential to influence others when predicting next-day returns. The LSTM model saw consistently better performance across algorithms but also showed the most variation where only IWSS_Equal displayed strong model results, unlike the other algorithms where many of the IWSS variables often perform consistently well. This suggests that the equal-weighted sentiment signal aligns well with LSTM's ability to extract temporal patterns, reinforcing its value in time-series forecasting

In summary, the analysis provides clear evidence that influence-weighted sentiment features enhance financial return forecasts more reliably than raw sentiment alone with improvements in explanatory power (R^2/ROOS), reduction in prediction error (RMSE/MSE) and superior directional accuracy by as much as 9% over raw sentiment alone and beating the 50% random chance threshold. Of the weighting strategies, IWSS_Equal consistently delivers these gains across OLS, Random Forest, XGBoost and LSTM Models, making it the preferred default for both research and real-world forecasting problems. Thus, in addressing Sub-RQ2, concerning to what extent influence-weighted sentiment variables improve next-day return forecasting accuracy over raw sentiment, these results clearly show that IWSS features yield statistically significant and economically meaningful enhancements in forecast accuracy across all tested algorithms, with improvements ranging from at least 1–2% up to over 11% in explanatory power, as well as marked gains in directional accuracy.

5. Theoretical and Practical Implications

Throughout Section 4, the results demonstrated the comparative strengths of FinBERT and RoBERTa for extracting investor sentiment from X posts and the quantitative advantages of applying influence-weighted sentiment using an equal-weighting scheme to next-day return forecasts. These insights contribute both to theoretical models of market dynamics as well as potential considerations for real-world implementation.

5.1.1 Contribution to sentiment extraction from texts with LLMs

The study makes a significant contribution to the rapidly growing literature on text extraction with large language models (LLMs), highlighting the critical role of task-specific fine-tuning for both linguistic style and domain-specific content for accurate sentiment analysis. When each pre-trained model (FinBERT and RoBERTa) was fine-tuned on financial tweets, there were improvements in accuracy compared to the zero-shot results. After fine-tuning, there was a 13% increase in accuracy for FinBERT and 4% for RoBERTa, alongside improvements in F1-score across all the sentiment classes. These results illustrate that although pretraining on massive text corpora provides a strong foundation for extracting sentiment with rather high accuracy for RoBERTa (61%), LLMs still require targeted adaptation to capture domain-specific language as well as the style of writing on social media. This echoes the study by Gururangan et al. (2020), who showed that continued pretraining and fine-tuning can significantly improve model task performance.

Further, most text in literature focuses on the domain-specificity of FinBERT, arguing that FinBERT's domain-specific pretraining provides a more accurate representation of financial language that will help extract investor sentiment (Araci, 2019, Kumar & Chaturvedi, 2024). However, the findings of this study contradict this, suggesting that the linguistic style of social media is just as important, if not more so, than finance-focused pre-training when extracting sentiment from X posts. RoBERTa not only outperforms FinBERT in zero-shot sentiment extraction by almost a 30% margin. Further, FinBERT managed to improve accuracy by a lot more than RoBERTa after being fine-tuned on a text corpus containing tweets. Thus, these results broaden the literature by showing that investor-sentiment extraction from social platforms should not be confined to FinBERT but that other transformer models can also be more effective. Thus, extracting sentiment from finance-related tweets requires fine-tuning models to their target text's linguistic style, and researchers and practitioners should consider designing a transformer model that trains on both social-media, oriented pre-training with financial-domain adaptation to account for platform-specific text style to maximize performance.

5.1.2 Contribution to Return Prediction

The results of the return forecasting highlight a key contribution to research. The introduction of a new sentiment variable, adjusted to factors in potential diffusion effects. Most of the existing literature (as concluded from the literature review) considers only the raw sentiment extracted from various media a reflection of investor sentiment when using this in forecasting, mostly by collecting tweets and for each post extracting a sentiment and aggregating the daily signal to find the overall sentiment for a day. However, basing sentiment on the number of social-media posts ignores the potential impact of posts from influencers which may have a higher impact on the true sentiment of investors. Further, it also ignores more quiet signals such as likes, retweets, and shares through which many investors express agreement, which are interactions omitted by only focusing on text. The results of this study when weighing sentiment on likes, reposts, tweets and follower count show that significant improvements on next day return forecasting compared to raw sentiment. Across multiple algorithms, the IWSS feature remains robust and consistently showed higher performance metrics, with higher R^2, ROOS, and directional accuracy as well as lower loss at a statistically significant level. Thus, this fills the research gap showing that by taking into consideration more quiet signals on how sentiment can spread across media which better reflects the investor sentiment.

5.1.3 Practical Implications

The finding that influence-weighted sentiment can outperform raw sentiment can have practical implications for trading. Specifically, incorporating the influence-weighted sentiment, particularly the equal-weight variant (IWSS_Equal) can perhaps give traders a meaningful edge over using raw sentiment, assuming that traders that incorporate investor sentiment use the methods mostly discussed in literature. The findings from this study demonstrate that under the LSTM framework, IWSS_Equal achieves 57% directional accuracy, which is a 9.5% improvement over raw sentiment and surpasses the 50 percent random-chance threshold. Even in Random Forest and XGBoost models, it sustains a 50–51% accuracy band, demonstrating robustness of the influence-weighted sentiment across algorithms. This is a significant result that could be utilized by traders, as in most financial markets, forecasting over the 50% random chance can generate profits, and 60% typically demonstrates very successful results (Yıldırım, Toroslu, & Fiore, 2021). This indicates that there is potential for profit if traders were to incorporate a similar model. Further, according to Grinold & Khan (1999) small predictive advantages can also compound into substantial gains over time, so using this variable also with other algorithms can also be advantageous for traders. However, it should be noted that the results only reflect next-day forecasting as this is delimited in the scope of this study and was not tested over different time horizons. Thus, even though the results suggest profitability potential, traders should test the variable across different horizons if they want to trade over longer horizons.

6. Limitations and Future Research

Throughout this study, the findings and discussion conclude that influence-weighted sentiment features can improve the predictive performance of short-term return forecasts. However, to properly assess the validity of these findings, internal and external validity will be assessed. The internal validity shows the extent the study's design and measurements allow for accurate and unbiased conclusions. External validity looks at the degree to which results generalize beyond the specific data, time frame, and context. Acknowledging threats to both internal (i.e., classification error, omitted variables) and external validity (i.e., sampling bias, limited market regimes) is essential not only for contextualizing our conclusions but also for navigating what could be important for future research.

6.1 External Validity

Firstly, there are some limitations regarding the external validity of the project. A key delimitation made for this study is that investor sentiment can be represented by X posts, and that users who post about stocks on X are themselves active market participants. Although this was made to facilitate the scope and feasibility of the study, it may introduce some sampling bias, as X may over-represent retail traders, influencers, as well as automated accounts, while under-representing institutional investors and long-term holders who may not make posts on X. This can lead to the sentiment signal reflecting only a subset of the market participants, which may not be a good representation of the broader market sentiment.

Secondly, the influence-weighting approach relies on engagement metrics such as likes, reposts, and replies under the assumption that greater reach and engagement can help represent sentiment better. However, this may lead to popularity bias, as these metrics can be artificially inflated by, for instance, buying followers and reposts could be driven by non-financial content regarding the S&P 500. As a result, more popular users without genuine market expertise may disproportionately influence the sentiment score. This could help explain why the other IWSS features when incorporated into models underperformed, if they put more weight on engagement metrics that could be generated by bots or inflated in some way like IWSS_SHAP (more weight on reposts). Although basic filters were applied to mitigate bot activity, a comprehensive analysis of automated or non-financial posts was beyond this study's scope. Overall, based on these issues, while the IWSS feature demonstrated improved forecasting accuracy over raw sentiment within our X-based sample, these findings may not generalize to the full market or other platforms.

A further constraint on the external validity of this study is due to the time horizon and market context. The analysis is limited to the period 2022-2024, which is a rather short time horizon, and excludes crisis episodes such as the Covid-19 pandemic. This decision was made deliberately to ensure the feasibility of the study, however, since next-day returns is based only a 2-year time window, it is not certain that the performance of the models, for example of the 57% directional accuracy for the LSTM model with IWSS_Equal will hold up during periods with extreme volatility, structural breaks or exogenous shocks. Moreover, this research relies on historical data, training models on past patterns and relationships. Financial markets can be continually shaped by changing macroeconomic conditions, and technological innovations. Thus, although the IWSS variable showed predictive value in this time frame, it may lose relevance if the market changes. Overall, the performance gains observed for the influence-weighted sentiment variable may not fully translate to different time periods, or when markets are subjected to exogenous shocks.

6.2 Internal Validity

A key internal validity limitation is that the RoBERTa model that was used to extract sentiment was not fine-tuned to the study's own X-post data set. Although it was fine-tuned to a similar dataset, the results of sentiment classification clearly showed that fine-tuning to specific datasets showed significant improvement. While relying on the pre-trained models were necessary as manual labelling was infeasible, the approach likely affected the accuracy and consistency of the sentiment measurements and potentially introduced some systematic classification errors into the constructed IWSS variable. For the best set up, either the whole dataset should have been labelled manually or enough of the dataset should have been fine-tuned to improve the accuracy, which on the TimKoornstra dataset used to finetune achieved only 64% accuracy. This was slightly mitigated as an brief evaluation was done on the dataset collected by this study, where 106 random X posts were manually annotated and compared to the predictions by the final RoBERTa model, however this is a very small amount of the total X posts, so even though it as 80% accurate, it may not be the most representative sample. This has some implications as the sentiment predictions may be biased or imprecise, as there may be over or under-emphasis on positive or negative signals. Further, models for return forecasting may have picked up on spurious correlation if the IWSS variable or raw sentiment picked up any misleading signals due to any potential misclassifications.

Another internal validity concern is the potential omitted variable bias. For feasibility and focus within the thesis time frame, this study deliberately delimited its base model to only three financial market predictors (volatility (VIX), trading volume, and lagged returns) in line with research by Gu et al. (2020) with the additional raw sentiment or IWSS feature to see if the influence-weighted sentiment can improve model performances. However, the results of the next-day forecasting in this study show a persistently low explanatory power which could be due to omitting important features like macro-economic factors, especially in OLS models. Excluding well-established return drivers such as the Fama-French factors, as well as macroeconomic factors like inflation rates means that unmodeled economic effects may be spuriously absorbed by the financial predictors or by the IWSS features and can distort their estimated effect. Moreover, limiting the predictor set prevents examination of interaction effects between sentiment and broader macroeconomic conditions. For instance, investor sentiment may interact with monetary policy regimes, or volatility cycles in ways that either amplify or dampen its forecasting power. Omitting these potential interactions risks underestimating the true value of a combined sentiment-macroeconomic framework.

6.3 Future Research

For future research, one way to expand our results is to explore sarcasm detection, emoji understanding, and informal-language handling to further enhance sentiment classification accuracy. For example, Bhargava et al. (2025) demonstrate that identifying and removing sarcasm drastically improved robustness and boosted sentiment model classification by up to 21%. Incorporating these techniques into our RoBERTa-based classifier would reduce misclassification of any potentially distinct linguistic styles particularly on X, to get a more accurate raw sentiment signal for modelling. This improves the IWSS feature as it would be more reliable, which could strengthen its predictive value. Further, future work can also address the bias introduced by assuming X posts sentiment is a good reflection of investor sentiment. Instead of only drawing sentiment using one source, perhaps drawing sentiment from a multitude of sources (combining news, discussion forums, and other textual data with X posts) can increase representativeness of data to strengthen the robustness and generalizability of the results in this study. For example, Mulakala et al. (2025) developed a multi-source approach that captures both retail investor buzz from Reddit, more formal news sentiment from Yahoo Finance news feeds, and market data. The added signals yielded more timely insights around key events like earnings reports and improved predictive accuracy, underscoring that incorporating multiple sentiment sources can strengthen models.

Some long-term extensions to the projects could include other ways to test the robustness of the IWSS features performance against raw sentiment. One way to test whether using the feature IWSS_Equal produced generalizable predictive signals would be to apply this methodology over different windows, especially during high-volatility periods, or over longer periods to test if the predictive value can be generalized across different periods. Another valuable avenue of future research could be to back test trading strategies driven by the IWSS features, in particular the LSTM model with the IWSS_Equal feature which produced a 57% directional accuracy. Although it is not within the scope of this research, it can provide valuable information for practical use. By simulating a long-short equity approach (going long on predicted up-moves and short on predicted down-moves), researchers (or traders) can assess whether this predictive edge can translate into profits. This extra analysis on top of the current results could more precisely show the economic value and robustness of influence-weighted sentiment features and how using them in models can show not only predictive enhancements in theory but also potentially in practice for real-world trading.

By systematically addressing these external and internal validity challenges, future research can strengthen the reliability and applicability of IWSS-based forecasting, bridging the gap between statistical improvements and real-world trading outcomes.

7. Conclusion

This study sets out to determine whether investor sentiment extracted from social media posts can be utilized more effectively for asset price prediction by accounting for user influence. Earlier studies conclude that sentiment demonstrates positive relationships and can produce high predictive accuracy when predicting returns. However, it overlooks how sentiment diffuses through social media which is where they are extracting investor sentiment from. Aggregating sentiment without accounting for engagement (likes, replies, reposts) can misrepresent true economic impact and risks producing results that are statistically significant but economically spurious.

Thus, the following research question was posed and then answered: **How does the predictive performance of influence-weighted sentiment variables for forecasting S&P 500 index returns compare to that of raw sentiment alone?**

The main question is addressed by two Sub-RQs:

1. Sub-RQ1: How can LLMs extract investor sentiment from X posts, and how can engagement-based indicators be incorporated into a novel sentiment variable?
2. Sub-RQ2**:** To what extent do influence-weighted sentiment variables improve next-day return forecasting accuracy over unweighted sentiment?

The findings reveal how investor sentiment from X posts can be reliably extracted via LLMs by fine-tuning transformer models like FinBERT and RoBERTa, and then incorporating engagement and influence (follower counts, likes, replies and reposts) to construct a novel influence-weighted sentiment variable, denoted as IWSS. This specific variable acknowledges that posts can have different weight by distributing greater impact to sentiment expressed by highly influential or widely engaged users and filtering out noise from low-impact opinions. This approach provided the ability to accurately classify posts and weigh each by its influence, ultimately building a daily sentiment variable adjusted for influence. This variable systematically outperformed raw sentiment in all evaluated forecasting models, including OLS regression, Random Forest, XGBoost, and a Long Short-Term Memory (LSTM) network.

On the sentiment-extraction side, fine-tuned RoBERTa achieved 64% accuracy score with fine-tuning on TimKoornstra's pre-labelled dataset, and moreover an impressive 80% on the dataset collected in this study to evaluate accuracy (albeit only on 106 samples). Then, by using the structure to first define a formulation that weighs follower count (reach) and engagement metrics (amplification) to reflect diffusion, allowed for the construction of the IWSS variable. The weights of the engagement score were extracted via different approaches including equal weights, ratio-based weights and weights based on impurity, permutation and SHAP feature importance, which resulted in some IWSS features putting more emphasis on deeper engagement metrics such as reposts. Based on this process, the raw sentiment signal was transformed into a more meaningful and predictive variable.

Empirical evidence showed how the inclusion of unweighted sentiment alone often yielded lower predictive performance, with higher prediction error, negative explanatory power for next-day returns, and never exceeded 50 % directional accuracy. In contrast, the inclusion of IWSS features improved the predictive outcomes for the best-performing models, achieving higher R^2 and ROOS (higher explanatory power), lower RMSE and MSE (higher predictive accuracy), and directional accuracy above 57%, compared to the maximum 48% for raw sentiment. The best performance was achieved by LSTM, which was most pronounced using the IWSS_Equal feature in combination with the base variables (current day return, volume and volatility). Diebold-Mariano tests confirmed that models containing the IWSS features were significantly better, indicating the improvements were conclusive at a statistically significant level compared to the model with the unweighted (raw) sentiment across multiple forecasting algorithms, directly addressing Sub-RQ2.

However, this study is not without limitations. Due to sentiment being drawn exclusively from X (formerly Twitter), results may not generalize to other platforms (e.g., Reddit, StockTwits) or to different asset classes. The assumption that engagement metrics fully capture "influence" also could be an issue due to bot activity. Future research should extend IWSS to multiple platforms (e.g., Reddit, StockTwits) to create more robust sentiment signals. Additional robustness checks could test whether IWSS holds across different time horizons and market regimes and potentially apply to trading algorithms to see if the variable can help turn profits.

Methodologically, this study provides a replicable pipeline: (1) fine-tune a domain-specific LLM on labeled financial text, (2) weight raw sentiment by engagement metrics to form IWSS, and (3) slot IWSS into both linear and non-linear forecasting models with rigorous time-based train/test splits and significance testing. This can readily be adapted by other researchers or practitioners interested in transforming large-scale unstructured text into predictive financial signals.

Overall, this study provides empirical support for the use of influence-weighted sentiment in next-day return forecasting. Utilizing the developed IWSS feature framework serves as a meaningful enhancement in sentiment-derived modeling. A key takeaway from this framework is its ability to bridge behavioral finance and data science, by treating the specific sentiment and how we measure it as a combined entity, creating a measurable difference in anticipating market movements.

References

Anger, Isabel & Kittl, Christian. (2011). Measuring influence on Twitter. I-KNOW. 31. 10.1145/2024288.2024326.

Antweiler, W., & Frank, M. Z. (2004). Is all that talk just noise? the information content of internet stock message boards. The Journal of Finance, 59(3), 1259–1294. https://doi.org/10.1111/j.1540-6261.2004.00662.x

Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. arXiv. https://doi.org/10.48550/arXiv.1908.10063

Azar, P., & Lo, A. W. (2016). The wisdom of twitter crowds: Predicting stock market movements using social media. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.2756815

Baker, M., & Wurgler, J. (2007). Investor sentiment in the stock market. Journal of Economic Perspectives, 21(2), 129–152. https://doi.org/10.1257/jep.21.2.129

Banz, R. W. (1981). The relationship between return and market value of common stocks. *Journal of Financial Economics*, *9*(1), 3–18. https://doi.org/10.1016/0304-405x(81)90018-0

Barabási, A.-L. (2018). The formula: The Universal Laws of Success. Little, Brown and Company.

Barbaglia, L., Consoli, S., Manzan, S., Tiozzo Pezzoli, L., & Toffano, M. B. (2023). Modeling investor attention with social media for stock returns. Economic Inquiry, 63(1), 125–143. https://doi.org/10.1111/ecin.13264

Barberis, N. (2013). The psychology of tail events: Progress and challenges. American Economic Review, 103(3), 611–616. https://doi.org/10.1257/aer.103.3.611

Barberis, N., & Huang, M. (2008). Stocks as lotteries: The implications of probability weighting for security prices. American Economic Review, 98(5), 2066–2100. https://doi.org/10.1257/aer.98.5.2066

Bass, F. M. (1969). A new product growth for model consumer durables. *Management Science*, *15*(5), 215–227. https://doi.org/10.1287/mnsc.15.5.215

Benartzi, S., & Thaler, R. (1995). Myopic loss aversion and the equity premium puzzle. Quarterly Journal of Economics, 110(1), 73–92. https://doi.org/10.2307/2118511

Bhargava, N., Radaideh, M. I., Kwon, O. H., Verma, A., & Radaideh, M. I. (2025). Emojis and financial communication: How emoji usage influences market sentiment (arXiv:2504.05603). arXiv. https://arxiv.org/abs/2504.05603

Bollen, J., Mao, H., & Zeng, X. (2011). Twitter mood predicts the stock market. Journal of Computational Science, 2(1), 1–8. https://doi.org/10.1016/j.jocs.2010.12.007

Bondt, W. F., & Thaler, R. (1985). Does the stock market overreact? *The Journal of Finance*, *40*(3), 793. https://doi.org/10.2307/2327804

Boyd, D., & Crawford, K. (2012). Critical questions for big data: Provocations for a cultural, technological, and scholarly phenomenon. Information, Communication & Society, 15(5), 662–679. https://doi.org/10.1080/1369118X.2012.678878

Brealey, R. A., Myers, S. C., & Allen, F. E. A. (2022). Principles of corporate finance (14th ed.). McGraw Hill.

Brown, G., & Cliff, M. T. (2001). Investor sentiment and asset valuation. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.292139

Bryman, A. (1984). The debate about quantitative and qualitative research: A question of method or epistemology? The British Journal of Sociology, 35(1), 75–92.

Buda, M., Maki, A., & Mazurowski, M. A. (2018). A systematic study of the class imbalance problem in Convolutional Neural Networks. *Neural Networks*, *106*, 249–259. https://doi.org/10.1016/j.neunet.2018.07.011

CardiffNLP. (2020). twitter-roberta-base-sentiment [Computer software]. Hugging Face. https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment

Cha, M., Haddadi, H., Benevenuto, F., & Gummadi, K. (2010). Measuring user influence in Twitter: The million follower fallacy. Proceedings of the International AAAI Conference on Web and Social Media, 4(1), 10–17. https://doi.org/10.1609/icwsm.v4i1.14033

Chen, X., Härdle, W., & Jeong, S. (2021). Predictive Directional Accuracy and Portfolio Performance Evaluation. Journal of Financial Econometrics, 19(2), 273–302.

Chen, X., Li, Y., Wang, Z., & Zhang, T. (2023). Alternative data in asset pricing: Evidence from machine learning and text analysis. arXiv preprint arXiv:2305.14368. https://arxiv.org/pdf/2305.14368

Cochrane, J. H. (2005). Asset pricing (Revised ed.). Princeton University Press.

Das, S., Behera, R. K., kumar, M., & Rath, S. K. (2018). Real-time sentiment analysis of Twitter streaming data for stock prediction. Procedia Computer Science, 132, 956–964. https://doi.org/10.1016/j.procs.2018.05.111

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (pp. 4171–4186). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423

Dhaoui, C., Webster, C., & Tan, L. (2017). Social media sentiment analysis: Lexicon versus machine learning. Journal of Consumer Marketing, 34(3), 1-12. https://doi.org/10.1108/JCM-03-2017-2141

Diallo, B.; Bagudu, A.; Zhang, Q. A Machine Learning Approach to the Fama-French Three- and Five-Factor Models 2019. Available online: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3440840

Diebold, F., & Mariano, R. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, *20*(1). https://doi.org/10.3386/t0169

Elmitwalli, S., & Mehegan, J. (2024). Sentiment analysis of COP9-related tweets: A comparative study of pre-trained models and traditional techniques. Frontiers in Big Data, 7. https://doi.org/10.3389/fdata.2024.1357926

Eyal, T., Liberman, N., Trope, Y., & Walther, E. (2004). The Pros and cons of temporally near and distant action. Journal of Personality and Social Psychology, 86(6), 781–795. https://doi.org/10.1037/0022-3514.86.6.781

Fama, E. F., & French, K. R. (2004). The capital asset pricing model: Theory and evidence. Journal of Economic Perspectives, 18(3), 25–46. https://doi.org/10.1257/0895330042162430

Fan, J., & Yao, Q. (2017). The elements of financial econometrics. Cambridge University Press.

Fatenaite, G. (2025). Is Web Scraping Legal? Oxylabs. https://oxylabs.io/blog/is-web-scraping-legal

Ferrara, E., Varol, O., Davis, C., Menczer, F., & Flammini, A. (2016). The rise of social bots. Communications of the ACM, 59(7), 96–104. https://doi.org/10.1145/2818717

Frijda, N. H., & Mesquita, B. (1994). The social roles and functions of emotions. Emotion and Culture: Empirical Studies of Mutual Influence., 51–87. https://doi.org/10.1037/10152-002

G2. (2025). 85+ Big Data Statistics To Map Growth in 2025. https://www.g2.com/articles/big-data-statistics

Ganti, A. (n.d.). Adjusted closing price: How it works, types, and examples. Investopedia. https://www.investopedia.com/terms/a/adjusted_closing_price.asp

GeeksforGeeks. (2025, March 3). Mean squared error: Definition and examples. https://www.geeksforgeeks.org/mean-squared-error/

Giachanou, A., & Crestani, F. (2016). Like it or not: A survey of Twitter sentiment analysis methods. ACM Computing Surveys (CSUR), 49(2), 28.

Glantz, S. A., Slinker, B. K., & Neilands, T. B. (2016). Primer of Applied Regression & Analysis of variance. McGraw-Hill Education.

Gleißner, W., & Ernst, D. (2019). Company valuation as result of risk analysis: Replication approach as an alternative to the CAPM. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3458862

Grinold, R. C., & Kahn, R. N. (1999). Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk. McGraw-Hill.

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.3220051

Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don't stop pretraining: Adapt language models to domains and tasks. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. https://doi.org/10.18653/v1/2020.acl-main.740

Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction (2nd ed.). Springer.

Henning, L., Tontodimamma, T., & Schütze, H. (2023). A survey of methods for addressing class imbalance in deep-learning based natural language processing. In Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL) (pp. 529–549). Association for Computational Linguistics. https://aclanthology.org/2023.eacl-main.38

Hong, H., & Stein, J. (1999). A unified theory of underreaction, momentum trading and overreaction in asset markets. The Journal of Finance, 54(6), 2143–2184.

Huang, Allen H., Hui Wang, & Yi Yang. FinBERT: A Large Language Model for Financial Sentiment Analysis from Financial Text. Contemporary Accounting Research (2022).

IBM. (2024a). What is random forest? IBM. Retrieved March 27, 2025, from https://www.ibm.com/think/topics/random-forest

IBM. (2024b). What is a transformer model? IBM. Retrieved March 27, 2025, from https://www.ibm.com/think/topics/transformer-model

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning: with Applications in R* (2nd ed.). New York, NY: Springer.

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for Stock Market Efficiency. The Journal of Finance, 48(1), 65. https://doi.org/10.2307/2328882

Jiang, T., & Zeng, A. (2023). Financial sentiment analysis using FinBERT with application in predicting stock movement. arXiv preprint arXiv:2306.02136

Kabir, M. R., Bhadra, D., Ridoy, M., & Milanova, M. (2025). LSTM–transformer-based robust hybrid deep learning model for Financial Time Series forecasting. Sci, 7(1). https://doi.org/10.3390/sci7010007

Kaeley, H., Qiao, Y., & Bagherzadeh, N. (2023). Support for Stock Trend Prediction Using Transformers and Sentiment Analysis. arXiv preprint arXiv:2305.14368.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–291. https://doi.org/10.2307/1914185

Kamil, I., & Shah, N. (2024). Comparison of lexicon-based method, machine learning, and ChatGPT on sentiment analysis of big cap and small cap companies in United State indexes. Journal of Theoretical and Applied Information Technology, 102(16), 6280–6287.

Karanikola, A., Davrazos, G., Liapis, C. M., & Kotsiantis, S. (2023). Financial sentiment analysis: Classic Methods vs. Deep Learning Models. Intelligent Decision Technologies, 17(4), 893–915. https://doi.org/10.3233/idt-230478

Kavlakoglu, E., & Russi, E. (2024). What is XGBoost? IBM. Retrieved March 27, 2025, from https://www.ibm.com/think/topics/xgboost

Kearney, C., & Liu, S. (2014). Textual sentiment in finance: A survey of methods and models. International Review of Financial Analysis, 33, 171–185. https://doi.org/10.1016/j.irfa.2014.02.006

Keskar, N. S., Mudigere, D., Nocedal, J., Smelyanskiy, M., & Tang, P. T. P. (2017). *On large-batch training for deep learning: Generalization gap and sharp minima*. In Proceedings of the 5th International Conference on Learning Representations (ICLR). https://arxiv.org/abs/1609.04836

Kirtac, K., & Germano, G. (2024). Sentiment trading with large language models. Finance Research Letters, 62, 105227. https://doi.org/10.1016/j.frl.2024.105227

Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. New York, NY: Springer.

Kuhn, T. S. (1970). *The structure of Scientific Revolutions*. Chicago University Press.

Kumar, S., & Chaturvedi, R. (2024). Evaluating the efficacy of distilled transformer models for sentiment analysis in financial texts: A comparative study. International Journal of Current Science, 14(2), 926–935. https://www.rjpn.org/ijcspub/papers/IJCSP24B1335

Lachana, I., & Schröder, D. (2024). Investor sentiment and stock returns: Wisdom of crowds or power of words? Evidence from Seeking Alpha and Wall Street Journal (Last revised December 17, 2024). SSRN. https://doi.org/10.2139/ssrn.3840866

Lachanski, M., & Pav, S. (2017). Shy of the Character Limit: "Twitter Mood Predicts the Stock Market" Revisited. Econ Journal Watch, 14(3), 302–345. https://econjwatch.org/articles/shy-of-the-character-limit-twitter-mood-predicts-the-stock-market-revisited

Lahboub, K., & Benali, M. (2024). Assessing the Predictive Power of Transformers, ARIMA, and LSTM in Forecasting Stock Prices of Moroccan Credit Companies. Journal of Risk and Financial Management, 17(7), 293. https://doi.org/10.3390/jrfm17070293

Lo, A. W. (2004). The adaptive markets hypothesis: Market efficiency from an evolutionary perspective. Journal of Portfolio Management, 30(5), 15–29. https://doi.org/10.3905/jpm.2004.442611

Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. In Advances in Neural Information Processing Systems (Vol. 30).

Machova, K., Mach, M., & Vasilko, M. (2022). Comparison of machine learning and sentiment analysis in detection of suspicious online reviewers on different type of data. Sensors, 22(1), 155. https://doi.org/10.3390/s22010155

Machová, K., Mach, M., & Vasilko, M. (2022). Comparison of machine learning and sentiment analysis in detection of suspicious online reviewers on different type of data. Sensors, 22(1), 155. https://doi.org/10.3390/s22010155

Milvus. (n.d.). What is root mean square error (RMSE) in time series forecasting. Milvus. https://milvus.io/docs/what-is-root-mean-square-error-rmse-in-time-series-forecasting

Molnar, C. (2022). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.). Leanpub.

Mulakala, S. V., Vangapally, U., Larkey, B., Henrichs, A., & Mahajan, V. (2025). Investigator effect: How investor recommendations influence stock returns (arXiv:2502.05403). arXiv. https://arxiv.org/abs/2502.05403

Nagabhushan, H. (2021). XGBoost versus Random Forest: Which one to choose? Medium. https://medium.com/geekculture/xgboost-versus-random-forest-898e42870f30

Najafabadi, M., Villanustre, F., Khoshgoftaar, T., Seliya, N., Wald, R., & Muharemagic, E. (2015). Deep learning applications and challenges in big data analytics. Journal of Big Data, 2, 1-21. https://doi.org/10.1186/s40537-014-0007-7

Nandwani, P., & Verma, R. (2021). A review on sentiment analysis and emotion detection from text. Social Network Analysis and Mining, 11(1), 81. https://doi.org/10.1007/s13278-021-00776-6

Ndikum, Philip. (2020). Machine Learning Algorithms for Financial Asset Price Forecasting. 10.48550/arXiv.2004.01504.

Nurollahian, S., Talegaonkar, I., Bell, A. Z., & Kogan, M. (2023). Factors affecting public's engagement with tweets by authoritative sources during crisis. Proceedings of the International ISCRAM Conference. https://doi.org/10.59297/avhj5856

Omar, A. B., Huang, S., Salameh, A. A., Khurram, H., & Fareed, M. (2022). Stock market forecasting using the random forest and deep neural network models before and during the COVID-19 period. Frontiers in Environmental Science, 10.

Owen, L., & Oktariani, F. (2020). SENN: Stock ensemble-based neural network for stock market prediction using historical stock data and sentiment analysis. 2020 International Conference on Data Science and Its Applications (ICoDSA), 1–7. IEEE.

Peress, J. (2012). The media and the diffusion of information in financial markets: Evidence from newspaper strikes. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.2026460

Perold, André, F. (2004). The Capital Asset Pricing Model. Journal of Economic Perspectives, 18(3), 3–24.

Probst, Philipp & Boulesteix, Anne-Laure & Wright, Marvin. (2018). Hyperparameters and Tuning Strategies for Random Forest. 10.48550/arXiv.1804.03515.

Pykes, K. (2025, February 11). Fuzzy String Matching in Python Tutorial. (DataCamp, Producer) Retrieved from DataCamp: https://www.datacamp.com/tutorial/fuzzy-string-python

Ranco, G., Aleksovski, D., Caldarelli, G., Grčar, M., & Mozetič, I. (2015). The effects of twitter sentiment on stock price returns. PLOS ONE, 10(9). https://doi.org/10.1371/journal.pone.0138441

Scikit-learn. (2023). Permutation Importance vs Random Forest Feature Importance (MDI). https://scikit-learn.org/stable/auto_examples/inspection/plot_permutation_importance.html

Sharma, A. (n.d.). *What are the limitations of using CAPM to calculate the cost of equity?* LinkedIn. https://www.linkedin.com/advice/1/what-limitations-using-capm-calculate-cost-equity

Shih, K.-H., Wang, Y.-H., Kao, I.-C., & Lai, F.-M. (2024). Forecasting ETF Performance: A Comparative Study of Deep Learning Models and the Fama-French Three-Factor Model. Mathematics, 12(19), 3158. https://doi.org/10.3390/math12193158

Shiller, R. J. (2003). From efficient markets theory to behavioral finance. Journal of Economic Perspectives, 17(1), 83–104. https://doi.org/10.1257/089533003321164967

Smailović, J., Grčar, M., Lavrač, N., & Žnidaršič, M. (2013). Predictive sentiment analysis of tweets: A stock market application. Lecture Notes in Computer Science, 77–88. https://doi.org/10.1007/978-3-642-39146-0_8

Su, Z. (2024). Applications of Bert in sentimental analysis. Applied and Computational Engineering, 92(1), 147–152. https://doi.org/10.54254/2755-2721/92/20241711

Suárez-Cetrulo, A. L., Quintana, D., & Cervantes, A. (2023). Machine learning for financial prediction under regime change using Technical Analysis: A Systematic Review. International Journal of Interactive Multimedia and Artificial Intelligence, 9(1). https://doi.org/10.9781/ijimai.2023.06.003

Sul, H. K., Dennis, A. R., & Yuan, L. (Ivy). (2016). Trading on Twitter: Using social media sentiment to predict stock returns. Decision Sciences, 48(3), 454–488. https://doi.org/10.1111/deci.12229

Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. The Journal of Finance, 62(3), 1139–1168. https://doi.org/10.2139/ssrn.685145

Thompson, P., & Loughran, M. (2014). Benchmarking sentiment analysis approaches for financial text. Journal of Behavioral Finance, 15(1), 76–104. https://doi.org/10.1080/15427560.2014.882150

Trope, Y., & Liberman, N. (2010). Construal-level theory of psychological distance. Psychological Review, 117(2), 440–463. https://doi.org/10.1037/a0018963

Varol, O., Ferrara, E., Davis, C. A., Menczer, F., & Flammini, A. (2017). Online human-bot interactions: Detection, estimation, and characterization. In Proceedings of the International AAAI Conference on Web and Social Media, 11(1). https://ojs.aaai.org/index.php/ICWSM/article/view/14809

Wang, J. (2022). The comparison of stock return prediction for random forest, ordinary least square, and XGBoost. BCP Business & Management FIBA 2022, 26, 686.

Wang, Q., & Yuan, Y. (2023). Stock price forecast: Comparison of LSTM, HMM, and Transformer. In J. Yen et al. (Eds.), Proceedings of the 2nd International Academic Conference on Blockchain, Information Technology and Smart Finance (ICBIS 2023) (pp. 126–136). Atlantis Press. https://doi.org/10.2991/978-94-6463-198-2_15

Wang, Qianzhun & Yuan, Yingqing. (2023). Stock Price Forecast: Comparison of LSTM, HMM, and Transformer. 10.2991/978-94-6463-198-2_15

Wang, T., Yuan, C., & Wang, C. (2020). Does applying deep learning in financial sentiment analysis lead to better classification performance? CUNY Academic Works, 40(2). https://academicworks.cuny.edu/qc_pubs/40

Wang, Yi. (2024). Application of ARIMA-GARCH-S model to short-term stock forecasting. Finance & Economics. 1. 10.61173/y2k63978.

Wigglesworth, R. (2019). How AI is reshaping the world of investing. Financial Times.

Wilson, T., Wiebe, J., & Hoffmann, P. (2005). Recognizing contextual polarity in phrase-level sentiment analysis. Proceedings of the Conference on Human Language Technology and Empirical Methods in Natural Language Processing - HLT '05, 347–354. https://doi.org/10.3115/1220575.1220619

Wooldridge, J. M. (2016). Introductory econometrics: A modern approach (6th ed.). Cengage Learning.

Yi Yang. (2020). finbert-tone [Computer software]. Hugging Face. https://huggingface.co/yiyanghkust/finbert-tone

Yin, Q. (2024). Three-class text sentiment analysis based on LSTM. arXiv. https://doi.org/10.48550/arXiv.2412.17347

Yıldırım, D. C., Toroslu, I. H., & Fiore, U. (2021). Forecasting directional movement of Forex data using LSTM with technical and macroeconomic indicators. Financial Innovation, 7, Article 1. https://doi.org/10.1186/s40854-020-00220-2

Zhu, X., Li, S., Srinivasan, K., & Lash, M. (2023, August 28). The effect of social media market and investor online word of mouth [Working paper]. SSRN.

Appendices

Appendix A: Hyperparameters

Appendix A.1: Random Forest Hyperparameter Tuning Grid

Hyperparameter	Description	Range Tested
n_estimators	Number of trees in the forest	50, 100
max_depth	Maximum depth of each decision tree	3, 5, 7
min_samples_split	Minimum number of samples required to split an internal node	5, 10
min_samples_leaf	Minimum number of samples required at a leaf node	3, 5, 7
max_features	Number of features to consider when looking for the best split	‘Sqrt’, 0.5

Appendix A.2: XGBoost best hyperparameters

```
{'Base+IWSS_Ratio': {'subsample': 0.6,
  'n_estimators': 100,
  'max_depth': 5,
  'learning_rate': 0.01,
  'colsample_bytree': 0.6},
 'Base+IWSS_Impurity': {'subsample': 0.6,
  'n_estimators': 100,
  'max_depth': 5,
  'learning_rate': 0.01,
  'colsample_bytree': 0.6},
 'Base+IWSS_Equal': {'subsample': 0.6,
  'n_estimators': 100,
  'max_depth': 5,
  'learning_rate': 0.01,
  'colsample_bytree': 0.6},
 'Base+IWSS_perm': {'subsample': 0.6,
  'n_estimators': 100,
  'max_depth': 5,
  'learning_rate': 0.01,
  'colsample_bytree': 0.6},
 'Base+IWSS_SHAP': {'subsample': 0.6,
  'n_estimators': 100,
  'max_depth': 5,
  'learning_rate': 0.01,
  'colsample_bytree': 0.6},
```

Appendix A.3: LSTM best hyperparameters

	Model	**dropout**	**l2**
4	Base+IWSS_Equal	0.1	0.00010
0	Base	0.3	0.00001
5	Base+IWSS_Impurity	0.2	0.00010
1	Base+Sentiment_score	0.2	0.00001

8	Base+IWSS_perm	0.1	0.00100
7	Base+IWSS_SHAP	0.1	0.00100
6	Base+IWSS_Ratio	0.1	0.00010

Appendix B: Exploratory analysis for sentiment classification

Appendix B.1 Descriptive Statistics

	Reply_Count	**Repost_Count**	**Like_Count**	**Follower_Count**	**Prediction**	**Confidence**
count	32230	32230	32230	32230	32230	32230
unique	NaN	NaN	NaN	NaN	3	NaN
top	NaN	NaN	NaN	NaN	neutral	NaN
freq	NaN	NaN	NaN	NaN	11182	NaN
mean	0.55861	0.5695	2.888303	4909.248402	NaN	0.883748
min	0	0	0	0	NaN	0.341799
25%	0	0	0	250	NaN	0.810676
50%	0	0	1	1081	NaN	0.963926

75%	1	1	2	3371	NaN	0.992687
max	496	179	928	510000	NaN	0.999387
std	3.328512	3.00577	13.955195	14672.39116	NaN	0.151448

	Follower Count		**Like Count**		**Reply Count**		**Repost Count**	
Transformation	Skew	Kurtosis	Skew	Kurtosis	Skew	Kurtosis	Skew	Kurtosis
Raw	11.06	228.06	29.95	1334.49	106.3	15320.38	32.58	1475.9
Log	0	0.04	1.4	2.21	2.07	6.73	2.41	8.94
Square root	2.97	14.18	3.87	38.09	3.42	53.06	3.66	34.66
Box-Cox	**0**	**-0.02**	**0.21**	**-1.26**	**1.11**	**-0.74**	**0.83**	**-1.23**
Yeo-Johnson	0	-0.04	0.34	-1.41	0.82	-1.29	1.01	-0.96

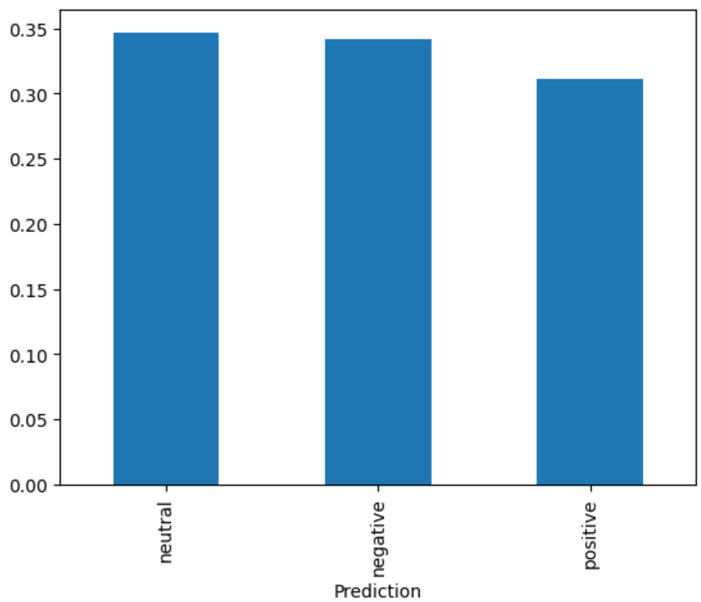
Appendix B.2 Zero-engagement posts:

Number of zeros in each column:	
Follower_Count	37
Like_Count	15325
Reply_Count	22133
Repost_Count	23317
View_Count	16054
Number of posts with zero engagement across all metrics: 9	
Number of posts with zero engagement across engagement metrics: 4946	

Appendix B.3 Multicollinearity

feature	VIF
const	1.0539
Like_Count	3.2175
Reply_Count	1.1651
Repost_Count	2.6531

Appendix B.4 Sentiment Variable Distribution and Correlation



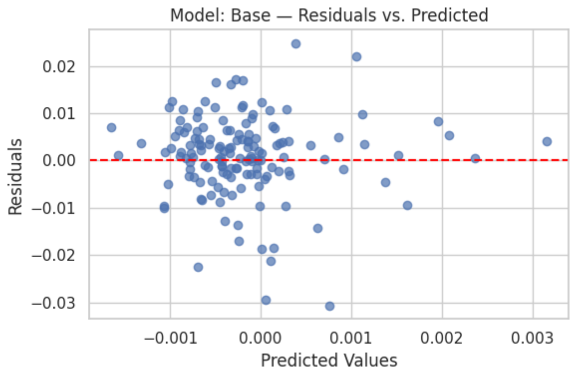
Feature	Correlation
Sentiment	1
Like_Count	0.021823
Reply_Count	0.014009
Repost_Count	0.003603
Follower_Count	-0.0534

Appendix C: Return forecasting and evaluation

Appendix C.1: Descriptive statistics and diagnostic plots

Base model

1) Linearity (Residuals vs. Predicted)



2) Multicollinearity (VIF)

return_t = 1.3228

Volume = 1.0223

VIX = 1.1431

3) Homoscedasticity (Breusch-Pagan Test)

Lagrange Multiplier stat = 8.686569298

p-value = 0.033761856

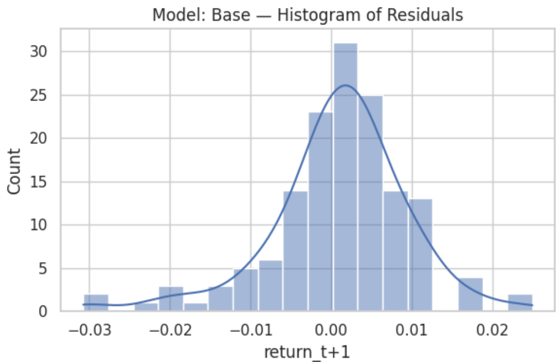
f-value = 2.993634104

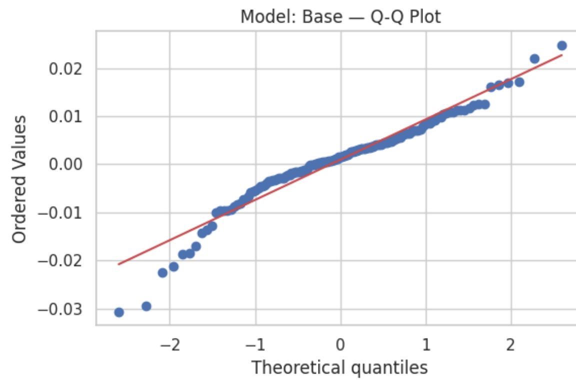
f p-value = 0.032949495

4) Autocorrelation (Durbin-Watson)

Durbin-Watson Statistic: 1.918 (ideal ~2.0)

5) Normality of Residuals





Shapiro-Wilk Test p-value: 0.0 (p > 0.05 suggests normality)

Conclusion Summary — Model: Base

Linearity: Check residuals plot visually. Look for random scatter.

Multicollinearity: ✅ No multicollinearity concern (all VIF ≤ 3)

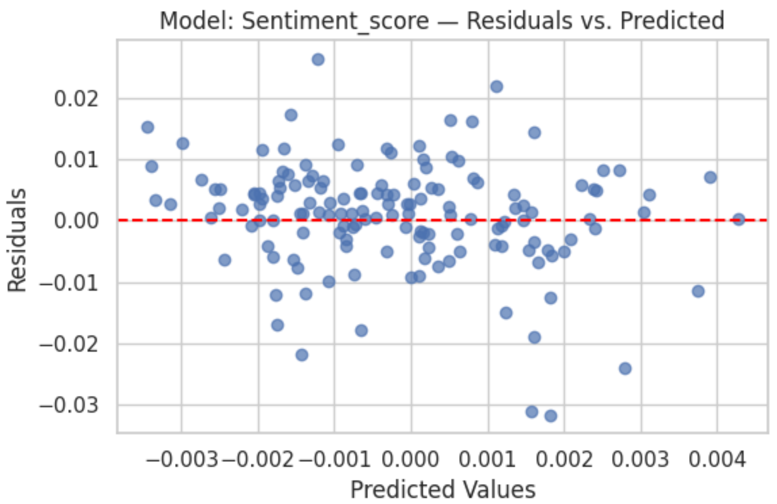
Homoscedasticity: ⚠️ Problem: p-value = 0.0338 (potential heteroscedasticity)

Autocorrelation: ✅ DW = 1.918 (no autocorrelation)

Normality: ⚠️ Problem: p-value = 0.0000 (non-normal residuals)

Sentiment_score model

1) Linearity (Residuals vs. Predicted)



2) Multicollinearity (VIF)

return_t = 1.223353

Volume = 1.024658

VIX = 1.214297

Sentiment_score = 1.225557

3) Homoscedasticity (Breusch-Pagan Test)

Lagrange Multiplier stat = 9.771840532

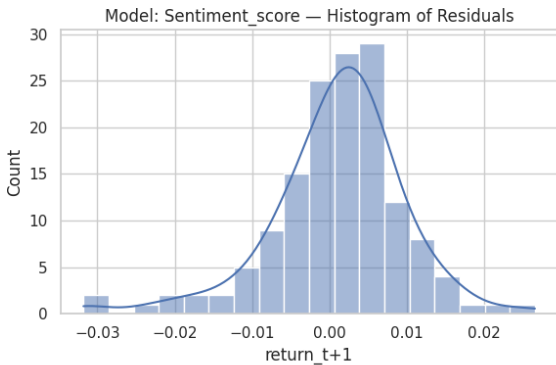
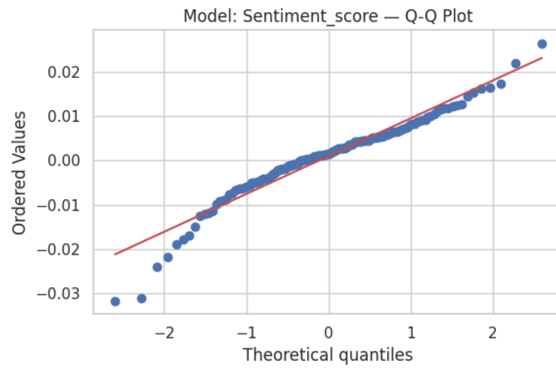
p-value = 0.044451474

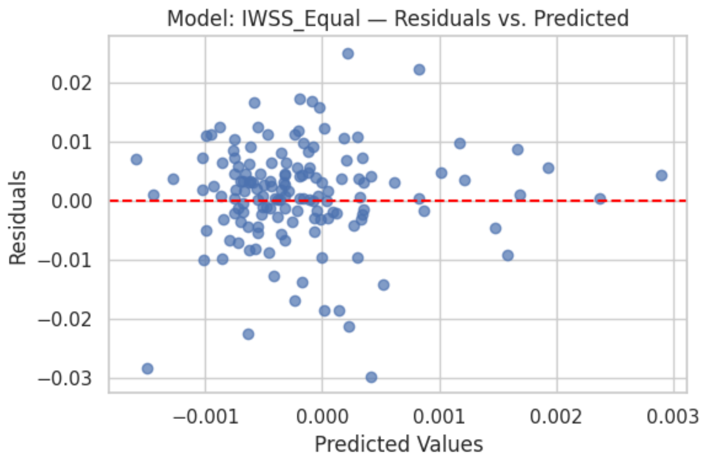
f-value= 2.527909288

f p-value = 0.043282754

4) Autocorrelation (Durbin-Watson)

Durbin-Watson Statistic: 1.935 (ideal ~2.0)

5) Normality of Residuals 
Shapiro-Wilk Test p-value: 0.0 ($p > 0.05$ suggests normality)
Conclusion Summary — Model: Sentiment_score Linearity: Check residuals plot visually. Look for random scatter. Multicollinearity: ✅ No multicollinearity concern (all VIF ≤ 3) Homoscedasticity: ⚠️ Problem: p-value = 0.0445 (potential heteroscedasticity) Autocorrelation: ✅ DW = 1.935 (no autocorrelation) Normality: ⚠️ Problem: p-value = 0.0000 (non-normal residuals)
IWSS_Equal model
1) Linearity (Residuals vs. Predicted)



2) Multicollinearity (VIF)

return_t = 1.134328

Volume = 1.037923

VIX = 1.159215

IWSS_Equal = 1.033201

3) Homoscedasticity (Breusch-Pagan Test)

Lagrange Multiplier stat = 16.82135569

p-value = 0.002093678

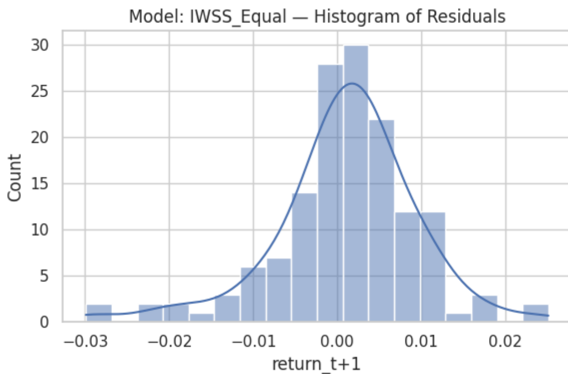
f-value = 4.587220356

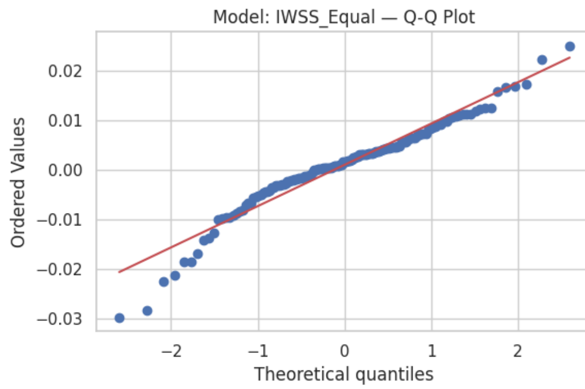
f p-value = 0.001633071

4) Autocorrelation (Durbin-Watson)

Durbin-Watson Statistic: 1.92 (ideal ~2.0)

5) Normality of Residuals





Shapiro-Wilk Test p-value: 0.0001 ($p > 0.05$ suggests normality)

Conclusion Summary — Model: IWSS_Equal

Linearity: Check residuals plot visually. Look for random scatter.

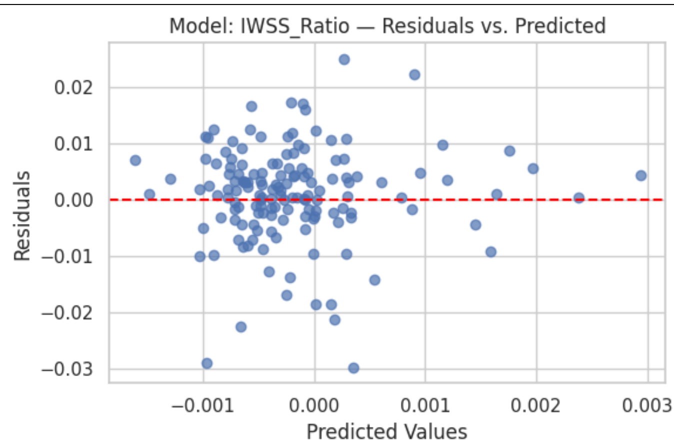
Multicollinearity: ✅ No multicollinearity concern (all VIF ≤ 3)

Autocorrelation: ✅ DW = 1.920 (no autocorrelation)

Normality: ⚠️ Problem: p-value = 0.0001 (non-normal residuals)

IWSS_Ratio model

1) Linearity (Residuals vs. Predicted)



2) Multicollinearity (VIF)

return_t = 1.133488

Volume = 1.043507

VIX = 1.155141

IWSS_Ratio = 1.03532

3) Homoscedasticity (Breusch-Pagan Test)

Lagrange Multiplier stat = 18.13933343

p-value = 0.001159068

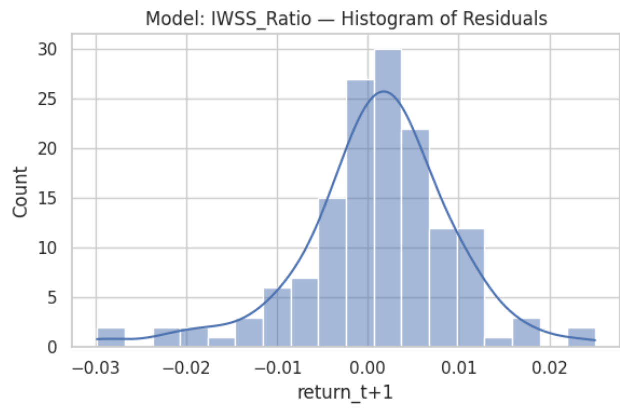
f-value = 4.997229596

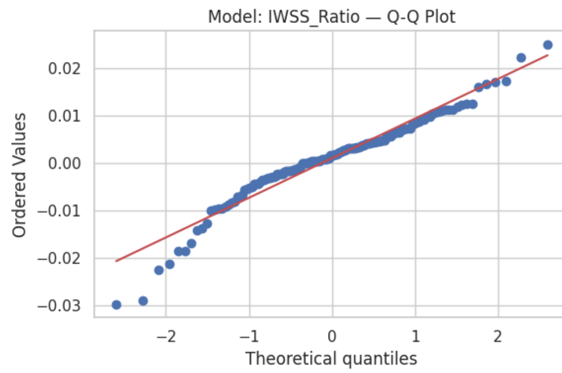
f p-value = 0.000848238

4) Autocorrelation (Durbin-Watson)

Durbin-Watson Statistic: 1.919 (ideal ~2.0)

5) Normality of Residuals





Shapiro-Wilk Test p-value: 0.0 (p > 0.05 suggests normality)

Conclusion Summary — Model: IWSS_Ratio

Linearity: Check residuals plot visually. Look for random scatter.

Multicollinearity: ✅ No multicollinearity concern (all VIF ≤ 3)

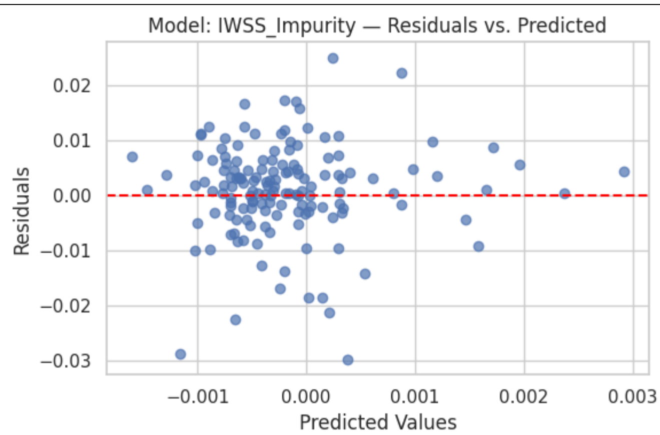
Homoscedasticity: ⚠️ Problem: p-value = 0.0012 (potential heteroscedasticity)

Autocorrelation: ✅ DW = 1.919 (no autocorrelation)

Normality: ⚠️ Problem: p-value = 0.0000 (non-normal residuals)

IWSS_Impurity model

1) Linearity (Residuals vs. Predicted)



2) Multicollinearity (VIF)

return_t = 1.134035

Volume = 1.040395

VIX = 1.157513

IWSS_Impurity = 1.034241

3) Homoscedasticity (Breusch-Pagan Test)

Lagrange Multiplier stat = 17.51574161

p-value = 0.001534123

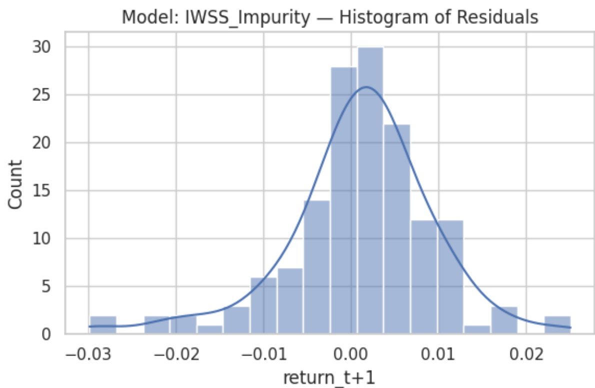
f-value = 4.802196304

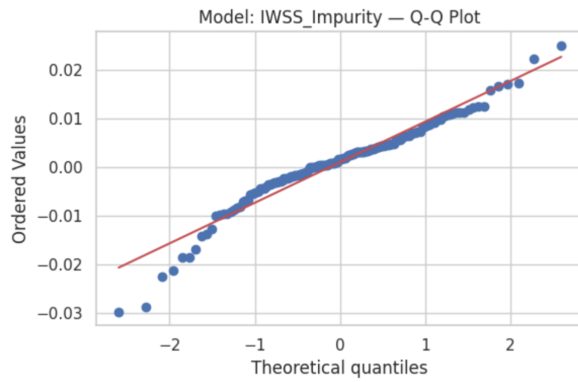
f p-value = 0.00115815

4) Autocorrelation (Durbin-Watson)

Durbin-Watson Statistic: 1.919 (ideal ~2.0)

5) Normality of Residuals





Shapiro-Wilk Test p-value: 0.0 (p > 0.05 suggests normality)

Conclusion Summary — Model: IWSS_Impurity

Linearity: Check residuals plot visually. Look for random scatter.

Multicollinearity: ✅ No multicollinearity concern (all VIF ≤ 3)

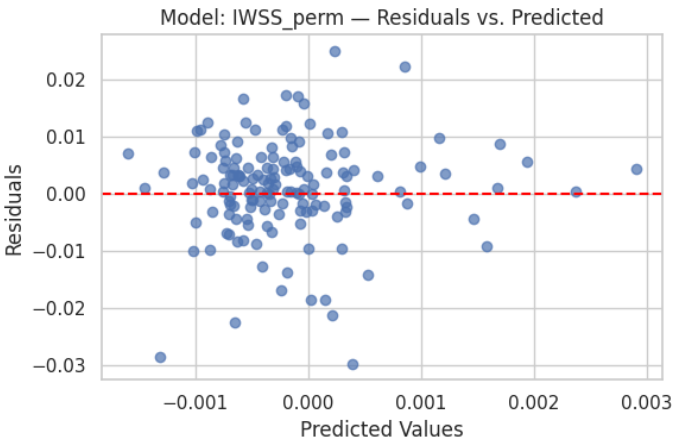
Homoscedasticity: ⚠️ Problem: p-value = 0.0015 (potential heteroscedasticity)

Autocorrelation: ✅ DW = 1.919 (no autocorrelation)

Normality: ⚠️ Problem: p-value = 0.0000 (non-normal residuals)

IWSS_perm model

1) Linearity (Residuals vs. Predicted)



2) Multicollinearity (VIF)

return_t = 1.134153

Volume = 1.039302

VIX = 1.158242

IWSS_perm = 1.033766

3) Homoscedasticity (Breusch-Pagan Test)

Lagrange Multiplier stat = 17.20197509

p-value = 0.001765853

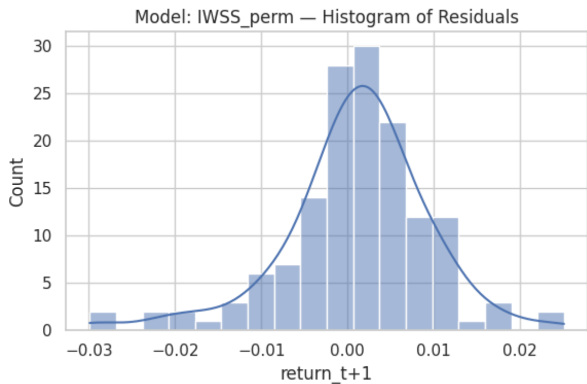
f-value = 4.704772019

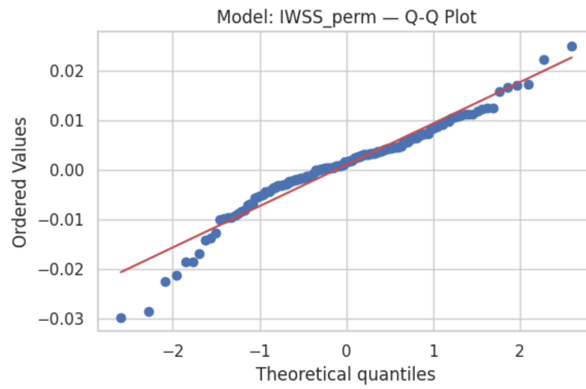
f p-value = 0.001353256

4) Autocorrelation (Durbin-Watson)

Durbin-Watson Statistic: 1.919 (ideal ~2.0)

5) Normality of Residuals





Shapiro-Wilk Test p-value: 0.0 (p > 0.05 suggests normality)

Conclusion Summary — Model: IWSS_perm

Linearity: Check residuals plot visually. Look for random scatter.

Multicollinearity: ✅ No multicollinearity concern (all VIF ≤ 3)

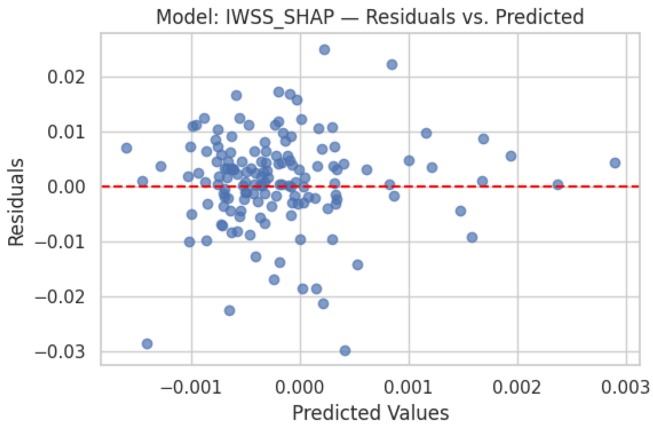
Homoscedasticity: ⚠️ Problem: p-value = 0.0018 (potential heteroscedasticity)

Autocorrelation: ✅ DW = 1.919 (no autocorrelation)

Normality: ⚠️ Problem: p-value = 0.0000 (non-normal residuals)

IWSS_SHAP model

1) Linearity (Residuals vs. Predicted)



2) Multicollinearity (VIF)

return_t = 1.133965

Volume = 1.039723

VIX = 1.157743

IWSS_SHAP = 1.033764

3) Homoscedasticity (Breusch-Pagan Test)

Lagrange Multiplier stat = 17.18061228

p-value = 0.00178283

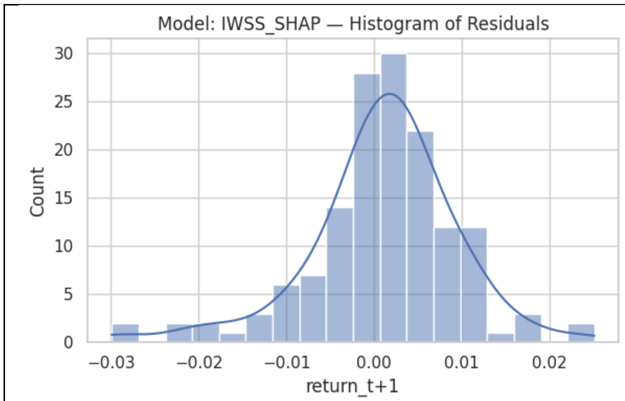
f-value = 4.698156006

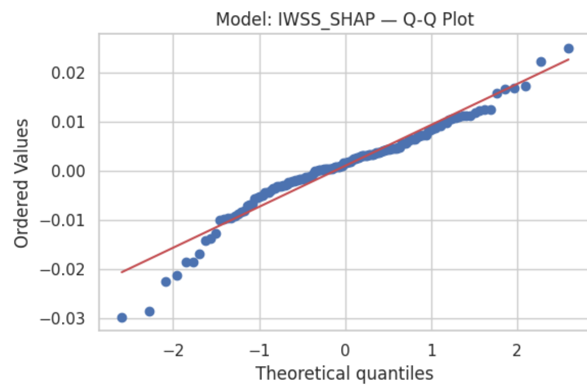
f p-value = 0.001367643

4) Autocorrelation (Durbin-Watson)

Durbin-Watson Statistic: 1.92 (ideal ~2.0)

5) Normality of Residuals





Shapiro-Wilk Test p-value: 0.0 (p > 0.05 suggests normality)

Conclusion Summary — Model: IWSS_SHAP

Linearity: Check residuals plot visually. Look for random scatter.

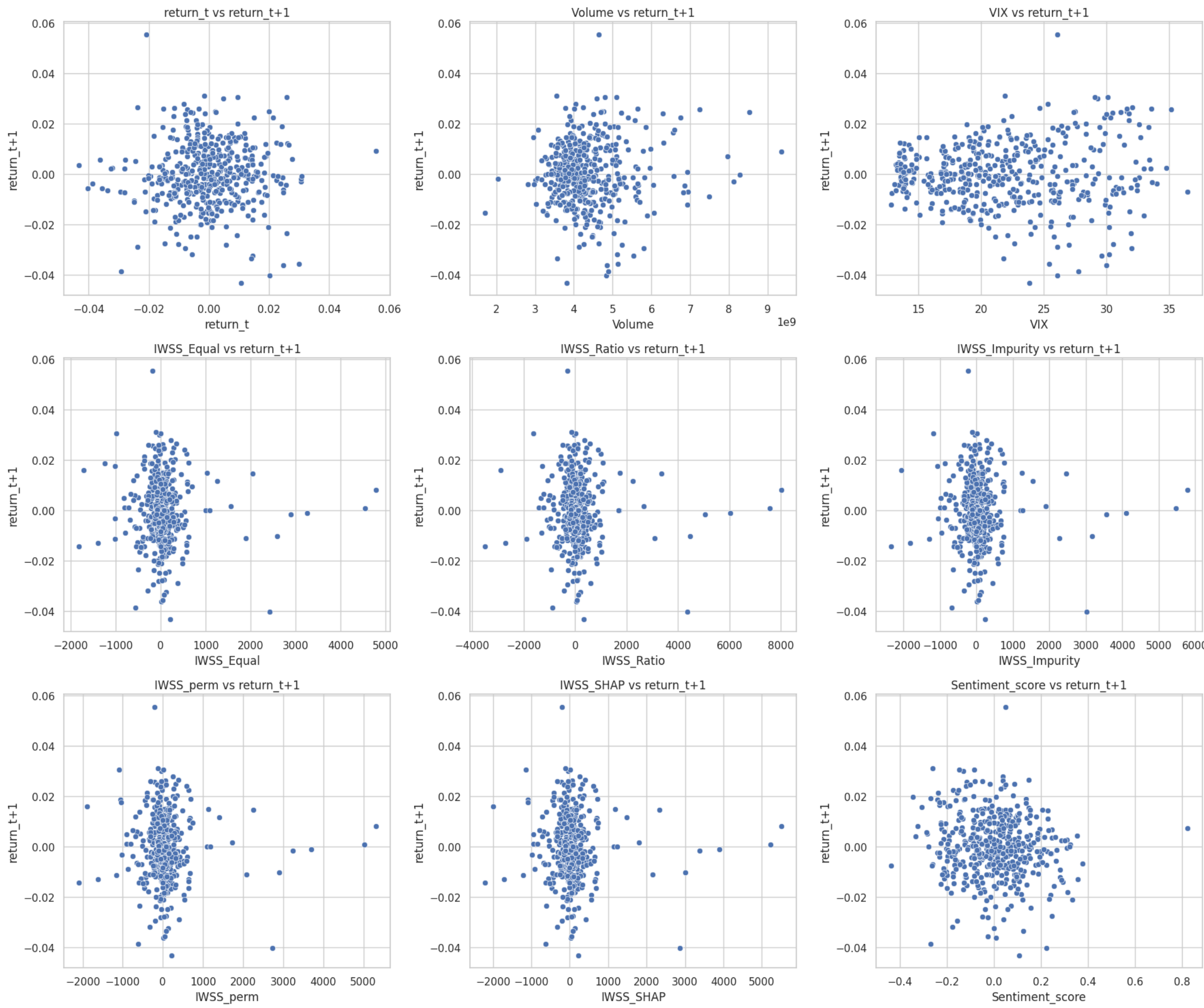
Multicollinearity: ✅ No multicollinearity concern (all VIF ≤ 3)

Homoscedasticity: ⚠️ Problem: p-value = 0.0018 (potential heteroscedasticity)

Autocorrelation: ✅ DW = 1.920 (no autocorrelation)

Normality: ⚠️ Problem: p-value = 0.0000 (non-normal residuals)

Appendix C.2: Relationship between predictors and next-day returns

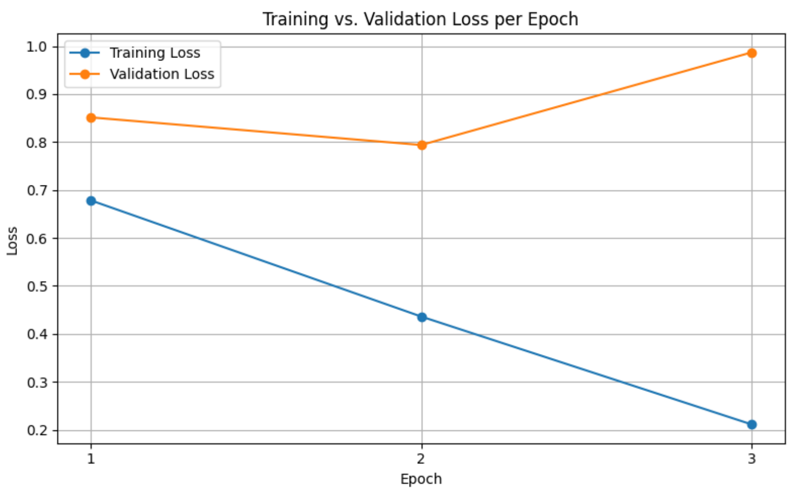


Appendix D: Sentiment Classification

Appendix D.1: Validation and Training Loss FinBERT

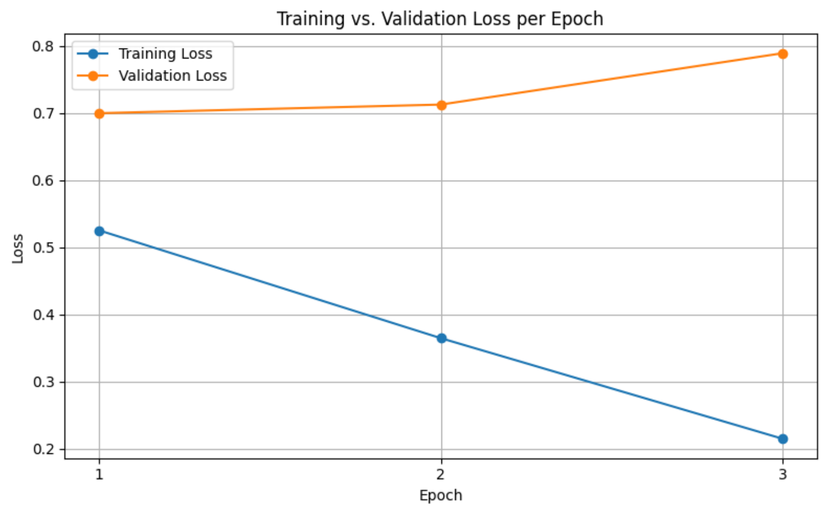
Epoch	Training Loss	Validation Loss	Balanced Accuracy	Accuracy
1	0.67850	0.85135	0.60781	0.63704

2	0.43610	0.79375	0.66385	0.67419
3	0.21110	0.98694	0.67440	0.66737



Appendix D.2: Validation and Training Loss RoBERTa

Epoch	**Training Loss**	**Validation Loss**	**Balanced Accuracy**	**Accuracy**
1	0.52540	0.69973	0.73005	0.72276
2	0.36470	0.71260	0.74600	0.74324
3	0.21480	0.78910	0.74303	0.74705



Model Variant	Subsample	Colsample by Tree	Max Depth	Learning Rate	N Estimators
Base	0.8	0.8	3	0.01	100
Base + Sentiment_score	0.8	0.8	3	0.01	100
Base + IWSS_Equal	0.6	0.6	5	0.01	100
Base + IWSS_Ratio	0.6	0.6	5	0.01	100
Base + IWSS_Impurity	0.6	0.6	5	0.01	100
Base + IWSS_perm	0.6	0.6	5	0.01	100

Base + IWSS_SHAP	0.6	0.6	5	0.01	100

	Correct predictions	Wrong predictions	Accuracy
Fine-tuned FinBERT	71	35	66.98%
Base FinBERT	17	89	16.04%
Base RoBERTa	64	42	60.38%
Fine-tuned RoBERTa	85	21	**80.19%**

Appendix E: Model results

Appendix E.1: OLS assumptions

Assumption	Purpose	Visual Check/Test	Test Description
Linearity	Ensures a linear relationship between the feature input and next-day returns.	Residuals vs. fitted values plot.	A random scatter suggests linearity; visible curves or patterns indicate misspecification (Wooldridge, 2013)

Independence of Errors	Residuals should not be correlated with each other.	Durbin-Watson test (time series)	Durbin-Watson tests for first-order autocorrelation. Values below 1.5 suggest positive autocorrelation. p-value < 0.05 implies significant autocorrelation. If errors are correlations (Durbin & Watson, 1950).
Homoscedasticity	Residual variance should be constant across predicted values.	Residuals vs. fitted values plot and Breusch-Pagan test	Funnel-shaped patterns suggest heteroscedasticity. Breusch-Pagan tests whether residual variance depends on predicted values. p-value < 0.05 indicates heteroscedasticity (Breusch & Pagan, 1979).
Normality of Residuals	Required for valid p-values and confidence intervals.	Histogram or Q-Q plot of residuals and Shapiro-Wilk test	Q-Q plot shows if residuals deviate from a normal distribution. Shapiro-Wilk formally tests for normality. p-value < 0.05 implies non-normality (Shapiro & Wilk, 1965)
No Perfect Multicollinearity	Ensures predictors are not exact linear combinations.	Variance Inflation Factor (VIF)	VIF quantifies how much a predictor is correlated with others. VIF > 5 is a red flag (Kutner et al., 2005).
Exogeneity (Zero Conditional Mean)	Predictors must not be correlated with the error term.	Residuals vs. each predictor	Patterns in residuals vs. predictors suggest omitted variable bias Wooldridge, 2010).

Appendix E.2: OLS Diebold-Mariano results

Model Compared	DM Statistic	p-value	Interpretation
Base + IWSS_Ratio	-3.211418	0.001625	IWSS better
Base + IWSS_Impurity	-3.209779	0.001634	IWSS better
Base + IWSS_perm	-3.206872	0.001649	IWSS better
Base + IWSS_SHAP	-3.204385	0.001662	IWSS better
Base + IWSS_Equal	-3.202146	0.001674	IWSS better

Comparison	DM Statistic	p-value	Interpretation
IWSS_perm vs. IWSS_SHAP	0.953101	0.342114	No significant difference
IWSS_Impurity vs. IWSS_SHAP	0.859910	0.391248	No significant difference

IWSS_Impurity vs. IWSS_perm	0.794520	0.428182	No significant difference
IWSS_Ratio vs. IWSS_SHAP	0.780922	0.436113	No significant difference
IWSS_Equal vs. IWSS_Impurity	-0.778293	0.437656	No significant difference
IWSS_Equal vs. IWSS_perm	-0.763591	0.446343	No significant difference
IWSS_Equal vs. IWSS_Ratio	-0.736064	0.462873	No significant difference
IWSS_Ratio vs. IWSS_perm	0.720821	0.472172	No significant difference
IWSS_Ratio vs. IWSS_Impurity	0.647441	0.518363	No significant difference
IWSS_Equal vs. IWSS_SHAP	-0.464399	0.643054	No significant difference

Appendix E.3: Random Forest Diebold-Mariano results

Model Compared	**DM Statistic**	**p-value**	**Interpretation**
Base + IWSS_Equal	-3.018763	0.002996	IWSS better
Base + IWSS_perm	-2.844202	0.005092	IWSS better

Base + IWSS_Impurity	-2.599585	0.010292	IWSS better
Base + IWSS_SHAP	-2.471250	0.014614	IWSS better
Base + IWSS_Ratio	-2.351493	0.020035	IWSS better

Comparison	**DM Statistic**	**p-value**	**Interpretation**	**Significant**
Base + IWSS_Equal vs. Sentiment_score	-3.018763	0.002996	IWSS better	True
Base + IWSS_perm vs. Sentiment_score	-2.844202	0.005092	IWSS better	True
Base + IWSS_Impurity vs. Sentiment_score	-2.599585	0.010292	IWSS better	True
Base + IWSS_SHAP vs. Sentiment_score	-2.471250	0.014614	IWSS better	True
Base + IWSS_Ratio vs. Sentiment_score	-2.351493	0.020035	IWSS better	True

Base + IWSS_Ratio vs. Base + IWSS_perm	2.076152	0.039633	Base + IWSS_Ratio better	True
Base + IWSS_Equal vs. Base + IWSS_SHAP	-2.000876	0.047260	Base + IWSS_SHAP better	True
Base + IWSS_Equal vs. Base + IWSS_Ratio	-1.877462	0.062450	Base + IWSS_Ratio better	False
Base + IWSS_Impurity vs. Base + IWSS_perm	1.875620	0.062704	Base + IWSS_Impurity better	False
Base + IWSS_Ratio vs. Base + IWSS_Impurity	1.453646	0.148191	Base + IWSS_Ratio better	False
Base + IWSS_perm vs. Base + IWSS_SHAP	-1.451825	0.148696	Base + IWSS_SHAP better	False
Base + IWSS_Equal vs. Base	-1.336677	0.183407	IWSS better	False
Base + IWSS_Equal vs. Base + IWSS_Impurity	-1.316314	0.190132	Base + IWSS_Impurity better	False
Base + IWSS_perm vs. Base	-1.192275	0.235088	IWSS better	False
Base + IWSS_Impurity vs. Base + IWSS_SHAP	-0.685527	0.494099	Base + IWSS_SHAP better	False

Base + IWSS_Ratio vs. Base	0.491320	0.623938	Base better	False
Base + IWSS_Impurity vs. Base	-0.490715	0.624365	IWSS better	False
Base + IWSS_Equal vs. Base + IWSS_perm	-0.461250	0.645305	Base + IWSS_perm better	False
Base + IWSS_SHAP vs. Base	0.238947	0.811482	Base better	False
Base + IWSS_Ratio vs. Base + IWSS_SHAP	0.179894	0.857485	Base + IWSS_Ratio better	False

Appendix E.4: XGBoost Diebold-Mariano results

Comparison	**DM Statistic**	**p-value**	**Interpretation**	**Significant**
Base+IWSS_Equal vs. Base+Sentiment_score	-1.358226	0.176487	IWSS better	False
Base+IWSS_Impurity vs. Base+Sentiment_score	-0.829265	0.408308	IWSS better	False

Base+IWSS_Ratio vs. Base+Sentiment_score	-0.873078	0.384054	IWSS better	False
Base+IWSS_SHAP vs. Base+Sentiment_score	-0.826495	0.409872	IWSS better	False
Base+IWSS_perm vs. Base+Sentiment_score	-0.496193	0.620505	IWSS better	False
Base+IWSS_Equal vs. Base	-0.467102	0.641123	IWSS better	False
Base+IWSS_Impurity vs. Base	0.494454	0.621728	Base better	False
Base+IWSS_Ratio vs. Base	0.316468	0.752099	Base better	False
Base+IWSS_SHAP vs. Base	0.374659	0.708458	Base better	False
Base+IWSS_perm vs. Base	0.887741	0.376140	Base better	False
Base+IWSS_Equal vs. Base+IWSS_Impurity	-1.477999	0.141562	Base+IWSS_Impurity better	False
Base+IWSS_Equal vs. Base+IWSS_Ratio	-0.859890	0.391259	Base+IWSS_Ratio better	False

Base+IWSS_Equal vs. Base+IWSS_SHAP	-1.213728	0.226811	Base+IWSS_SHAP better	False
Base+IWSS_Equal vs. Base+IWSS_perm	-1.727297	0.086229	Base+IWSS_perm better	False
Base+IWSS_Impurity vs. Base+IWSS_Ratio	0.266060	0.790568	Base+IWSS_Impurity better	False
Base+IWSS_Impurity vs. Base+IWSS_SHAP	0.197963	0.843349	Base+IWSS_Impurity better	False
Base+IWSS_Impurity vs. Base+IWSS_perm	-1.060313	0.290753	Base+IWSS_perm better	False
Base+IWSS_Ratio vs. Base+IWSS_SHAP	-0.131350	0.895679	Base+IWSS_SHAP better	False
Base+IWSS_Ratio vs. Base+IWSS_perm	-1.135159	0.258169	Base+IWSS_perm better	False
Base+IWSS_SHAP vs. Base+IWSS_perm	-1.544032	0.124745	Base+IWSS_perm better	False

Appendix E.5: LSTM Diebold-Mariano results

Comparison	DM Statistic	p-value	Better	Significant
Base+IWSS_SHAP vs Sentiment_score	-2.844607	0.005086	Base+IWSS_SHAP	True
Base+IWSS_perm vs Sentiment_score	-2.813702	0.005573	Base+IWSS_perm	True
Base+IWSS_Equal vs Sentiment_score	-2.662950	0.008615	Base+IWSS_Equal	True
Base+IWSS_Ratio vs Base+IWSS_SHAP	2.528619	0.012514	Base+IWSS_Ratio	True
Base+IWSS_Ratio vs Base+IWSS_perm	2.461283	0.015009	Base+IWSS_Ratio	True
Base+IWSS_Equal vs Base+IWSS_Ratio	-2.405715	0.017392	Base+IWSS_Ratio	True
Base+IWSS_Equal vs Base+IWSS_Impurity	-1.824793	0.070076	Base+IWSS_Impurity	False
Base+IWSS_Ratio vs Base	1.693619	0.092470	Base	False

Base+IWSS_Ratio vs Sentiment_score	-1.583898	0.115381	Base+IWSS_Ratio	False
Base+IWSS_Impurity vs Sentiment_score	-1.149007	0.252432	Base+IWSS_Impurity	False
Base+IWSS_Impurity vs Base	0.978842	0.329277	Base	False
Base+IWSS_SHAP vs Base+IWSS_perm	-0.938323	0.349628	Base+IWSS_perm	False
Base+IWSS_Equal vs Base	-0.879334	0.380665	Base+IWSS_Equal	False
Base+IWSS_Equal vs Base+IWSS_perm	-0.845119	0.399427	Base+IWSS_perm	False
Base+IWSS_Equal vs Base+IWSS_SHAP	-0.795947	0.427355	Base+IWSS_SHAP	False
Base+IWSS_Impurity vs Base+IWSS_SHAP	0.751350	0.453652	Base+IWSS_Impurity	False
Base+IWSS_Impurity vs Base+IWSS_perm	0.712089	0.477546	Base+IWSS_Impurity	False
Base+IWSS_perm vs Base	0.315158	0.753092	Base	False

Base+IWSS_Impurity vs Base+IWSS_Ratio	-0.247980	0.804498	Base+IWSS_Ratio	False
Base+IWSS_SHAP vs Base	0.244576	0.807128	Base	False

Appendix F: Code

The link to the github repository is here: https://github.com/SebastianG-J/Master-Thesis